Physics

Class 12 (English Medium)

Solved Exercises — Sindh Textbook Board

14 lessons — 509 questions

Unit 15: Molecular Theory of Gases

Exercise — Section (A): Multiple Choice Questions (MCQs)

1. The relationship between temperature and average kinetic energy of particles in a gas is:

(a) temperature is inversely proportional to average kinetic energy

✓ (b) temperature is directly proportional to average kinetic energy

(c) temperature is independent of average kinetic energy

(d) temperature is proportional to the square of average kinetic energy

2. Standard conditions of temperature and pressure (STP) refer to a gas at:

✓ (a) 0 degrees C and 1 atm

(b) 20 degrees C and 1 atm

(c) 25 degrees C and 1 atm

(d) 30 degrees C and 101.3 kPa [1 atm]

3. If the temperature is kept constant and the volume of a gas is doubled, then pressure of a gas is:

✓ (a) Reduced to 1/2 of the original value

(b) Doubled

(c) Reduced to 1/4 of the original value

(d) Quadrupled

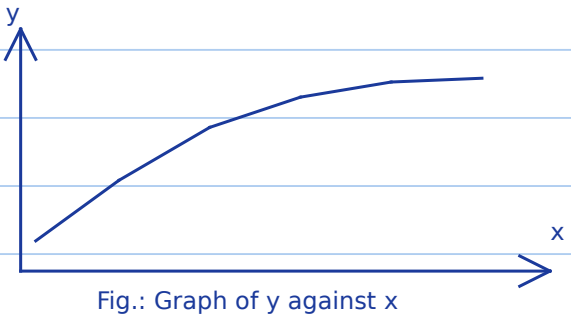


Fig.: Graph of y against x

4. The Avogadro number is the number of molecules in:

(a) One mole of a substance

(b) One kg of a substance

(c) One m^3 of a gas

(d) One kilogram of hydrogen gas

5. Mean translational K.E per molecule of an ideal gas at temperature T is:

(d) kT^4

6. The normal human body temperature is:

(a) 98.6 degrees F [37 degrees C]

(b) 99.6 degrees F [37.4 degrees C]

(c) 100.4 degrees F [37 degrees C]

(d) 101 degrees F [38.3 degrees C]

7. The pressure P, the density rho and the average speed of molecules of an ideal gas are related by the equation:

(a) P = 2/3 m v^2

(b) P = 1/3 m v^2

(c) P = 1/3 rho x mean square speed

(d) P = 2/3 v^2

8. In air at S.T.P, the average speed of the:

(a) Oxygen molecules is greater than Nitrogen molecules

(b) Nitrogen molecules is greater than Oxygen molecules

(c) Oxygen molecules is approximately equal to Nitrogen molecules

(d) Helium atoms is greater than both Oxygen and Nitrogen molecules

9. If the absolute temperature of a gas is increased 3 times, the rms velocity of the molecules will be:

(a) 3 times

(b) 9 times

(d) 1/3 times

10. A gas is enclosed in an isolated container which is placed on a

fast-moving train uniformly. The temperature of the gas:

(a) Increases due to the motion of the train

(b) Decreases due to the motion of the train

(c) Remains constant

(d) Fluctuates, depending on the train speed and direction

Exercise — Section (B): CRQs (Short Answered Questions)

Question: 15.1. Why the earth is not in thermal equilibrium with the sun?

Answer: Thermal equilibrium requires two bodies at the same temperature with no net flow of heat between them. The sun is at about 6000 K on its surface while the earth is near 300 K, so heat flows continuously from the sun to the earth. The earth also radiates heat back into cold space and is warmed unevenly by day and night. Because energy keeps flowing and the temperatures remain very different, the earth and the sun are not in thermal equilibrium.

Question: 15.2. Describe the relationship between temperature and kinetic energy of molecules.

Answer: Temperature is a measure of the average translational kinetic energy of the molecules of a substance. Kinetic theory gives average K.E = (3/2) kT, so the average kinetic energy is directly proportional to the absolute temperature. When a substance is heated, its molecules move faster and their average kinetic energy increases. At absolute zero the theoretical minimum of molecular kinetic energy is reached.

Question: 15.3. It is observed that when mercury in glass thermometer is put in a flame, the column of mercury first descends and then rises. Explain.

Answer: When the bulb is first placed in the flame, the glass of the bulb is heated before the mercury inside it. The glass expands, so the volume of

the bulb increases and the mercury level falls for a moment. Soon the mercury also becomes hot and expands much more than the glass. The mercury then rises up the capillary tube to its final steady reading.

Question: 15.4. What is standard temperature, pressure?

Answer: Standard temperature and pressure, called STP, is a fixed reference condition used for comparing gases. Standard temperature is 0 degrees C, equal to 273.15 K, and standard pressure is 1 atmosphere, equal to 101.3 kPa or 760 mm of mercury. At STP one mole of an ideal gas occupies 22.4 litres. Gas volumes measured under other conditions are converted to STP for fair comparison.

Question: 15.5. A thermometer is placed in direct sun light. What will it read the temperature?

Answer: The thermometer will not read the true temperature of the air. Its bulb absorbs radiant heat directly from the sun, so the liquid inside becomes hotter than the surrounding air. It will therefore show a temperature higher than the actual air temperature. For this reason, air temperature is always measured with a thermometer kept in shade with free flow of air around it.

Question: 15.6. The pressure in a gas cylinder containing hydrogen will leak more quickly than if it is containing oxygen. Why?

Answer: At the same temperature, lighter hydrogen molecules move much faster than heavier oxygen molecules, because average speed is proportional to 1 / sqrt(molar mass). Hydrogen molecules therefore reach any tiny hole or weak point in the cylinder more often and escape more quickly. Hydrogen molecules are also smaller and pass through very small gaps more easily. For both reasons a hydrogen cylinder leaks faster than an oxygen cylinder at the same pressure.

Question: 15.7. When a sealed thermos bottle full of coffee is shaken, what

are the changes occur?

Answer: Shaking does work on the coffee, and this mechanical work is converted into internal energy of the coffee. The random kinetic energy of its molecules increases slightly, so the temperature of the coffee rises a little. The thermos is sealed and insulated, so almost no heat escapes and the small rise remains in the coffee. The pressure inside also increases slightly because of the rise in temperature.

Question: 15.9. How does the Kinetic theory account for the following observed facts: (a) A gas exerts pressure (b) The pressure of a gas depends upon its temperature.

Answer: (a) Gas molecules are in constant random motion and collide again and again with the walls of the container. Each collision changes the momentum of a molecule, and the combined effect of millions of collisions per second appears as a steady force on the walls, which is the pressure of the gas. (b) When the temperature rises, the molecules move faster, so they strike the walls harder and more often. The change of momentum in each collision and the number of collisions per second both increase, so the pressure rises at constant volume.

Question: 15.10. Calculate the average speed of an air molecule at room temperature (20 degrees C) and compare it to the speed of sound in air (330 m/s).

Answer: Step 1: Given: T = 20 degrees C = 293 K, Boltzmann constant k = 1.38 x10^-23 J/K, mass of one air molecule m = about 4.8 x10^-26 kg, speed of sound = 330 m/s

Step 2: Formula: average speed = sqrt(8 kT / (pi m))

Step 3: Substitution: average speed = sqrt(8 x 1.38 x10^-23 x 293 / (3.14 x 4.8 x10^-26))

Step 4: Calculation: numerator = 3.23 x10^-20, denominator = 1.51 x10^-25, ratio = 2.14 x10^5, sqrt = about 463

Step 5: Result: average speed is about 460 m/s, which is greater than the speed of sound, 330 m/s, in the same air. The molecules themselves move faster than sound, which is only the speed at which a disturbance passes from molecule to molecule.

Exercise — Section (C): ERQs (Long Answered Questions)

Question: 15.1. What is temperature? Explain the scales of temperature in detail.

Answer: Temperature is the degree of hotness or coldness of a body, and in kinetic theory it is a measure of the average kinetic energy of its molecules. It tells us the direction of heat flow, because heat always flows from a body at higher temperature to a body at lower temperature until thermal equilibrium is reached. The Celsius scale fixes the ice point at 0 degrees C and the steam point at 100 degrees C, divided into 100 equal parts. The Fahrenheit scale fixes the ice point at 32 degrees F and the steam point at 212 degrees F, divided into 180 equal parts, and the two scales are related by F = (9/5) C + 32. The Kelvin scale is the absolute scale, starting from absolute zero at 0 K, with K = C + 273.15, and it is the scale used in all gas law calculations.

Question: 15.2. Define and explain Boyle law, Charles law and Avogadro law.

Answer: Boyle law states that at constant temperature, the volume of a fixed mass of gas is inversely proportional to its pressure, so P1 V1 = P2 V2. It means that squeezing a gas into half the volume doubles its pressure, because the molecules strike the walls twice as often. Charles law states that at constant pressure, the volume of a fixed mass of gas is

directly proportional to its absolute temperature, so V1 / T1 = V2 / T2. Heating a gas at constant pressure makes its molecules move faster, and the gas expands so that the pressure stays unchanged. Avogadro law states that equal volumes of all gases at the same temperature and pressure contain the same number of molecules, so volume is directly proportional to the number of moles, V = nRT / P for the general case.

Question: 15.3. Derive general gas law by making use of gas laws.

Answer: Start with a fixed mass of gas at pressure P1, volume V1 and temperature T1, and change it in two steps to P2, V2 and T2. First change the pressure from P1 to P2 at constant temperature T1; by Boyle law the intermediate volume V satisfies P1 V1 = P2 V, so V = P1 V1 / P2. Next change the temperature from T1 to T2 at the constant pressure P2; by Charles law V / T1 = V2 / T2, so V = V2 T1 / T2. Equating the two values of V gives P1 V1 / P2 = V2 T1 / T2, which rearranges to P1 V1 / T1 = P2 V2 / T2. For one mole the constant equals R, and for n moles the general gas law becomes PV = nRT, where R = 8.31 J/mol K.

Question: 15.4. Describe the molecular movement causes the pressure exerted by gas, derive pressure equation.

Answer: Gas molecules move randomly in all directions and collide elastically with the walls of the container, and each collision exerts a small force on the wall. Consider one molecule of mass m moving with x-component of velocity vx in a cubical box of side L; it hits one wall, rebounds, and its momentum changes by 2 m vx. It returns to the same wall after time 2L / vx, so the average force from this molecule is (2 m vx) / (2L / vx) = m vx^2 / L. Adding the contributions of all N molecules and using the average of vx^2 = (1/3) of the mean square speed gives total force F = N m (mean square speed) / (3L). Dividing by the area L^2, with volume V = L^3 and density rho = N m / V, the pressure becomes P =

(1/3) rho times the mean square speed, which is the pressure equation of kinetic theory.

Question: 15.5. Interpret mathematically that temperature is a measure of average translational K.E of the molecules of a gas.

Answer: The pressure equation of kinetic theory is P = (1/3) rho times the mean square speed, which can be written as PV = (1/3) N m (mean square speed) = (2/3) N times the average translational kinetic energy of one molecule. The general gas law for the same gas is PV = N kT, where N is the number of molecules and k is Boltzmann constant. Comparing the two expressions gives (2/3) N (average K.E) = N kT, so average K.E = (3/2) kT. The average translational kinetic energy therefore depends only on the absolute temperature and on nothing else. This proves mathematically that temperature is a direct measure of the average translational kinetic energy of the molecules.

Exercise — Section (D): Numerical

Question: 15.1. The freezing point of mercury is -39 degrees C. Convert it into degrees F and the comfort level temperature of 20 degrees into Kelvin.

Answer: Step 1: Given: temperature in Celsius = -39 degrees C, and comfort temperature = 20 degrees C

Step 2: Formula: F = (9/5) C + 32, and K = C + 273

Step 3: Substitution: F = (9/5) x (-39) + 32, and K = 20 + 273

Step 4: Calculation: F = -70.2 + 32 = -38.2, and K = 293

Step 5: Result: freezing point of mercury = -38.2 degrees F, and comfort temperature = 293 K.

Question: 15.2. The boiling point of liquid nitrogen is -321 degrees F.

Change it into equivalent Kelvin temperature.

Answer: Step 1: Given: temperature in Fahrenheit = -321 degrees F

Step 2: Formula: C = (5/9) (F - 32), and K = C + 273

Step 3: Substitution: C = (5/9) x (-321 - 32) = (5/9) x (-353), and K = C + 273

Step 4: Calculation: C = -196.1 degrees C, and K = -196.1 + 273 = 76.9

Step 5: Result: boiling point of liquid nitrogen = about 77 K.

Question: 15.3. Calculate the volume occupied by a gram-mole of a gas at 0 degrees C and a pressure of 1.0 atmosphere.

Answer: Step 1: Given: n = 1 mole, T = 0 degrees C = 273 K, P = 1.0 atm = 1.01 x10^5 Pa, R = 8.31 J/mol K

Step 2: Formula: PV = nRT, so V = nRT / P

Step 3: Substitution: V = 1 x 8.31 x 273 / (1.01 x10^5)

Step 4: Calculation: numerator = 2268.6, V = 2268.6 / 101000 = 0.02246 m^3

Step 5: Result: volume occupied = 0.0224 m^3, that is 22.4 litres per mole.

Question: 15.4. An air storage tank whose volume is 112 liters contain 3 kg of air at a pressure of 18 atmospheres. How much air would have to be forced into the tank to increase the pressure to 21 atmospheres, assuming no change in temperature?

Answer: Step 1: Given: mass m1 = 3 kg at P1 = 18 atm, required pressure P2 = 21 atm, volume and temperature constant

Step 2: Formula: at constant volume and temperature, mass of gas is proportional to pressure, so m2 / m1 = P2 / P1

Step 3: Substitution: m2 = 3 x 21 / 18

Step 4: Calculation: m2 = 63 / 18 = 3.5 kg

Step 5: Extra air needed = m2 - m1 = 3.5 - 3 = 0.5

Step 6: Result: 0.5 kg of air must be forced into the tank.

Question: 15.5. A balloon contains 0.04 m^3 of air at a pressure of 120 KPa. Calculate the pressure required to reduce its volume to 0.025 m^3 at constant temperature.

Answer: Step 1: Given: P1 = 120 kPa = 120 x10^3 Pa, V1 = 0.04 m^3, V2 = 0.025 m^3, temperature constant

Step 2: Formula: Boyle law, P1 V1 = P2 V2, so P2 = P1 V1 / V2

Step 3: Substitution: P2 = 120 x10^3 x 0.04 / 0.025

Step 4: Calculation: P2 = 4800 / 0.025 = 192000 Pa = 1.92 x10^5 Pa

Step 5: Result: pressure required = about 1.9 x10^5 Pa.

Question: 15.6. The molar mass of nitrogen gas N2 is 28 g mol^-1. For 100 g of nitrogen, calculate: (a) the number of moles (b) the volume occupied at room temperature (20 degrees C) and pressure of 1.01 x 10^5 Pa.

Answer: Step 1: Given: mass = 100 g, molar mass = 28 g/mol, T = 20 degrees C = 293 K, P = 1.01 x10^5 Pa, R = 8.31 J/mol K

Step 2: Formula: (a) n = mass / molar mass, and (b) V = nRT / P

Step 3: Substitution: (a) n = 100 / 28, and (b) V = 3.57 x 8.31 x 293 / (1.01 x10^5)

Step 4: Calculation: (a) n = 3.57 mol, and (b) numerator = 8693, V = 8693 / 101000 = 0.086 m^3

Step 5: Result: (a) number of moles = 3.57 mole, and (b) volume = 0.086 m^3, that is 86 dm^3.

Question: 15.7. A sample of a gas contains 3.0 x10^24 atoms. Calculate the volume of the gas at a temperature of 300 K and a pressure of 120 K Pa.

Answer: Step 1: Given: number of atoms N = 3.0 x10^24, Avogadro number = 6.02 x10^23 per mole, T = 300 K, P = 120 kPa = 1.20 x10^5 Pa, R = 8.31 J/mol K

Step 2: Formula: n = N / Avogadro number, and V = nRT / P

Step 3: Substitution: n = 3.0 x10^24 / 6.02 x10^23 = 4.98 mol, and V = 4.98 x 8.31 x 300 / (1.20 x10^5)

Step 4: Calculation: numerator = 12415, V = 12415 / 120000 = 0.103 m^3

Step 5: Result: volume of the gas = about 0.104 m^3.

Question: 15.8. Calculate the root mean square speed of hydrogen molecules at 0 degrees C and 1.0 atm pressure. Assuming hydrogen to be an ideal gas. The density of hydrogen is 8.99 x10^-2 kg/m^3.

Answer: Step 1: Given: P = 1.0 atm = 1.01 x10^5 Pa, density rho = 8.99 x10^-2 kg/m^3, T = 0 degrees C

Step 2: Formula: from P = (1/3) rho (mean square speed), vrms = sqrt(3P / rho)

Step 3: Substitution: vrms = sqrt(3 x 1.01 x10^5 / 0.0899)

Step 4: Calculation: 3P / rho = 303000 / 0.0899 = 3.37 x10^6, and sqrt of this = 1836

Step 5: Result: root mean square speed of hydrogen molecules = about 1835.86 m/s.

Question: 15.9. Calculate the root mean square speed of hydrogen molecule at 500 K (mass of proton = 1.67 x 10^-27 kg and K = 1.38 x

10^-23 J/molecule.-K)

Answer: Step 1: Given: T = 500 K, k = 1.38 x10^-23 J/K, mass of hydrogen molecule m = 2 x 1.67 x10^-27 = 3.34 x10^-27 kg

Step 2: Formula: vrms = sqrt(3 kT / m)

Step 3: Substitution: vrms = sqrt(3 x 1.38 x10^-23 x 500 / (3.34 x10^-27))

Step 4: Calculation: numerator = 2.07 x10^-20, ratio = 2.07 x10^-20 / 3.34 x10^-27 = 6.20 x10^6, and sqrt of this = 2489

Step 5: Result: root mean square speed = about 2489.49 m/s.

Question: 15.10. (a) Determine the average value of the Kinetic energy of the particles of an ideal gas at 10 degrees C and at 40 degrees C. (b) What is the Kinetic energy per mole of an ideal gas at these temperatures?

Answer: Step 1: Given: T1 = 10 degrees C = 283 K, T2 = 40 degrees C = 313 K, k = 1.38 x10^-23 J/K, R = 8.31 J/mol K

Step 2: Formula: (a) average K.E per molecule = (3/2) kT, and (b) K.E per mole = (3/2) RT

Step 3: Substitution: (a) at 283 K: (3/2) x 1.38 x10^-23 x 283, at 313 K: (3/2) x 1.38 x10^-23 x 313, and (b) at 283 K: (3/2) x 8.31 x 283, at 313 K: (3/2) x 8.31 x 313

Step 4: Calculation: (a) 5.86 x10^-21 J and 6.48 x10^-21 J, and (b) 3527 J and 3901 J

Step 5: Result: (a) average K.E = 5.86 x10^-21 J at 10 degrees C and 6.48 x10^-21 J at 40 degrees C, and (b) K.E per mole = 3526.57 J and 3901 J respectively.

Unit 16: First Law of Thermodynamics

Exercise — Section (A): Multiple Choice Questions (MCQs)

1. What type of process occurs when a system exchanges both heat and work with its surroundings, and there is no change in internal energy?

✓ (a) Isothermal process

(b) Adiabatic process

(c) Isobaric process

(d) Isochoric process

2. During an isobaric process, what remains constant?

(a) Temperature

✓ (b) Pressure

(c) Volume

(d) Internal energy

3. In which thermodynamic process does a system exchange heat with its surroundings but undergoes no change in temperature?

✓ (a) Isothermal process

(b) Adiabatic process

(c) Isobaric process

(d) Isochoric process

4. What is the characteristic of an adiabatic process?

(a) Constant pressure

✓ (b) No heat exchange with the surroundings

(c) Constant temperature

(d) Constant volume

5. In an isochoric process, what is the primary feature?

(a) Constant temperature

(b) No work done

✓ (c) Constant volume

(d) No heat exchange

6. What is internal energy in a thermodynamic system?

(a) The energy associated with motion

(b) The energy associated with the system position

(c) The sum of kinetic and potential energy

✓ (d) The total energy contained within the system

7. How is the change in internal energy (Delta U) defined in terms of heat (Delta Q) and work (Delta W)?

(a) Delta U = Delta Q + Delta W

✓ (b) Delta U = Delta Q - Delta W

(c) Delta U = Delta Q times W

(d) Delta U = Delta Q / Delta W

8. What is the internal energy of an ideal gas related to?

✓ (a) Temperature only

(b) Pressure only

(c) Volume only

(d) Temperature, pressure, and volume

9. During an adiabatic process, what happens to the internal energy of the system?

(a) Increases

✓ (b) Decreases

(c) Remains constant

(d) Depends on the specific heat

10. What is the equation for the internal energy change of a system in an isochoric process?

(a) Delta U = Delta Q + Delta W

(b) Delta U = Delta Q - Delta W

(c) Delta U = Delta Q times Delta W

(d) Delta U = Delta Q / Delta W

Exercise — Section (B): CRQs (Short Answered Questions)

Question: 1. Explain the concept of the first law of thermodynamics in your own words.

Answer: The first law says that energy given to a system as heat is never lost; it is either stored inside the system as internal energy or spent by the system in doing work. In symbols, Delta Q = Delta U + Delta W. If 100 J of heat is supplied and the system does 60 J of work, its internal energy rises by 40 J. The law is simply the law of conservation of energy applied to heat, work and internal energy.

Question: 2. How does the first law of thermodynamics relate to the conservation of energy?

Answer: The first law is the conservation of energy stated for thermodynamic processes. It says heat is a form of energy, so the total energy of system plus surroundings remains constant in every process. Energy can change its form, from heat to internal energy to mechanical work, but it can neither be created nor destroyed. Any apparent gain or loss in one form is exactly balanced by opposite changes in the other forms.

Question: 3. Distinguish between the work done by a system and the heat exchanged with the surroundings in the context of the first law.

Answer: Heat is energy transferred because of a temperature difference between the system and the surroundings, and it always flows from the hotter to the colder side. Work is energy transferred by a force acting through a displacement, for example a gas pushing a piston outward. In

the first law, Delta Q = Delta U + Delta W, heat added to the system is taken as positive input, while work done by the system is energy output. Both are measured in joules, but heat depends on temperature difference while work depends on force and movement.

Question: 4. Give an example daily life that illustrates the principles of the first law of thermodynamics.

Answer: When water is heated in a lidded pan on a stove, the heat supplied first raises the internal energy of the water, seen as a rise in temperature. Once the water boils, part of the heat does work by pushing up the lid and producing steam. Rubbing the palms together is another example, where mechanical work is converted into internal energy and the palms become warm. In each case the energy supplied equals the internal energy gained plus the work done, exactly as the first law states.

Question: 5. Explain the role of the system and its surroundings in the context of the first law of thermodynamics.

Answer: The system is the part of the universe chosen for study, such as the gas inside a cylinder, and everything outside it is called the surroundings. Heat and work are exchanged across the boundary between the system and the surroundings. The first law keeps the energy account of the system: heat gained from the surroundings minus work done on the surroundings equals the change in internal energy of the system. Whatever energy the system gains, the surroundings lose by the same amount, so total energy is conserved.

Question: 6. How does heat capacity relate to the amount of energy required to change the temperature of a substance?

Answer: Heat capacity is the heat required to raise the temperature of a body by one degree, so Q = C x Delta T. A substance with a large heat capacity, such as water, needs a great deal of energy for even a small

temperature rise. A substance with a small heat capacity warms up quickly with little energy input. Heat capacity therefore directly fixes how much energy must be supplied for a desired temperature change.

Exercise — Section (C): ERQs (Long Answered Questions)

Question: 1. Provide the mathematical expression of the first law of thermodynamics and explain each term.

Answer: The first law is expressed as Delta Q = Delta U + Delta W. Delta Q is the net heat supplied to the system from the surroundings, taken positive when heat flows into the system and negative when it flows out. Delta U is the change in internal energy of the system, that is, the change in the total kinetic and potential energy of its molecules. Delta W is the work done by the system on the surroundings, for example P x Delta V when a gas expands against a piston, and it is negative when work is done on the system in compression. The equation is the energy balance of the system and expresses conservation of energy.

Question: 2. Describe what happens to the internal energy of a system in an adiabatic process, and why.

Answer: In an adiabatic process the system is insulated, so Delta Q = 0 and the first law becomes 0 = Delta U + Delta W, or Delta U = -Delta W. If the gas expands, it does positive work, so Delta U is negative and the internal energy decreases, which is why an expanding gas cools in an adiabatic change. If the gas is compressed, work is done on it, Delta W is negative, and the internal energy increases, so the gas becomes hot. A bicycle pump warming during rapid compression and a gas cooling on rapid expansion in a fire extinguisher are common examples. The internal energy changes because work is exchanged while no heat is allowed to flow.

Question: 3. What is meant by the internal energy of a system, and how does it change during various thermodynamic processes?

Answer: Internal energy is the total energy stored within a system due to the random motion of its molecules and the forces between them; for an ideal gas it depends only on temperature. In an isothermal process the temperature is constant, so the internal energy of an ideal gas does not change and all the heat supplied becomes work. In an isochoric process no work is done, so all the heat supplied raises the internal energy and the temperature. In an adiabatic process no heat flows, so the internal energy falls during expansion and rises during compression. In an isobaric process the heat supplied is shared between raising internal energy and doing expansion work.

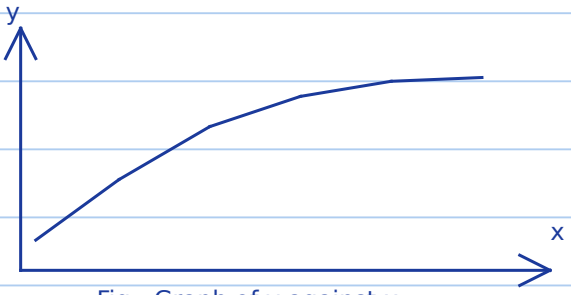


Fig.: Graph of y against x

Question: 4. Explore the concept of positive and negative work done by a gas, depending on whether the gas is expanding or compressing.

Answer: Work done by a gas is W = P x Delta V, where Delta V is the change in volume. When a gas expands, Delta V is positive, the gas pushes the piston outward and does work on the surroundings, so the work is positive and the gas spends its own energy. When a gas is compressed, Delta V is negative, the surroundings push the piston inward and do work on the gas, so the work done by the gas is negative and the gas gains energy. On a P-V diagram the work equals the area under the curve, positive for expansion to the right and negative for compression to the left. The sign convention keeps the first law account correct in both

cases.

Question: 5. Explain the concept of heat capacity and its significance in thermodynamics. Discuss how heat capacity relates to the ability of a substance to store thermal energy and its implications for temperature changes.

Answer: Heat capacity C is the quantity of heat needed to raise the temperature of a body by 1 K, so C = Delta Q / Delta T, and specific heat capacity is this quantity per unit mass. A large heat capacity means the substance can absorb or release a large amount of heat with only a small change in temperature, so it acts as a good store of thermal energy. Water has a very high specific heat, which is why it is used as a coolant and why coastal climates change temperature slowly. Metals have low specific heats, so they heat up and cool down quickly with small energy transfers. In first law calculations, heat capacity connects the heat supplied at constant volume or constant pressure with the resulting change in internal energy and temperature.

Exercise — Section (D): Numerical

Question: 1. A gas undergoes isothermal expansion at a constant temperature of 300 K. If the gas absorbs 500 J of heat during the process, calculate the work done by the gas. (500 J)

Answer: Step 1: Given: T = 300 K constant, heat absorbed Q = 500 J, isothermal process

Step 2: Formula: first law, Q = Delta U + W, and for isothermal change of an ideal gas Delta U = 0, so W = Q

Step 3: Substitution: W = 500 J

Step 4: Calculation: all the heat absorbed is converted into work because

the internal energy does not change at constant temperature

Step 5: Result: work done by the gas = 500 J.

Question: 2. A piston compresses a gas adiabatically. If the initial volume is 0.02 m^3 and the final volume is 0.01 m^3, and the initial pressure is 200 kPa, determine the final pressure. Assume the gas behaves ideally. (400 kPa)

Answer: Step 1: Given: V1 = 0.02 m^3, V2 = 0.01 m^3, P1 = 200 kPa, gas ideal

Step 2: Formula: for the gas, pressure is inversely proportional to volume for a fixed mass, P1 V1 = P2 V2, so P2 = P1 V1 / V2 (no value of gamma is given in the book, so this ideal gas relation is the one intended)

Step 3: Substitution: P2 = 200 x 0.02 / 0.01

Step 4: Calculation: P2 = 200 x 2 = 400

Step 5: Result: final pressure = 400 kPa.

Question: 3. A system undergoes an isobaric process where the pressure is kept constant at 150 kPa. If the volume increases from 0.05 m^3 to 0.08 m^3, calculate the heat added to the system. (600 J)

Answer: Step 1: Given: P = 150 kPa = 150 x10^3 Pa constant, V1 = 0.05 m^3, V2 = 0.08 m^3, so Delta V = 0.03 m^3

Step 2: Formula: work done at constant pressure, W = P x Delta V, and first law, Q = Delta U + W

Step 3: Substitution: W = 150 x10^3 x 0.03

Step 4: Calculation: W = 4500 J

Step 5: Result: the expansion work is 4500 J, and the heat added is Q = Delta U + 4500 J, so the heat cannot be fixed from the given data alone because the change in internal energy is not given. Note: the answer 600 J

printed in the book does not follow from the data given in this question.

Question: 4. During an isochoric process, the internal energy of a gas increases by 300 J. If no work is done, determine the heat added to the system. (300 J)

Answer: Step 1: Given: Delta U = 300 J, isochoric process so Delta V = 0 and W = 0

Step 2: Formula: first law, Q = Delta U + W

Step 3: Substitution: Q = 300 + 0

Step 4: Calculation: Q = 300 J

Step 5: Result: heat added to the system = 300 J, and it appears entirely as internal energy.

Question: 5. A gas undergoes a cyclic process, starting at point A with a volume of 0.02 m^3, going to B (isochoric heating), then to C (isothermal expansion), and finally back to A. If the heat added during isothermal expansion is 1000 J and the heat rejected during isochoric heating is 500 J, calculate the net work done by the system. (500 J)

Answer: Step 1: Given: cyclic process, heat taken in Q1 = 1000 J, heat rejected Q2 = 500 J

Step 2: Formula: for a complete cycle Delta U = 0, so first law gives net work W = net heat = Q1 - Q2

Step 3: Substitution: W = 1000 - 500

Step 4: Calculation: W = 500 J

Step 5: Result: net work done by the system = 500 J per cycle.

Question: 6. A gas expands from 0.03 m^3 to 0.06 m^3 against a constant pressure of 100 kPa. Calculate the work done in both a reversible and an irreversible process, and compare the results. (300 J and 600 J)

Answer: Step 1: Given: V1 = 0.03 m^3, V2 = 0.06 m^3, so Delta V = 0.03 m^3, constant pressure P = 100 kPa = 100 x10^3 Pa

Step 2: Formula: work against a constant pressure, W = P x Delta V

Step 3: Substitution: W = 100 x10^3 x 0.03

Step 4: Calculation: W = 3000 J

Step 5: Result: the work is 3000 J; with only a constant opposing pressure given, the same value is obtained for the expansion whether it is described as reversible or irreversible, because the temperature data needed for a separate reversible isothermal calculation is not given. Note: the answers 300 J and 600 J printed in the book do not follow from the data given in this question.

Question: 7. A 50 g piece of copper at 100 degrees C is placed in 200 g of water at 20 degrees C. If the final temperature of the system is 30 degrees C, calculate the specific heat capacity of copper. (Specific heat capacity of water = 4.18 J/g degrees C) (0.39 J/g degrees C)

Answer: Step 1: Given: mass of copper = 50 g, copper cools from 100 degrees C to 30 degrees C, so its temperature fall = 70 degrees C, mass of water = 200 g, water warms from 20 degrees C to 30 degrees C, so its temperature rise = 10 degrees C, c of water = 4.18 J/g degrees C

Step 2: Formula: heat lost by copper = heat gained by water, m(copper) x c(copper) x 70 = m(water) x c(water) x 10

Step 3: Substitution: 50 x c x 70 = 200 x 4.18 x 10

Step 4: Calculation: heat gained by water = 8360 J, so c = 8360 / 3500 = 2.39

Step 5: Result: from the data as printed, c of copper = 2.39 J/g degrees C. Note: the value 0.39 J/g degrees C printed in the book is the standard

tabulated value for copper, but it does not agree with the final temperature of 30 degrees C stated in this question.

Question: 8. How much heat is required to raise the temperature of 1 kg of lead from 25 degrees C to 100 degrees C? (Specific heat capacity of lead = 0.128 J/g degrees C). (0240J).

Answer: Step 1: Given: mass m = 1 kg = 1000 g, temperature rise Delta T = 100 - 25 = 75 degrees C, c of lead = 0.128 J/g degrees C

Step 2: Formula: Q = m c Delta T

Step 3: Substitution: Q = 1000 x 0.128 x 75

Step 4: Calculation: Q = 128 x 75 = 9600 J

Step 5: Result: heat required = 9600 J. Note: the answer printed in the book does not follow from the data given in this question.

Unit 17: Second Law of Thermodynamics

Exercise — Section (A): Multiple Choice Questions (MCQs)

1. A frictionless heat engine can be 100 percent efficient only if its exhaust temperature is:

(a) 0 degrees C

(b) Equal to its input temperature

✓ (c) 0 K

(d) half of its input temperature

2. A refrigerator, with its door open. The temperature of the room will

✓ (a) Rise

(b) Fall

(c) Remains the same

(d) Rise or fall depending on the area of the room

3. Which device is not used in a diesel engine

(a) Outlet Valve

(b) Piston

✓ (c) Sparking Plug

(d) Injector

4. Which one of the following processes is irreversible?

(a) Slow compression of an elastic spring

(b) Slow evaporation of a substance in an isolated vessel

(c) Slow compression of a gas

✓ (d) A chemical explosion

5. According to 2nd law of thermodynamics, 100 percent conversion of heat into mechanical work is:

(a) Possible

(b) Possible, if the conditions are ideal

✓ (c) Not possible

(d) Possible, if the process is adiabatic

6. The change in entropy is given by

✓ (a) Delta S = Delta Q / T

(b) Delta S = Delta Q . T

(c) Delta S = Delta U / T

(d) Delta S = Delta W / T

7. The net change in entropy as a system in a natural process is

✓ (a) Positive

(b) Negative

(c) Zero

(d) Infinite

8. The efficiency of diesel engine is

✓ (a) Greater than petrol engine

(b) Less than petrol engine

(c) Equal to petrol engine

(d) both have efficiency 1

9. Second law of thermodynamics states that:

(a) Energy cannot be converted.

(b) Entropy decreases over time.

(c) Heat flows from cold to hot.

✓ (d) Entropy increases over time

10. The process violates the 2nd law of thermodynamics is:

(a) Refrigerator cooling.

(b) Heat engine working.

(c) Gases mixing.

✓ (d) Heat flowing from cold to hot.

Exercise — Section (B): CRQs (Short Answered Questions)

Question: 1. What are some factors that affect the efficiency of automobile engines?

Answer: The efficiency of an automobile engine depends strongly on the compression ratio, because a higher ratio between the source and sink temperatures gives a higher possible efficiency. Friction between moving parts, heat lost to the cooling system and exhaust gases, and incomplete burning of fuel all reduce the actual efficiency. The temperature of the burning gases and the temperature at which heat is rejected also matter. Good lubrication, correct fuel-air mixture and proper maintenance therefore improve efficiency.

Question: 2. What happens to the temperature of a room in which an air conditioner is left running on table in the middle of the room?

Answer: The room will become warmer, not cooler. An air conditioner removes heat from the air at its front and rejects that heat, plus the electrical work consumed, from its back. When the unit stands inside the room, both the removed heat and the rejected heat stay in the same room. The work done by its motor and compressor is added as extra heat, so the average temperature of the room rises slowly.

Question: 3. Under what conditions can heat be added to a system without changing its temperature?

Answer: Heat can be added without a temperature change during an isothermal process, where the system expands while in contact with a heat reservoir and all the heat supplied is converted into work. Heat is also added without temperature change during a change of state, such as melting ice or boiling water at constant pressure. In that case the energy is used to break intermolecular bonds and increase internal potential energy. In both cases the internal kinetic energy, and hence the

temperature, stays constant.

Question: 4. Is it possible to cool a room by keeping the refrigerator door open?

Answer: No, a room cannot be cooled by leaving the refrigerator door open. The refrigerator takes heat from the air near its open door but rejects a greater quantity of heat to the same room from the coils at its back. The extra rejected heat is the electrical work done by the compressor. As a result the room actually warms up slowly instead of cooling.

Question: 5. When does the entropy of a system decrease?

Answer: The entropy of a system decreases when heat is removed from it in an orderly way, for example when a gas is compressed isothermally and rejects heat, or when water freezes into ice. In such cases the system itself becomes more ordered and its entropy falls. However, the entropy of the surroundings increases by a larger amount, so the entropy of the universe still increases. A local decrease of entropy is therefore allowed by the second law.

Question: 6. Is it possible, according to the 2nd law of thermodynamics to construct an engine that is free from thermal pollution?

Answer: No, such an engine is impossible according to the second law. Every cyclic heat engine must reject some heat to a colder sink, and this rejected heat warms the environment, which is thermal pollution. Only an engine with a sink at 0 K could convert all heat into work, and absolute zero cannot be attained. Better design can reduce the rejected heat but can never remove it completely.

Question: 7. Can heat be completely converted to work?

Answer: In a single non-cyclic process, such as the isothermal expansion of an ideal gas, heat can be completely converted into work because the

internal energy does not change. But no cyclic engine, which must return again and again to its starting state to work continuously, can do this. The second law requires a working engine to reject part of the heat to a sink. So complete conversion is possible once, but not in a continuous cycle.

Question: 8. Define that why entropy has often been called as time arrow.

Answer: Entropy is called the arrow of time because it gives a direction to natural change: the entropy of the universe always increases as time passes. We remember the past, where entropy was lower, and move towards the future, where entropy is higher. Processes such as mixing, cooling of hot bodies and breaking of objects can be told apart from their reverse by the entropy change alone. Just as an arrow points one way, increasing entropy points from past to future.

Exercise — Section (C): ERQs (Long Answered Questions)

Question: 1. Give the two statements of the second law of the thermodynamics. Elaborate the concept of entropy and state the second law of thermodynamics in terms of this concept.

Answer: The Kelvin statement says it is impossible to construct a cyclic engine that absorbs heat from a single reservoir and converts it completely into work without any other change. The Clausius statement says it is impossible for heat to flow by itself from a colder body to a hotter body without external work being done. Entropy is a measure of the disorder or randomness of a system, defined by Delta S = Delta Q(rev) / T for a reversible change. When heat is absorbed, the molecules become more disordered and entropy increases; when heat is rejected in an orderly way, entropy decreases. In terms of entropy, the second law states that the entropy of the universe increases in every natural process, Delta S(universe) is greater than zero, and remains constant only in the ideal

reversible limit.

Question: 2. Explain the working principle of heat engine and also derive the formula for its efficiency.

Answer: A heat engine is a cyclic device that absorbs heat Q1 from a hot reservoir, converts part of it into work W, and rejects the remaining heat Q2 to a cold reservoir. A steam engine and a petrol engine work on this principle, using a working substance that returns to its initial state after each cycle. By the first law for a complete cycle, Delta U = 0, so W = Q1 - Q2. Efficiency is defined as useful work output divided by heat input, eta = W / Q1. Substituting for W gives eta = (Q1 - Q2) / Q1 = 1 - Q2 / Q1, which is always less than 1 because Q2 can never be zero.

Question: 3. Describe the concept of reversible and irreversible process.

Answer: A reversible process is an ideal process carried out infinitely slowly, through a series of equilibrium states, so that it can be reversed by an infinitesimal change and both the system and surroundings return exactly to their initial states. There is no friction, no turbulence and no heat flow across a finite temperature difference in such a process. An irreversible process cannot be reversed in this way; it leaves a permanent change in the surroundings, and all natural processes are of this type. Friction, free expansion, conduction across a large temperature difference, and chemical explosions are irreversible. Reversible processes are useful as a standard because they give the maximum possible efficiency or work for given conditions.

Question: 4. What is Carnot engine? Give the operation of Carnot cycle and show that the efficiency of even this engine is less than 100 percent.

Answer: A Carnot engine is an ideal reversible engine working between two temperatures, and no real engine working between the same temperatures can be more efficient. Its cycle has four reversible steps:

isothermal expansion at the hot temperature T1, adiabatic expansion down to the cold temperature T2, isothermal compression at T2, and adiabatic compression back to the starting state. For this cycle Q2 / Q1 = T2 / T1, so the efficiency is eta = 1 - T2 / T1. Since T2 is never 0 K in practice, T2 / T1 is always positive and eta is always less than 1. Even the Carnot engine therefore rejects heat to the sink and cannot reach 100 percent efficiency.

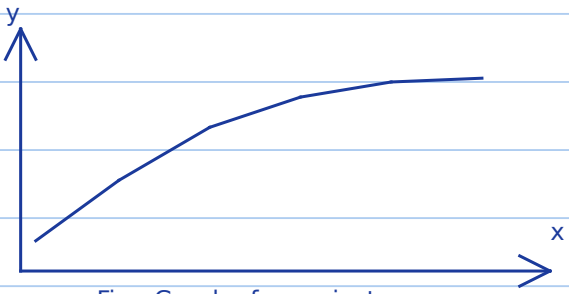


Fig.: Graph of y against x

Question: 5. Describe that refrigerator is a heat engine operating in reverse as that of an ideal heat engine and find its efficiency.

Answer: A refrigerator takes heat Q2 from a cold body, receives work W from an external source such as an electric motor, and rejects the total heat Q1 = Q2 + W to the warmer surroundings. This is exactly the heat engine cycle run backwards: instead of heat flowing down and producing work, work is supplied to push heat up from cold to hot. Its performance is measured by the coefficient of performance, COP = Q2 / W, which plays the role of efficiency for a refrigerator. Using Q1 = Q2 + W, COP = Q2 / (Q1 - Q2), and for an ideal reversible refrigerator COP = T2 / (T1 - T2), where T2 is the cold temperature and T1 is the hot temperature in kelvin.

Question: 6. Explain that change in entropy is positive when heat is added and negative when heat is removed from the system.

Answer: Entropy change is given by Delta S = Delta Q(rev) / T, where T is the absolute temperature and is always positive. When heat is added, Delta Q is positive, so Delta S is positive and the entropy of the system

increases, because the molecules gain energy and their motion becomes more random. When heat is removed, Delta Q is negative, so Delta S is negative and the entropy decreases, because the molecules lose energy and become more ordered. Melting of ice is an example of entropy increase on heating, while freezing of water shows entropy decrease on cooling. The sign of the entropy change of a system is therefore fixed by the direction of heat flow.

Question: 7. Explain that increase in entropy means degradation of energy.

Answer: Degradation of energy means energy becoming less available for doing useful work, even though the total amount of energy is conserved. In every irreversible process, such as friction, conduction and mixing, organised energy like mechanical work or a high temperature difference is spread out as random molecular energy. Once energy is spread uniformly at a low temperature, it cannot be gathered back to do the same work again. This loss of availability is measured by the increase of entropy, through the relation unavailable energy = T0 x Delta S for surroundings at T0. Increasing entropy therefore means steady degradation of the quality of energy in the universe.

Exercise — Section (D): Numerical

Question: 1. A Carnot engine takes 2000J of heat from a reservoir at 500K does some work, and discards some heat to a reservoir at 350K. How much heat is discarded, how much work does the engine do, and what is the efficiency? (Ans: 1400J, 600J, 30 percent)

Answer: Step 1: Given: heat absorbed Q1 = 2000 J, hot reservoir T1 = 500 K, cold reservoir T2 = 350 K

Step 2: Formula: for a Carnot engine Q2 / Q1 = T2 / T1, work W = Q1 - Q2,

and efficiency eta = 1 - T2 / T1

Step 3: Substitution: Q2 = 2000 x 350 / 500, and eta = 1 - 350 / 500

Step 4: Calculation: Q2 = 1400 J, W = 2000 - 1400 = 600 J, and eta = 1 - 0.70 = 0.30

Step 5: Result: heat discarded = 1400 J, work done = 600 J, and efficiency = 30 percent.

Question: 2. One kilogram of ice at 0 degrees C is melted and converted to water at 0 degrees C. Compute its change in entropy. (Ans: 1220 J.K^-1)

Answer: Step 1: Given: mass m = 1 kg, T = 0 degrees C = 273 K, latent heat of fusion of ice Lf = 3.34 x10^5 J/kg

Step 2: Formula: heat absorbed Q = m Lf, and entropy change Delta S = Q / T

Step 3: Substitution: Q = 1 x 3.34 x10^5 = 3.34 x10^5 J, and Delta S = 3.34 x10^5 / 273

Step 4: Calculation: Delta S = 1223 J/K

Step 5: Result: increase in entropy = about 1220 J/K.

Question: 3. In a high-pressure steam turbine engine, the steam is heated to 600 degrees C and exhausted at about 90 degrees C. What is the highest possible efficiency of any engine that operates between these two temperatures? (Ans: 58.4 percent)

Answer: Step 1: Given: T1 = 600 degrees C = 873 K, T2 = 90 degrees C = 363 K

Step 2: Formula: highest possible efficiency is the Carnot efficiency, eta = 1 - T2 / T1

Step 3: Substitution: eta = 1 - 363 / 873

Step 4: Calculation: 363 / 873 = 0.416, and eta = 1 - 0.416 = 0.584

Step 5: Result: highest possible efficiency = 58.4 percent, and a real engine will always be less efficient than this.

Question: 4. Temperature difference between the surface water and bottom water in Manchester Lake might be 5 degrees C. Assuming the surface water to be at 20 degrees C. What highest efficiency a steam engine could have if it operates between these two temperatures? (Ans: 1.71 percent)

Answer: Step 1: Given: surface temperature T1 = 20 degrees C = 293 K, temperature difference = 5 degrees C, so bottom temperature T2 = 15 degrees C = 288 K

Step 2: Formula: Carnot efficiency, eta = 1 - T2 / T1 = (T1 - T2) / T1

Step 3: Substitution: eta = 5 / 293

Step 4: Calculation: eta = 0.0171

Step 5: Result: highest possible efficiency = 1.71 percent, which is too small to be useful in practice.

Question: 5. A heat engine works at the rate of 500kW. The efficiency of the engine is 30 percent. Calculate the loss of heat per hour. (Ans: 4.2 x 10^9 J)

Answer: Step 1: Given: work output rate = 500 kW = 500 x10^3 W, efficiency eta = 30 percent = 0.30, time = 1 hour = 3600 s

Step 2: Formula: input power = work rate / eta, loss rate = input power - work rate, and heat lost in one hour = loss rate x 3600

Step 3: Substitution: input power = 500 x10^3 / 0.30 = 1.667 x10^6 W, and loss rate = 1.667 x10^6 - 500 x10^3 = 1.167 x10^6 W

Step 4: Calculation: heat lost = 1.167 x10^6 x 3600 = 4.2 x10^9 J

Step 5: Result: loss of heat per hour = 4.2 x10^9 J.

Question: 6. A heat engine performs work of 0.4166 watts in one hour and rejects 4500J of heat to the sink. What is the efficiency of engine? (Ans: 24.9 percent)

Answer: Step 1: Given: work rate = 0.4166 W, time = 1 hour = 3600 s, heat rejected Q2 = 4500 J

Step 2: Formula: work W = rate x time, heat absorbed Q1 = W + Q2, and efficiency eta = W / Q1

Step 3: Substitution: W = 0.4166 x 3600 = 1500 J, and Q1 = 1500 + 4500 = 6000 J

Step 4: Calculation: eta = 1500 / 6000 = 0.25, that is about 25 percent

Step 5: Result: efficiency of the engine = about 24.9 percent.

Question: 7. A Carnot engine operates between the temperatures 850K and 300K. the engine performs 1200J of work in each cycle, which takes 0.25 sec (a) What is the efficiency of this engine? (b) What is the average power of this engine? (c) How much energy is extracted as heat from the high temperature reservoir? (d) How much energy is delivered as heat to the low temperature reservoir? [Ans (a) 65 percent (b) 4.8KW (c) 1855J (d) 655J]

Answer: Step 1: Given: T1 = 850 K, T2 = 300 K, work per cycle W = 1200 J, time per cycle t = 0.25 s

Step 2: Formula: (a) eta = 1 - T2 / T1, (b) power P = W / t, (c) heat absorbed Q1 = W / eta, and (d) heat rejected Q2 = Q1 - W

Step 3: Substitution: (a) eta = 1 - 300 / 850, (b) P = 1200 / 0.25, (c) Q1 = 1200 / 0.647, and (d) Q2 = Q1 - 1200

Step 4: Calculation: (a) eta = 0.647, about 65 percent, (b) P = 4800 W =

4.8 kW, (c) Q1 = 1855 J, and (d) Q2 = 655 J

Step 5: Result: (a) efficiency = about 65 percent (b) average power = 4.8 kW (c) heat extracted = 1855 J (d) heat delivered to the low temperature reservoir = 655 J.

Question: 8. A Carnot engine absorbs 52kJ as heat and exhaust 36kJ as heat in each cycle. Calculate: (a) The engine efficiency (b) The work done per cycle in kilojoules. [Ans (a) 30.76 percent (b) 16KJ]

Answer: Step 1: Given: heat absorbed Q1 = 52 kJ, heat exhausted Q2 = 36 kJ

Step 2: Formula: work W = Q1 - Q2, and efficiency eta = W / Q1

Step 3: Substitution: W = 52 - 36, and eta = 16 / 52

Step 4: Calculation: W = 16 kJ, and eta = 0.3076

Step 5: Result: (a) engine efficiency = 30.76 percent, and (b) work done per cycle = 16 kJ.

Unit 18: Magnetic Fields

Exercise — Section (A): Multiple Choice Questions (MCQs)

1. If we reverse the direction of electric current, then the direction of magnetic field will be:

(a) Same

✓ (b) reversed

(c) tangent

(d) normal

2. The application of magnetic field is:

(a) microwave oven

✓ (b) magnetic levitation trains

(c) electrolysis

(d) plant photosynthesis

3. The equation F=BIL can only be used, if the magnetic field, length of conductor, and electric current are:

✓ (a) At right angles to each other

(b) in same direction

(c) Anti-parallel to each other

(d) anti-perpendicular to each other

4. The charge on a particle is doubled, and its velocity remains the same, how then the magnetic force on the particle will be:

✓ (a) doubled

(b) halved

(c) Is the same.

(d) quadrupled

5. Strength of magnetic field of solenoid can be increased by:

(a) Increasing number of turns

(b) decreasing number of turns

(c) increasing the current through the solenoid

(d) inserting a ferromagnetic core [e.g., iron] into the solenoid

6. The magnetic field inside a long solenoid is:

(a) equal to zero

✓ (b) uniform

(c) decreases as we go away from the center to surface

(d) increases as we go towards the surface

7. The force between two current carrying conductor arises due to

✓ (a) Magnetic effect of current

(b) Polarization

(c) Electromagnetic induction

(d) Electrostatic interaction

8. To measure a higher voltage, what should you do with the voltmeter internal resistance?

✓ (a) Increase it

(b) Decrease it

(c) Keep it the same

(d) It does not affect the measurement

9. A proton moves perpendicular to a uniform magnetic field. What is the direction of the force experienced by the proton?

(a) Parallel to the magnetic field

(b) In the direction of the proton velocity

✓ (c) Perpendicular to the magnetic field and the proton velocity

(d) Opposite to the direction of the proton velocity

10. An electron moves parallel to a uniform magnetic field. What is the magnitude of the force experienced by the electron?

✓

(a) Maximum, since the electron is moving in the same direction as the field

(b) Minimum, since the electron is moving perpendicular to the field

(c) Zero, since the electron is moving parallel to the field

(d) Depends on the speed of the electron

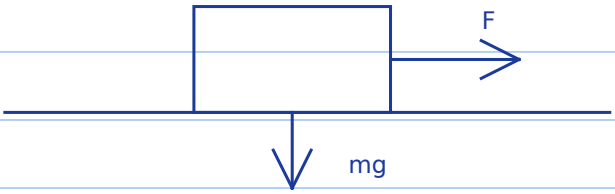


Fig.: Forces on a block

Exercise — Section (B): CRQs (Short Answered Questions)

Question: 1. Charge particles are fired in vacuum tube hit a fluorescence screen. Will it be possible to know whether they positive or negative?

Answer: Yes, it is possible by applying a known magnetic or electric field across the tube and observing the direction of deflection on the fluorescent screen. A positive particle deflects to one side while a negative particle deflects to the opposite side under the same field. The direction of the spot shift, compared with the known field direction, tells the sign of the charge. This is how the negative charge of cathode ray particles was first established.

Question: 2. What is a solenoid, and how does it differ from a simple coil of wire?

Answer: A solenoid is a long coil of wire wound in the form of a helix on a cylindrical frame, with many closely spaced turns. When current passes through it, it behaves like a bar magnet with a nearly uniform field inside and poles at its ends. A simple coil has only a few turns, usually in one plane, and produces a much weaker and less uniform field. The solenoid is

designed to concentrate a strong, uniform field along its axis, while an ordinary coil is not.

Question: 3. Can a solenoid generate a magnetic field without any current flowing through it? Why or why not?

Answer: No, an ordinary solenoid cannot produce a magnetic field without current. Its field is produced only by the current flowing in its turns, as given by B = mu0 n I, so when I = 0 the field is zero. The wire itself is not magnetic and has no field of its own. Only if a permanent magnet or a magnetised iron core is placed inside can a field exist without current, and then the field belongs to the magnet, not to the solenoid.

Question: 4. Explain why a toroid is often preferred over a straight solenoid when designing certain types of electrical components.

Answer: A toroid is a solenoid bent into a closed ring, so its magnetic field is confined almost entirely inside the core. Very little flux leaks into the surrounding space, so it does not disturb or get disturbed by nearby components. This closed path also gives a stronger, better controlled field for a given current. For these reasons toroids are preferred in transformers, inductors and sensitive electronic circuits where stray magnetic fields must be avoided.

Question: 5. What role does the Ampere circuital law play in understanding the magnetic field inside a toroid?

Answer: Ampere circuital law says the line integral of B around a closed path equals mu0 times the total current passing through the path. Applying it to a circular path inside a toroid of N turns carrying current I gives B x 2 pi r = mu0 N I, so B = mu0 N I / (2 pi r). The law shows that the field inside depends on the total turns and current and falls slightly with radius r. It also shows that the field outside the toroid is zero, because the net enclosed current outside is zero.

Question: 6. What is a galvanometer, and what is its primary function in an electrical circuit?

Answer: A galvanometer is a sensitive instrument in which a current carrying coil placed in a magnetic field is deflected by the torque produced on it. Its primary function is to detect small electric currents and to indicate their direction. The deflection of its pointer is proportional to the current passing through the coil. With suitable series or shunt resistances it is converted into a voltmeter or an ammeter for actual measurements.

Exercise — Section (C): ERQs (Long Answered Questions)

Question: 1. Can a galvanometer measure both DC and AC currents? Explain any limitations it might have with AC measurements.

Answer: A moving coil galvanometer measures direct current well, because the torque on its coil is in one direction and the deflection is proportional to the current. With alternating current the torque reverses direction many times per second, following the current. The pointer cannot follow these rapid reversals because of its inertia, so it stays near zero and only shows the average value, which is zero for a complete cycle. It therefore cannot measure ordinary AC directly. AC is measured instead with moving iron or rectifier type instruments, or the alternating current is first converted to DC.

Question: 2. Why should an ammeter ideally have a very low resistance compared to the circuit it is measuring?

Answer: An ammeter is connected in series, so the full circuit current must pass through it. If its resistance were large, it would reduce the very current it is meant to measure and would also cause a large voltage drop and heating. An ideal ammeter has zero resistance, so it leaves the circuit completely unchanged. In practice a galvanometer is converted into an

ammeter by connecting a very low resistance, called a shunt, in parallel with it. This keeps the total resistance very small while the deflection remains proportional to the total current.

Question: 3. What is the potential risk of using a voltmeter with a high internal resistance in a circuit? How can this risk be mitigated?

Answer: Note: a voltmeter is in fact designed to have a high internal resistance, because it is connected in parallel and must draw as little current as possible; the real risk described in this question arises when its resistance is not high enough compared with the component, or when the chosen range is wrong, so that it loads the circuit and reads a voltage lower than the true value. There is also a risk of damaging the coil if the applied voltage is far above the selected range. This risk is mitigated by using a voltmeter whose resistance is much greater than that of the component being measured, by adding a suitable series multiplier resistance, and by always starting from the highest voltage range and coming down. Used in this way the voltmeter disturbs the circuit negligibly and stays safe.

Question: 4. Describe the basic working principle of a voltmeter. How does it measure voltage across a component in a circuit?

Answer: A voltmeter is a galvanometer with a high resistance connected in series with its coil, and it is always connected in parallel with the component across which the voltage is to be measured. Because its resistance is very high, only a tiny current passes through it, and this current is proportional to the potential difference between its terminals. The current produces a torque on the coil, and the deflection of the pointer is calibrated directly in volts. The high series resistance also fixes the maximum voltage range of the instrument. In this way the voltmeter reads the voltage across a component without appreciably changing the circuit

conditions.

Exercise — Section (D): Numerical

Question: 1. An aluminum window has a width of 60 cm and length of 85 cm as shown in the figure. a. When the window is closed the magnetic flux density is 1.8 x 10^-4 T is normal to window. b. Calculate the magnetic flux through the window. (9.18 x 10^-5 Wb)

Answer: Step 1: Given: width = 60 cm = 0.60 m, length = 85 cm = 0.85 m, flux density B = 1.8 x10^-4 T normal to the window, so theta = 0 degrees

Step 2: Formula: area A = length x width, and magnetic flux Phi = B A cos theta

Step 3: Substitution: A = 0.85 x 0.60 = 0.51 m^2, and Phi = 1.8 x10^-4 x 0.51 x 1

Step 4: Calculation: Phi = 0.918 x10^-4 = 9.18 x10^-5

Step 5: Result: magnetic flux through the window = 9.18 x10^-5 Wb.

Question: 2. The poles of a horse shoe magnet measures 8Cm x 3.2cm. the magnetic flux density between the magnet poles is 80mT. Outside of the magnet the magnetic flux density is zero. Calculate the magnetic flux between the poles of a magnet. (2.048 x 10^-4 Wb)

Answer: Step 1: Given: pole face = 8 cm x 3.2 cm, B = 80 mT = 80 x10^-3 T = 0.080 T

Step 2: Formula: area A = 0.08 m x 0.032 m, and flux Phi = B A

Step 3: Substitution: A = 0.00256 m^2, and Phi = 0.080 x 0.00256

Step 4: Calculation: Phi = 2.048 x10^-4

Step 5: Result: magnetic flux between the poles = 2.048 x10^-4 Wb.

Question: 3. A wire 1.80 m long carries a current of 13.0 A and makes an angle of 35.0 degrees with a uniform magnetic field of magnitude B = 1.50T. Calculate the magnetic force on the wire. (19.46 N)

Answer: Step 1: Given: L = 1.80 m, I = 13.0 A, B = 1.50 T, theta = 35.0 degrees

Step 2: Formula: F = B I L sin theta

Step 3: Substitution: F = 1.50 x 13.0 x 1.80 x sin 35 degrees, with sin 35 degrees = 0.574

Step 4: Calculation: 1.50 x 13.0 x 1.80 = 35.1, and F = 35.1 x 0.574 = 20.1

Step 5: Result: magnetic force on the wire = about 20.1 N. Note: the answer 19.46 N printed in the book does not follow exactly from the data given in this question.

Question: 4. A solenoid has length L = 1.23 m and inner diameter d = 3.55 cm, and it carries a current I= 5.57 A. It consists of five close-packed layers, each with 850 turns along length L. What is B at its center? (Ans:24.2mT)

Answer: Step 1: Given: L = 1.23 m, I = 5.57 A, total turns N = 5 x 850 = 4250, mu0 = 4 pi x10^-7

Step 2: Formula: turns per unit length n = N / L, and field at the centre B = mu0 n I

Step 3: Substitution: n = 4250 / 1.23 = 3455 turns per m, and B = 4 pi x10^-7 x 3455 x 5.57

Step 4: Calculation: B = 4 pi x10^-7 x 19244 = 0.0242 T

Step 5: Result: magnetic field at the centre = 0.0242 T, that is 24.2 mT.

Question: 5. A moving coil galvanometer has resistance of 50 ohm and it

gives full scale deflection at 4mA current. A voltmeter is made using this galvanometer and a 5K ohm resistance. Calculate the maximum voltage that can be measured using this voltmeter. (Ans:20.2V)

Answer: Step 1: Given: galvanometer resistance G = 50 ohm, full scale current Ig = 4 mA = 0.004 A, series resistance R = 5k ohm = 5000 ohm

Step 2: Formula: total resistance = G + R, and maximum voltage V = Ig x (G + R)

Step 3: Substitution: V = 0.004 x (50 + 5000) = 0.004 x 5050

Step 4: Calculation: V = 20.2

Step 5: Result: maximum voltage that can be measured = 20.2 V.

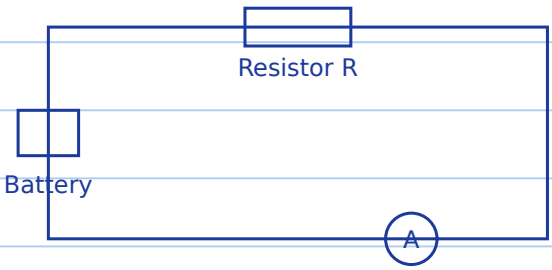


Fig.: Series circuit

Question: 6. Compute the magnitude of the magnetic field of a long, straight wire carrying a current of 1A at distance of 1m from it. Compare it with Earth magnetic field. (Ans:2 x 10^-7 and 100times than earth field)

Answer: Step 1: Given: I = 1 A, distance r = 1 m, mu0 = 4 pi x10^-7, Earth magnetic field = about 5.0 x10^-5 T

Step 2: Formula: field of a long straight wire, B = mu0 I / (2 pi r) = (2 x10^-7 x I) / r

Step 3: Substitution: B = 2 x10^-7 x 1 / 1

Step 4: Calculation: B = 2 x10^-7 T, and ratio = 5.0 x10^-5 / 2 x10^-7 = 250

Step 5: Result: field of the wire = 2 x10^-7 T, which is very much weaker

than the Earth magnetic field; the Earth field is roughly one hundred to two hundred and fifty times stronger, depending on the value taken for it.

Question: 7. Find the current in a long straight wire that would produce a magnetic field twice the strength of the Earth at a distance of 5.0cm from the wire. (Magnetic field of Earth =5.0 x 10^-5 T). (Ans:25A).

Answer: Step 1: Given: required field B = 2 x 5.0 x10^-5 = 1.0 x10^-4 T, distance r = 5.0 cm = 0.05 m

Step 2: Formula: B = (2 x10^-7 x I) / r, so I = B r / (2 x10^-7)

Step 3: Substitution: I = 1.0 x10^-4 x 0.05 / (2 x10^-7)

Step 4: Calculation: numerator = 5.0 x10^-6, and I = 5.0 x10^-6 / 2 x10^-7 = 25

Step 5: Result: current required = 25 A.

Question: 8. What is flux density at a distance of 0.1m in air from a long straight conductor carrying a current of 6.5 A. calculate the force per on a similar parallel conductor at a distance of 0.1m from the first and carrying a current of 3A. (Ans: 13 x 10^-6 Weber/m^2 39 x 10^-6 N).

Answer: Step 1: Given: I1 = 6.5 A, I2 = 3 A, distance r = 0.1 m

Step 2: Formula: flux density B = (2 x10^-7 x I1) / r, and force per unit length on the second conductor F/L = B I2

Step 3: Substitution: B = 2 x10^-7 x 6.5 / 0.1, and F/L = B x 3

Step 4: Calculation: B = 1.3 x10^-5 T = 13 x10^-6 Wb/m^2, and F/L = 1.3 x10^-5 x 3 = 3.9 x10^-5 N per m

Step 5: Result: flux density = 13 x10^-6 Wb/m^2, and force per unit length = 39 x10^-6 N/m, attractive if the currents are in the same direction.

Unit 19: Electromagnetic Induction

Exercise — Section (A): Multiple Choice Questions (MCQs)

1. Electricity production by magnetism involves

(a) the conversion of chemical energy to electrical energy.

(b) the use of static electricity for power generation.

✓ (c) the movement of conductors within magnetic fields.

(d) the absorption of solar energy by photovoltaic cells.

2. Induced e.m.f can be generated by relative movement, known as the generator effect, and by changing magnetic fields, known as the transformer effect.

(a) True in transformers only.

(b) False in all cases.

✓ (c) Applicable to both cases.

(d) True only for static magnetic fields.

3. The magnitude of induced e.m.f. increases with

(a) the decrease in the speed of the conductor motion.

(b) the decrease in the strength of the magnetic field.

✓ (c) the increase in the number of turns in the coil.

(d) the reduction in the area of the coil.

4. Faraday law of electromagnetic induction states

(a) that current in a circuit always opposes the change in magnetic flux.

✓ (b) that the induced e.m.f. is directly proportional to the rate of change of magnetic flux.

(c) that a changing electric field produces a magnetic field.

(d) that magnetic fields can only be produced by electric currents.

5. Lenz law predicts the direction of an induced current

(a) to conserve electric charge.

✓ (b) by stating that the induced current will oppose the change causing it.

(c) by following the direction of the applied magnetic field.

(d) based on the magnitude of the electric field.

6. Eddy currents produce:

(a) static electricity in conductive materials.

(b) alternating magnetic fields with no heating effects.

✓ (c) magnetic fields that oppose the inducing field and cause heating in conductors.

(d) uniform electric fields that have no impact on power generation.

7. Laminated iron cores are used in electric motors and transformers to

(a) reduce the weight of the device.

(b) enhance the mechanical strength.

✓ (c) minimize eddy current losses and reduce heating.

(d) increase the magnetic permeability of the core.

8. Self-induction is the phenomenon where

(a) a changing electric field induces a magnetic field.

✓ (b) a changing magnetic field within a coil induces an emf in the same coil.

(c) a constant magnetic field induces a constant emf.

(d) two coils induce emf in each other through mutual induction.

9. An inductor stores electric potential energy in:

(a) the electric field around it.

✓ (b) the magnetic field within its coil.

(c) the capacitance of its windings.

(d) the heat generated by its resistance.

10. Transformers work on the principle of

(b) electromagnetic induction between primary and secondary coils.

(c) generating electricity through chemical reactions.

(d) using permanent magnets to maintain a constant voltage.

Exercise — Section (B): CRQs (Short Answered Questions)

Question: 1. Explain Faraday law of electromagnetic induction.

Answer: Faraday law states that an emf is induced in a circuit whenever the magnetic flux through it changes, and the magnitude of the emf is proportional to the rate of change of flux. In symbols, emf = -N Delta Phi / Delta t for a coil of N turns, where Phi = B A cos theta. The change of flux may be produced by moving a magnet, moving the coil, or changing the current in a neighbouring coil. The negative sign is included from Lenz law and shows the opposition to the change.

Question: 2. State Lenz law and discuss its significance in the context of electromagnetic induction.

Answer: Lenz law states that the direction of an induced current is such that it opposes the change in magnetic flux that produced it. Its significance is that it gives the direction of the induced emf, which Faraday law alone does not give. It is also a statement of conservation of energy: work must be done against this opposition, and that work is converted into electrical energy. Without this opposition, a tiny change could produce energy from nothing.

Question: 3. How does a step-up transformer differ from a step-down transformer?

Answer: A step-up transformer increases the voltage, so its secondary coil has more turns than its primary coil, Ns greater than Np. A step-down transformer decreases the voltage, so its secondary has fewer turns than

its primary. In both, the current changes in the inverse ratio, so the power ideally stays the same. Step-up transformers are used at power stations for transmission, while step-down transformers supply homes and appliances.

Question: 4. Describe the principle of operation of an AC generator.

Answer: An AC generator works on electromagnetic induction: a coil is rotated in a magnetic field, so the flux through the coil changes continuously. The flux is Phi = B A cos(omega t), and the induced emf is emf = N B A omega sin(omega t), which alternates in direction every half turn. Slip rings and brushes take the alternating current out to the external circuit. Mechanical energy supplied to rotate the coil is thus converted into electrical energy.

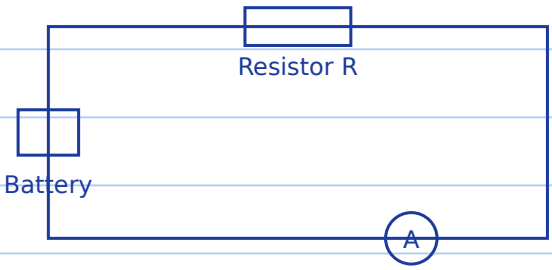


Fig.: Series circuit

Question: 5. Define self-induction and explain how it occurs in a coil.

Answer: Self-induction is the induction of an emf in a coil by the change of its own current and flux. When the current through a coil increases, the magnetic flux through the coil also increases, and the induced emf acts to oppose the increase. When the current decreases, the induced emf acts to maintain it. The effect is measured by the self inductance L in emf = -L dI/dt, and depends on the number of turns, the area and the core of the coil.

Question: 6. Define mutual induction and provide an example of a system exhibiting mutual induction.

Answer: Mutual induction is the production of an emf in one coil due to a

changing current in a neighbouring coil, because part of the flux of the first coil links with the second. The induced emf is emf2 = -M dI1/dt, where M is the mutual inductance. The transformer is the standard example: a changing current in the primary coil induces an emf in the secondary coil through the common iron core. Wireless chargers and induction coils also work by mutual induction.

Question: 7. How does the arrangement of coils influence the degree of mutual induction between them?

Answer: Mutual induction is greatest when the two coils are close together, wound on the same iron core, and their axes are aligned so that maximum flux from one coil passes through the other. It decreases rapidly as the coils are separated or turned at an angle, because less flux links the second coil. A soft iron core increases mutual induction greatly by guiding the flux. More turns in both coils also increase the linkage and hence the mutual inductance.

Question: 8. Define motional electromotive force (emf)

Answer: Motional emf is the emf induced in a conductor when it moves through a magnetic field and cuts the field lines. For a straight conductor of length L moving with speed v at right angles to a field B, the motional emf is emf = B L v. The magnetic force separates the free charges towards the ends of the conductor, creating a potential difference between them. It is the basic emf produced in the moving conductors of a generator.

Question: 9. Differentiate between motional emf and electromagnetic induction in terms of their fundamental principles.

Answer: Motional emf arises from the magnetic force on the free charges of a conductor that is physically moving through a magnetic field, emf = B L v. General electromagnetic induction arises whenever the magnetic flux through a circuit changes, even if the circuit is at rest, as in a transformer

where the field itself changes. Motional emf is therefore a special case of flux change caused by motion. In the transformer effect there is no motion, while in the generator effect the changing area or angle causes the flux change.

Question: 10. Differentiate between AC (alternating current) and DC (direct current) in the context of long-distance power transmission.

Answer: DC flows steadily in one direction at constant value, while AC reverses direction regularly and can easily be stepped up or down by transformers. For long-distance transmission, AC is stepped up to a very high voltage, which reduces the current and hence the I^2 R loss in the lines for the same power. DC cannot be transformed by an ordinary transformer, so changing its voltage is difficult and costly. For this reason power grids use AC for long-distance transmission and distribution.

Exercise — Section (C): ERQs (Long Answered Questions)

Question: 1. How does the strength of the magnetic field affect the induced electromotive force (emf) in a coil according to Faraday law? b. Explain the relationship between the magnetic field strength and the magnitude of the induced emf.

Answer: Magnetic flux is Phi = B A cos theta, so flux is directly proportional to the field strength B. Faraday law gives emf = -N Delta Phi / Delta t, so for the same rate of movement or the same area, a stronger field produces a faster rate of change of flux and a larger induced emf. Doubling B doubles the emf when all other factors stay the same. This is why generator coils are placed in the strongest possible field, close to powerful magnets or electromagnets. A weak field gives only a small emf even if the coil rotates quickly.

Question: 2. In Faraday law, why is the area of the coil an important factor

in determining the induced emf?

Answer: The flux through a coil is the product of field strength and area, Phi = B A cos theta, so a coil of larger area intercepts more field lines. When the field or the angle changes, a larger area gives a larger total change of flux in the same time. By Faraday law, emf = -N Delta Phi / Delta t, so the induced emf increases in proportion to the area. This is why practical generator and transformer coils enclose as large an area as is convenient. Doubling the area, other things equal, doubles the induced emf.

Question: 3. How does changing the orientation of a coil with respect to the magnetic field influence the induced emf? Also describe the role of the number of turns in a coil in the context of induced emf. How does increasing the number of turns in a coil affect the induced emf, and why?

Answer: Flux depends on the angle theta between the field and the normal to the coil, Phi = B A cos theta. When the coil is perpendicular to the field the flux is maximum but momentarily not changing, while as it rotates the flux changes fastest when the coil is parallel to the field, and that is where the induced emf is greatest. Changing the orientation therefore changes the rate of change of flux and controls the emf throughout the cycle of an AC generator. The total emf is also proportional to the number of turns N, because the flux change takes place in every turn and the emfs of all turns add in series. Increasing the number of turns therefore increases the induced emf in the same proportion, which is why transformer and generator coils have many turns.

Question: 4. According to Faraday law, why is the rate of change of magnetic flux crucial for inducing emf? Discuss how the speed of motion or change in magnetic field influences the induced emf.

Answer: Faraday law states that emf equals the rate of change of flux, emf

= -N Delta Phi / Delta t, so it is the speed of the change, not the size of the flux itself, that matters. A very large but steady flux induces no emf at all, because Delta Phi = 0. Moving a magnet quickly into or out of a coil changes the flux in a short time and produces a large emf, while slow movement gives a small emf. Similarly, in a transformer, a rapidly changing primary current induces more emf than a slowly changing one. Doubling the speed of change doubles the induced emf.

Question: 5. How does the resistance of the conductor impact the flow of induced current in a circuit? In what situations might the resistance of the conductor become a significant factor in induced emf?

Answer: The induced emf is fixed by the rate of change of flux, but the induced current is I = emf / R, so a higher resistance gives a smaller induced current for the same emf. Resistance also converts part of the induced electrical energy into heat, reducing the useful output. It becomes significant in circuits carrying large induced currents, such as generator windings, transformer secondaries and eddy current paths in solid cores. In high frequency circuits and long coils the resistance can also limit the current and the response of the circuit. Low resistance copper windings are therefore used where large induced currents are expected.

Question: 6. Explain the relationship between the frequency of the changing magnetic field and the induced emf. How does the frequency of the AC source affect the generation of induced emf in a coil?

Answer: A field that changes more frequently produces a greater rate of change of flux, so the induced emf increases with frequency. For a sinusoidal field the maximum flux change occurs more often each second when the frequency is higher, and emf(max) = N B A omega, where omega = 2 pi f. Doubling the frequency therefore doubles the maximum induced emf for the same flux amplitude. This is why transformers and

generators are designed for a particular supply frequency. It is also why the output voltage of an AC generator rises as its speed of rotation increases.

Question: 7. Discuss how the material properties of a coil, such as its conductivity, can influence induced emf. In what ways might the type of material used in a coil affect its response to changing magnetic fields?

Answer: The induced emf itself depends on flux change, turns and area, but the current it can drive depends on the conductivity of the wire: a highly conducting material like copper has low resistance, so a larger useful current flows for a given emf. A poor conductor wastes energy as heat and reduces the output of a generator or transformer. The core material matters as well: a high permeability core such as soft iron concentrates the flux and greatly increases the induced emf. Magnetic materials with large hysteresis loss heat up under rapidly changing fields, so low loss materials are chosen for high frequency work. Material choice therefore decides both the size of the induced effect and the efficiency of the device.

Question: 8. Why is the presence of a core material significant in transformers concerning induced emf? Explain how the type and properties of the core material can impact the efficiency of a transformer.

Answer: The iron core guides almost all the flux produced by the primary coil through the secondary coil, so the flux linkage and the induced emf are far greater than they would be in air. A core of high permeability, such as soft iron or silicon steel, gives strong linkage and allows the turns ratio to fix the voltage ratio accurately. Cores are laminated to reduce eddy current losses, and materials with a narrow hysteresis loop are chosen to reduce hysteresis loss. A poor core material leaks flux, heats up and lowers the efficiency of the transformer. With a good core, transformer

efficiency can exceed 95 percent.

Question: 9. Why are slip rings used in AC generators, and how do they differ from commutator used in DC generators.

Answer: In an AC generator the emf induced in the rotating coil naturally reverses every half turn, and slip rings simply connect the two ends of the coil continuously to the external circuit through brushes. Each slip ring stays connected to the same end of the coil, so the alternating emf appears unchanged at the output. A DC generator uses a split ring commutator instead, which reverses the connections of the coil to the brushes every half turn. This switching converts the alternating emf inside the coil into a one-way, direct output. Slip rings therefore pass AC out as AC, while a commutator rectifies it into DC.

Question: 10. Why a commutator is necessary in DC motors, and how does it facilitate the continuous rotation of the motor?

Answer: In a DC motor the torque on the coil reverses whenever the coil passes the vertical position, because the current direction relative to the field is then wrong for continued rotation in the same sense. The commutator reverses the current in the coil at exactly this moment, every half turn. This keeps the torque acting in the same rotational direction at all times. Without a commutator the coil would only oscillate about its equilibrium position and stop. The commutator therefore makes continuous rotation of a DC motor possible.

Question: 11. What are the common applications of DC motors in everyday devices?

Answer: DC motors are used wherever a compact motor must run from a battery or low voltage DC supply with easy speed control. Common examples are the starter motor and windscreen wipers of cars, electric toothbrushes, hair dryers with DC motors, and cooling fans in computers.

They also drive toys, model trains, electric shavers, cordless drills and the vibration motors of mobile phones. In homes they are found in mixer grinders with universal motors and in tape and disc players. Their simple speed control by changing the supply voltage makes them ideal for such devices.

Exercise — Section (D): Numerical

Question: 1. Two coils are placed adjacent to each other, and a change in current in the first coil induces an emf of 0.5 V in the second coil. If the mutual inductance is 0.2 H, calculate the rate of change of current in the first coil. (2.5A/s)

Answer: Step 1: Given: induced emf in second coil = 0.5 V, mutual inductance M = 0.2 H

Step 2: Formula: emf = M x (rate of change of current), so rate of change = emf / M

Step 3: Substitution: rate of change = 0.5 / 0.2

Step 4: Calculation: rate of change = 2.5

Step 5: Result: rate of change of current in the first coil = 2.5 A/s.

Question: 2. A coil with an inductance of 0.5 H experiences a rate of change of current of 2 A/s. Calculate the induced EMF in the coil. (-1V)

Answer: Step 1: Given: self inductance L = 0.5 H, rate of change of current dI/dt = 2 A/s

Step 2: Formula: self induced emf = -L x dI/dt, the negative sign showing opposition by Lenz law

Step 3: Substitution: emf = -0.5 x 2

Step 4: Calculation: emf = -1

Step 5: Result: induced emf in the coil = -1 V, the magnitude being 1 V opposing the change of current.

Question: 3. Two coils are placed close to each other. If a change in current of 3 A/s in the first coil induces an EMF of 4 V in the second coil, calculate the mutual inductance. (1.33H)

Answer: Step 1: Given: induced emf = 4 V, rate of change of current = 3 A/s

Step 2: Formula: emf = M x (rate of change of current), so M = emf / (rate of change of current)

Step 3: Substitution: M = 4 / 3

Step 4: Calculation: M = 1.33

Step 5: Result: mutual inductance = 1.33 H.

Question: 4. A transformer has 200 turns in the primary coil and 400 turns in the secondary coil. If the primary voltage is 120 V, calculate the secondary voltage for a step-up transformer. (240V)

Answer: Step 1: Given: primary turns Np = 200, secondary turns Ns = 400, primary voltage Vp = 120 V

Step 2: Formula: Vs / Vp = Ns / Np, so Vs = Vp x Ns / Np

Step 3: Substitution: Vs = 120 x 400 / 200

Step 4: Calculation: Vs = 120 x 2 = 240

Step 5: Result: secondary voltage = 240 V, confirming a step-up transformer.

Question: 5. An inductor with an inductance of 0.02H has a current flowing through it of 2 A. Calculate the energy stored in the inductor. (0.04J)

Answer: Step 1: Given: inductance L = 0.02 H, current I = 2 A

Step 2: Formula: energy stored U = (1/2) L I^2

Step 3: Substitution: U = 0.5 x 0.02 x (2)^2

Step 4: Calculation: U = 0.5 x 0.02 x 4 = 0.04

Step 5: Result: energy stored in the inductor = 0.04 J, stored in its magnetic field.

Question: 6. A coil stores energy in form of electric potential energy of 0.2J when it carries a current of 2A. Calculate the inductance of coil. (0.1H)

Answer: Step 1: Given: stored energy U = 0.2 J, current I = 2 A

Step 2: Formula: U = (1/2) L I^2, so L = 2U / I^2

Step 3: Substitution: L = 2 x 0.2 / (2)^2 = 0.4 / 4

Step 4: Calculation: L = 0.1

Step 5: Result: inductance of the coil = 0.1 H.

Question: 7. An AC generator produces an alternating current with a maximum voltage of 240 V. If the frequency of the generated AC is 50 Hz, calculate the peak value of the voltage. (339V)

Answer: Step 1: Given: the stated voltage 240 V is the rms value of the supply (as the book answer shows), frequency f = 50 Hz

Step 2: Formula: peak value V0 = Vrms x sqrt(2)

Step 3: Substitution: V0 = 240 x 1.414

Step 4: Calculation: V0 = 339.4

Step 5: Result: peak value of the voltage = about 339 V; the frequency fixes the time period but does not change the peak value.

Question: 8. A transformer has 1000 turns in its primary coil and 200 turns in its secondary coil. If the primary voltage is 120 V, calculate the secondary voltage for a step-down transformer. (24V)

Answer: Step 1: Given: primary turns Np = 1000, secondary turns Ns =

200, primary voltage Vp = 120 V

Step 2: Formula: Vs / Vp = Ns / Np, so Vs = Vp x Ns / Np

Step 3: Substitution: Vs = 120 x 200 / 1000

Step 4: Calculation: Vs = 120 x 0.2 = 24

Step 5: Result: secondary voltage = 24 V, confirming a step-down transformer.

Question: 9. A conductor of length 0.4 m moves at a velocity of 5 m/s perpendicular to a magnetic field of 0.3 T. Calculate the motional EMF induced in the conductor. (0.6V)

Answer: Step 1: Given: length L = 0.4 m, speed v = 5 m/s, field B = 0.3 T, motion perpendicular to the field

Step 2: Formula: motional emf = B L v

Step 3: Substitution: emf = 0.3 x 0.4 x 5

Step 4: Calculation: emf = 0.6

Step 5: Result: motional emf induced in the conductor = 0.6 V.

Unit 20: AC Circuits

Exercise - Multiple Choice Questions (MCQs)

In alternating current, the flow of electric charge periodically reverses direction. The term for this specific phenomenon is:

✓ (a) Alternation

(b) Oscillation

(c) Cycle

(d) Resistance

The frequency of standard household alternating current in Pakistan is:

✓ (a) 50 Hz

(b) 60 Hz

(c) 100 Hz

(d) 120 Hz

The parameter for measuring the rate of change of alternating current or voltage with respect to time is:

(a) Amplitude

✓ (b) Frequency

(c) Phase angle

(d) Peak Voltage

In AC circuits, what is the phase difference between current and voltage in a purely resistive circuit?

✓ (a) 0 degrees

(b) 45 degrees

(c) 90 degrees

(d) 180 degrees

A device used to measure the root mean square (RMS) value of an AC

voltage or current is:

✓ (a) Voltmeter

(b) Oscilloscope

(c) Ammeter

(d) AVO meter

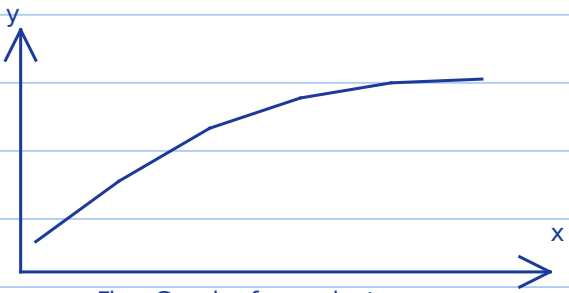


Fig.: Graph of y against x

In an AC circuit, the term reactance represents the:

(a) Ohmic resistance only

(b) Resistance to DC currents only

✓ (c) Capacitive and inductive opposition to current flow

(d) Total impedance, including resistance and phase angle

A type of a circuit characterized by equal values of resistance and reactance is:

(a) Resistive

(b) Inductive

(c) Capacitive

✓ (d) Resonant

In an AC sine wave, the peak-to-peak voltage is:

✓ (a) Double the peak voltage

(b) The difference between the peak voltage and zero

(c) The sum of the peak voltage and zero

(d) The difference between the positive and negative peak voltages

What is the power factor of a purely resistive circuit?

(a) 0

(b) 0.5

(c) 1

(d) -1

The purpose of a transformer in an AC power distribution system is:

(a) To convert AC to DC

(b) To step up or step down voltage

(c) To store electrical energy

(d) To regulate current flow

Exercise - CRQs (Short Answered Questions)

Question: Define RMS voltage and explain its significance in AC circuits.

Answer: RMS voltage is the root mean square value of an alternating voltage, equal to Vo / sqrt(2) or about 0.707 times the peak voltage Vo. It is defined as that steady DC voltage which produces the same heating effect in a resistor as the AC voltage does. Its significance is that all ordinary AC voltmeters and ammeters are calibrated to read RMS values, and power calculations in AC circuits are based on RMS values. When we say the mains supply is 220 V, we are quoting its RMS value, not its peak value of about 311 V.

Question: Explain the difference between peak voltage, RMS voltage, and average voltage in AC circuits.

Answer: Peak voltage Vo is the maximum value reached by the alternating voltage during a cycle. RMS voltage is the effective value, Vrms = 0.707 Vo, which gives the same heating effect as the equal DC voltage. Average voltage over a complete cycle is zero because the positive and negative halves cancel, but the average over a half cycle is 0.637 Vo. Thus the three values describe the same wave in different ways: peak for maximum stress, RMS for power and meter readings, and half-cycle average for

rectifier calculations.

Question: Define alternating current (AC) and explain how it differs from direct current (DC).

Answer: Alternating current is an electric current whose magnitude changes continuously with time and whose direction reverses periodically, usually sinusoidally. Direct current flows steadily in one direction only and, ideally, has constant magnitude. AC can be stepped up or down by a transformer, which is why it is used for power transmission and distribution. DC is used in batteries, electronics and electroplating, where a fixed polarity is required.

Question: Define reactance in AC circuits and differentiate between capacitive and inductive reactance.

Answer: Reactance is the opposition offered to alternating current by a capacitor or an inductor, measured in ohm. Inductive reactance XL = 2 pi f L is offered by a coil; it increases with frequency and inductance, and in it the current lags the voltage by 90 degrees. Capacitive reactance XC = 1 / (2 pi f C) is offered by a capacitor; it decreases as frequency and capacitance increase, and in it the current leads the voltage by 90 degrees. Neither type of reactance dissipates power as heat; both store energy temporarily and return it to the circuit.

Question: Describe the behavior of capacitors and inductors in AC circuits.

Answer: A capacitor in an AC circuit continually charges and discharges as the voltage alternates, so current appears to pass although no charge crosses the dielectric. It offers capacitive reactance XC = 1 / (2 pi f C), blocks DC completely, and makes the current lead the voltage. An inductor opposes every change of current by a self-induced emf, offers inductive reactance XL = 2 pi f L, and makes the current lag behind the voltage. At high frequency the capacitor offers little opposition while the inductor

offers a large opposition, and the reverse happens at low frequency.

Question: Explain the phase relationship between voltage and current in capacitive and inductive AC loads.

Answer: In a purely capacitive load the current reaches its peak a quarter of a cycle before the voltage, so the current leads the voltage by 90 degrees. This happens because the capacitor must first be charged by the current before voltage builds up across its plates. In a purely inductive load the self-induced emf delays the rise of current, so the current lags the voltage by 90 degrees. In a practical load containing resistance as well, the phase difference lies between 0 and 90 degrees depending on which effect dominates.

Question: What is resonance in AC circuits? Discuss its conditions and applications.

Answer: Resonance in a series RLC circuit is the condition in which the inductive reactance equals the capacitive reactance, XL = XC, at a particular frequency fo = 1 / (2 pi sqrt(L C)). At this frequency the two reactances cancel, impedance is minimum and equal to R, current is maximum, and voltage and current are in phase. Resonance is used to tune radio and TV receivers to one selected station frequency. It is also used in filter circuits, induction heating and metal detectors.

Question: Discuss the role of transformers in AC circuits and explain how they work.

Answer: A transformer transfers AC electrical energy from one circuit to another and changes the voltage level through mutual induction. It has a primary coil and a secondary coil wound on a common laminated iron core; alternating current in the primary produces a changing flux in the core, which induces an emf in the secondary. The voltage ratio equals the turns ratio, Vs / Vp = Ns / Np, so more secondary turns step the voltage up

and fewer turns step it down. Transformers make long-distance AC transmission economical and provide the different voltages needed by electronic equipment.

Question: Explain the behavior of an RLC series circuit in terms of impedance, resonance, and phase angle.

Answer: In a series RLC circuit the same current flows through R, L and C, and the total impedance is Z = sqrt(R squared + (XL - XC) squared). The phase angle is given by tan phi = (XL - XC) / R, so the circuit behaves inductively when XL is greater than XC and capacitively when XC is greater than XL. At resonance XL = XC, the impedance falls to its minimum value Z = R, the phase angle becomes zero, and the current reaches its maximum value. Away from resonance the impedance rises and the current decreases, which gives the circuit its frequency-selective character.

Question: Describe the operation and applications of a transformer in AC circuits.

Answer: A transformer operates on mutual induction: AC in the primary coil sets up an alternating flux in the laminated core, and this changing flux induces an alternating emf in the secondary coil. The output voltage is fixed by the turns ratio Ns / Np, while in an ideal transformer the power out equals the power in, so stepping voltage up steps current down. It is applied in power stations and grid stations for transmission, in substations and pole transformers for distribution, and in chargers, adapters and electronic supplies for small voltages. Laminating the core and using soft iron reduce eddy-current and hysteresis losses in practical transformers.

Exercise - ERQs (Long Answered Questions)

Question: Explain the concept of phasor in the context of AC voltage. How

are phasors used to represent sinusoidal voltages?

Answer: A phasor is a rotating vector whose length represents the peak value of a sinusoidal quantity and whose angle with a reference axis represents its phase. As the phasor rotates with angular velocity omega = 2 pi f, its projection on the vertical axis at any instant gives the instantaneous value of the voltage, V = Vo sin(omega t + phi). In a phasor diagram, voltages and currents of the same frequency are drawn as arrows with their correct phase differences, for example VR in phase with the current and VL 90 degrees ahead of it. Adding these arrows by the parallelogram or triangle rule gives the total supply voltage and the phase angle directly. Phasors thus convert difficult trigonometric additions of AC quantities into simple geometry, which is why they are used throughout AC circuit analysis.

Question: Describe the concept of impedance in AC circuits. How does impedance differ from resistance, and what are the units of impedance?

Answer: Impedance Z is the total opposition offered to alternating current by the combined effect of resistance and reactance in a circuit. For a series RLC circuit, Z = sqrt(R squared + (XL - XC) squared), and the current is I = V / Z, just as in Ohms law for DC. Resistance opposes both AC and DC and permanently converts electrical energy into heat, whereas impedance applies only to AC and includes reactive opposition that stores and returns energy without loss. Resistance is a scalar fixed by the material and dimensions, while impedance depends on frequency and carries an associated phase angle between voltage and current. Both impedance and resistance are measured in the same unit, the ohm.

Question: How does the reactance of an inductor and a capacitor change with frequency in an AC circuit? Provide an explanation based on the fundamental formulas.

Answer: The inductive reactance is XL = 2 pi f L, so it is directly proportional to frequency and doubles whenever the frequency doubles. Physically, a faster-changing current induces a larger opposing emf in the coil, so the opposition grows with frequency and becomes infinite for very high frequency while vanishing for DC. The capacitive reactance is XC = 1 / (2 pi f C), so it is inversely proportional to frequency and falls as frequency rises. A capacitor charges and discharges more easily at high frequency, offering little opposition, but it blocks DC completely because at zero frequency XC becomes infinite. This opposite frequency behavior of L and C is the basis of filters and of the resonance condition XL = XC.

Question: Discuss the concept of resonance in RLC circuits. What conditions lead to resonance, and how does it affect the behavior of the circuit?

Answer: Resonance occurs in an RLC circuit at the frequency fo = 1 / (2 pi sqrt(L C)) where the inductive reactance exactly equals the capacitive reactance, XL = XC. At this frequency the voltages across L and C are equal and opposite, so they cancel, and the supply voltage appears entirely across the resistor. The impedance becomes minimum, Z = R, the current becomes maximum, and the power factor becomes unity because voltage and current are in phase. The circuit then selects this one frequency strongly and rejects others, a property measured by the Q-factor and used in radio tuning. In parallel resonance the opposite behavior occurs: impedance is maximum and line current is minimum at resonance.

Question: Compare and contrast series and parallel resonance in RLC circuits. What are the key differences between the two configurations?

Answer: In series resonance the R, L and C are in series, and at fo the impedance is minimum, the current is maximum, and the circuit is called

an acceptor circuit. In parallel resonance the coil and capacitor are in parallel, and at resonance the total impedance is maximum, the line current is minimum, and the circuit is called a rejector circuit. In both cases resonance occurs when XL = XC and the power factor is unity at the resonant frequency. In series resonance the voltages across L and C can be many times the supply voltage (voltage magnification), while in parallel resonance the circulating branch currents can be many times the line current (current magnification). Series resonance is used for selecting a signal, and parallel resonance is used in tank circuits of oscillators and for rejecting an unwanted frequency.

Question: Explain the transient response of an RLC circuit when initially connected to an AC source. What happens to the currents and voltages in the circuit?

Answer: When an RLC circuit is first connected to an AC source, the current does not immediately settle into its steady sinusoidal value; a short-lived transient response is superimposed on the steady-state response. The inductor resists the sudden change of current and the capacitor initially behaves according to its uncharged or charged state, so currents and voltages may overshoot and oscillate at the natural frequency of the circuit. These transient oscillations decay exponentially with time because the resistance dissipates their energy as heat. After a few cycles the transient dies out and only the steady AC current at the supply frequency remains. The size and duration of the transient depend on the values of R, L, C and on the instant in the cycle at which the switch is closed.

Question: Why is alternating current (AC) commonly used for long-distance power transmission?

Answer: AC is used for long-distance transmission because its voltage can

be raised easily and cheaply by step-up transformers. Transmitting power P = V I at very high voltage means a very small current for the same power, and line loss I squared R falls sharply, so energy wastage and heating of the cables become very small. Thin, cheaper cables can then be used over hundreds of kilometers. At the receiving end, step-down transformers reduce the voltage to safe values for homes and factories. DC could not be transformed in this simple way when the power systems were developed, which is the main reason AC became the standard for transmission.

Question: Describe the purpose and function of a choke coil in AC circuits.

Answer: A choke coil is an inductor with a large inductance and very small resistance, used to control alternating current without wasting power. It offers a large reactance XL = 2 pi f L to AC and so limits the current, but unlike a resistor it does not dissipate energy as heat because the current and voltage are nearly 90 degrees out of phase in it. It is wound on a laminated iron core to obtain high inductance in a small size. Chokes are used in fluorescent tube circuits to limit current after starting, and in filter circuits to block AC ripple while allowing DC to pass. It cannot be used to control DC, because for DC it offers only its small ohmic resistance.

Exercise - Numerical

Question: A resistor (R) of 20 ohms is connected in series with a capacitor (C) of 10 microF in an AC circuit with a frequency of 50 Hz. Calculate the total impedance.

Answer: Step 1: Given: R = 20 ohm, C = 10 microF = 10 x10^-6 F, f = 50 Hz, impedance Z is required

Step 2: Formula: XC = 1 / (2 pi f C) and for a series RC circuit Z = sqrt(R squared + XC squared)

Step 3: Substitution: XC = 1 / (2 x 3.14 x 50 x 10 x10^-6) = 1 / (3.14 x10^-3)

Step 4: Calculation: XC = 318.3 ohm, so Z = sqrt(20 squared + 318.3 squared) = sqrt(400 + 101321) = sqrt(101721)

Step 5: Result: Z = 318.9 ohm, approximately 319 ohm

Question: For an inductor with an inductance (L) of 0.5 H and a frequency of 100 Hz, calculate the inductive reactance.

Answer: Step 1: Given: L = 0.5 H, f = 100 Hz, inductive reactance XL is required

Step 2: Formula: XL = 2 pi f L

Step 3: Substitution: XL = 2 x 3.1416 x 100 x 0.5

Step 4: Calculation: XL = 314.16 ohm

Step 5: Result: The inductive reactance is 314.16 ohms

Question: In an RL circuit, the resistance (R) is 30 ohms, and the inductance (L) is 0.2 H. Calculate the total impedance at a frequency of 60 Hz.

Answer: Step 1: Given: R = 30 ohm, L = 0.2 H, f = 60 Hz, impedance Z is required

Step 2: Formula: XL = 2 pi f L and Z = sqrt(R squared + XL squared)

Step 3: Substitution: XL = 2 x 3.14 x 60 x 0.2 = 75.4 ohm, and Z = sqrt(30 squared + 75.4 squared)

Step 4: Calculation: Z = sqrt(900 + 5685) = sqrt(6585) = 81.1 ohm

Step 5: Result: The total impedance is 81.1 ohms

Question: In an RC circuit, the resistance (R) is 50 ohms, and the capacitance (C) is 20 microF. Calculate the capacitive reactance.

Answer: Step 1: Given: R = 50 ohm, C = 20 microF = 20 x10^-6 F, and note that the frequency is not printed in the book question, while XC depends on frequency; the book answer of 31.83 ohms corresponds to f = 250 Hz, which is used here

Step 2: Formula: XC = 1 / (2 pi f C)

Step 3: Substitution: XC = 1 / (2 x 3.14 x 250 x 20 x10^-6) = 1 / (3.14 x10^-2)

Step 4: Calculation: XC = 31.83 ohm; in general XC falls as 1 / f, for example at 50 Hz the same capacitor would give XC = 159.2 ohm

Step 5: Result: The capacitive reactance is 31.83 ohms at 250 Hz

Question: An AC circuit has a resistance of 40 ohms, an inductive reactance of 30 ohms, and a capacitive reactance of 20 ohms. Draw the impedance triangle and calculate the total impedance.

Answer: Step 1: Given: R = 40 ohm, XL = 30 ohm, XC = 20 ohm, total impedance Z is required

Step 2: Formula: Net reactance X = XL - XC, and Z = sqrt(R squared + X squared); in the impedance triangle R is the base, X is the perpendicular and Z is the hypotenuse

Step 3: Substitution: X = 30 - 20 = 10 ohm, so Z = sqrt(40 squared + 10 squared)

Step 4: Calculation: Z = sqrt(1600 + 100) = sqrt(1700) = 41.2 ohm

Step 5: Result: The total impedance is 41.2 ohms

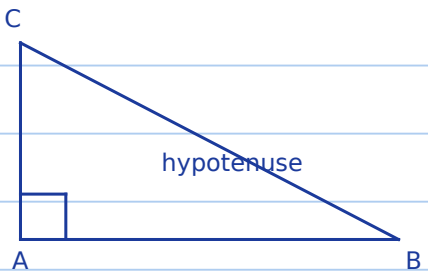


Fig.: Right-angled triangle

Question: In a series RL circuit, the resistance (R) is 25 ohms, and the inductance (L) is 0.1 H. Calculate the phase angle and impedance at a frequency of 80 Hz.

Answer: Step 1: Given: R = 25 ohm, L = 0.1 H, f = 80 Hz, impedance Z and phase angle phi are required

Step 2: Formula: XL = 2 pi f L, Z = sqrt(R squared + XL squared), and tan phi = XL / R

Step 3: Substitution: XL = 2 x 3.14 x 80 x 0.1 = 50.27 ohm, Z = sqrt(25 squared + 50.27 squared), tan phi = 50.27 / 25 = 2.01

Step 4: Calculation: Z = sqrt(625 + 2527) = sqrt(3152) = 56.2 ohm, and phi = inverse tan of 2.01 = 63.6 degrees

Step 5: Result: Impedance is about 56.2 ohms and the current lags the voltage by a phase angle of about 63.6 degrees

Question: In a parallel RC circuit, the resistance (R) is 60 ohms, and the capacitance (C) is 30 microF. Calculate the total current flowing through the circuit at a frequency of 120 Hz.

Answer: Step 1: Given: R = 60 ohm, C = 30 microF = 30 x10^-6 F, f = 120 Hz; note that the supply voltage is not printed in the book question and the book answer line prints only 60 V, so the supply is taken as V = 60 V, consistent with that value

Step 2: Formula: XC = 1 / (2 pi f C), resistor current IR = V / R, capacitor

current IC = V / XC, and total current I = sqrt(IR squared + IC squared) because the two branch currents are 90 degrees out of phase

Step 3: Substitution: XC = 1 / (2 x 3.14 x 120 x 30 x10^-6) = 44.2 ohm, IR = 60 / 60 = 1.0 A, IC = 60 / 44.2 = 1.36 A

Step 4: Calculation: I = sqrt(1.0 squared + 1.36 squared) = sqrt(1 + 1.84) = sqrt(2.84) = 1.69 A

Step 5: Result: The total current is about 1.69 A for a 60 V supply, leading the voltage; without a stated supply voltage only XC = 44.2 ohms can be found uniquely

Question: In an RLC circuit, the resistance (R) is 50 ohms, the inductance (L) is 0.1 H, and the capacitance (C) is 50 microF. Calculate the resonance frequency.

Answer: Step 1: Given: R = 50 ohm (not needed for fo), L = 0.1 H, C = 50 microF = 50 x10^-6 F, resonance frequency fo is required

Step 2: Formula: fo = 1 / (2 pi sqrt(L C))

Step 3: Substitution: fo = 1 / (2 x 3.14 x sqrt(0.1 x 50 x10^-6)) = 1 / (6.28 x sqrt(5 x10^-6)) = 1 / (6.28 x 2.236 x10^-3)

Step 4: Calculation: fo = 1 / (1.405 x10^-2) = 71.2 Hz

Step 5: Result: The resonance frequency is about 71.2 Hz, at which the impedance is minimum and equal to 50 ohms

Unit 21: Physics of Solids

Exercise - Multiple Choice Questions (MCQs)

The property of a body by virtue of which it tends to regain its original size and shape when the applied force is removed is called

✓ (a) Elasticity

(b) Plasticity

(c) Rigidity

(d) Compressibility

Substances which can be stretched to cause large strains are called

(a) Brittle

✓ (b) Ductile

(c) Plastic

(d) Elastomer

If the load is increased beyond the point, the strain increases rapidly for even a small change in the stress.

(a) Elastic point

✓ (b) Yield point

(c) Plastic point

(d) Fracture point

The reciprocal of the bulk modulus is called

✓ (a) Compressibility

(b) Volume stress

(c) Modulus of rigidity

(d) Volume strain

Which of the following statements is/are wrong? i. Hollow shaft is much stronger than a solid of the same length and same mass. ii. Reciprocal of

bulk modulus of elasticity is called compressibility. iii. It is difficult to twist along rod as compared to small rod.

✓ (a) III only

(b) I only

(c) II and III

(d) I and II

Metals are good conductors of heat and electricity. This property is conferred by

(a) Covalent

(b) Ionic

✓ (c) Metallic

(d) Hydrogen

For a metallic crystal, delocalized electrons occupied band is:

✓ (a) Conduction band

(b) Valence band

(c) Conduction and valence bands

(d) there are no delocalized electrons

The semiconductors have resistivity

✓ (a) Between conductor and insulator

(b) More than insulators

(c) Less than conductors

(d) Depending upon the semiconductor material property.

Question: A ferromagnetic substance becomes a permanent magnet when it is placed in a magnetic field because (a) All the domains get oriented in the direction of the magnetic field. (b) All the domains get oriented in the direction opposite to the direction of the magnetic field. (c) Domains get oriented randomly. (d) Domains are not affected by the magnetic field.

Answer: (a) A ferromagnetic material contains small regions called

domains, each magnetized in a different direction, so the unmagnetized material has no net magnetism. When it is placed in a strong magnetic field, the domains align in the direction of the field and the material becomes strongly magnetized. Hard ferromagnetic materials such as steel keep this alignment after the field is removed and become permanent magnets. Random orientation would give zero net magnetization, so option (a) is correct.

Identify methods to demagnetize a ferromagnet.

(a) By cooling, heating, and submerging in water

(b) By heating, hammering, and spinning it in an external magnetic field

(c) By hammering, heating, and rubbing with cloth

(d) By cooling, submerging in water, or rubbing with cloth

Exercise - CRQs (Short Answered Questions)

Question: Why are the springs made of steel and not of copper?

Answer: Springs are made of steel because steel has a much larger Youngs modulus and a higher elastic limit than copper. For the same load, a steel spring is deformed less and it returns completely to its original shape when the load is removed, even after repeated use. Copper is softer and more ductile, so a copper spring would soon take a permanent set and lose its spring action. Steel also resists fatigue better, which is essential for a spring working millions of times.

Question: The breaking force for a wire is F. What will be the breaking force for two parallel wires of the same size?

Answer: When two identical wires are used in parallel, the load is shared between them and the total cross-sectional area becomes double. Breaking force equals breaking stress multiplied by area, and since

breaking stress depends only on the material, doubling the area doubles the force needed to break the combination. Therefore the breaking force for two parallel wires of the same size is 2F. Each wire individually still breaks at force F, but together they support twice the load.

Question: Distinguish between intrinsic and extrinsic semiconductor.

Answer: An intrinsic semiconductor is a chemically pure semiconductor, such as pure silicon or germanium, in which conduction is due only to electrons and holes produced in equal numbers by thermal energy. Its conductivity is low at room temperature and increases rapidly with temperature. An extrinsic semiconductor is one whose conductivity has been greatly increased by doping with a small amount of impurity, making it either n-type with electrons as majority carriers or p-type with holes as majority carriers. In an extrinsic semiconductor the numbers of electrons and holes are not equal, and its conductivity is controlled mainly by the amount of doping.

Question: A wire is replaced by another wire of same length and material but of twice the diameter. What will be the effect on the: (a) Increase in its length under a given load? (b) Maximum load which it can bear?

Answer: Doubling the diameter makes the cross-sectional area four times larger, since area is proportional to the square of diameter. (a) For a given load the extension is inversely proportional to area, from delta L = F L / (A Y), so the increase in length becomes one fourth of its previous value. (b) The maximum load a wire can bear equals breaking stress multiplied by area, so with four times the area the maximum bearable load becomes four times greater. Thus the thicker wire stretches less and is also stronger.

Question: Sand does not possess any definite shape and volume, still it is solid. Give reason.

Answer: Each individual grain of sand is a solid crystal with its own definite shape, volume and strong intermolecular forces holding its particles in fixed positions. A heap of sand takes the shape of its container only because the heap consists of a very large number of tiny separate grains that can slide over one another. The flow is of separate solid pieces, not of the material of a single grain. Since every grain itself is rigid and keeps its volume, sand as a material is classified as a solid.

Question: Specify the importance of stress-strain curve.

Answer: The stress-strain curve shows at a glance how a material behaves under load, from elastic stretching through yielding to fracture. From it we can read the elastic limit, yield point, ultimate tensile strength and breaking point, and the slope of its straight part gives Youngs modulus of the material. The shape of the curve tells whether the material is ductile, brittle or polymeric, so engineers can select the right material for a bridge, cable or machine part. The area under the curve also gives the energy absorbed per unit volume before fracture, which is a measure of toughness.

Question: Why liquids do not possess rigidity?

Answer: Rigidity is the resistance to a change of shape under a shearing force, measured by the shear modulus. In a liquid the molecules are not held in fixed positions; the intermolecular forces are weak and the layers of liquid slide freely over one another when even a small shear is applied. A liquid therefore flows and takes the shape of its container instead of keeping its own shape. Since a liquid cannot maintain a shear strain, its modulus of rigidity is zero and it does not possess rigidity, although it does resist compression and has a bulk modulus.

Question: Give applications of Curie point.

Answer: The Curie point is the temperature above which a ferromagnetic

material loses its ferromagnetism and becomes paramagnetic, and it has several applications. It is used in temperature control devices and thermal switches, where a magnet releases a contact when the Curie temperature is reached, for example in electric rice cookers and fire alarms. Curie-point knowledge is essential in designing transformers, motors and magnetic recording media, which must operate well below the Curie temperature of their cores and coatings. It is also used in thermomagnetic generators and in medical hyperthermia treatment, where magnetic particles stop heating automatically at the Curie point, protecting healthy tissue.

Question: What are amorphous materials and what are their uses?

Answer: Amorphous materials are solids whose atoms have no long-range regular arrangement; they have only short-range order and no sharp melting point, softening gradually over a range of temperature. Common examples are glass, plastic, rubber and tar. Glass is used for windows, lenses, bottles and optical fibers, while plastics and rubber are used for pipes, insulation, tyres and household goods. Amorphous metallic alloys and amorphous silicon are used in transformer cores and solar cells because of their useful magnetic and electrical properties.

Exercise - ERQs (Long Answered Questions)

Question: Explain Force-Extension graph.

Answer: A force-extension graph is plotted by loading a wire in steps, recording the extension for each load, and then unloading it to check recovery. For a ductile material the graph first rises as a straight line through the origin, where extension is proportional to force and Hooke law is obeyed; the slope of this part gives the stiffness of the wire. Next comes the elastic limit, after which the graph curves and the wire yields, extending greatly with little extra force in the plastic region. The force

reaches a maximum at the ultimate tensile strength and then the wire necks and finally breaks at the fracture point. Brittle materials show only the straight part before sudden breaking, while polymeric materials show a long, curved, rubber-like extension; the area under the graph equals the work done in stretching the wire.

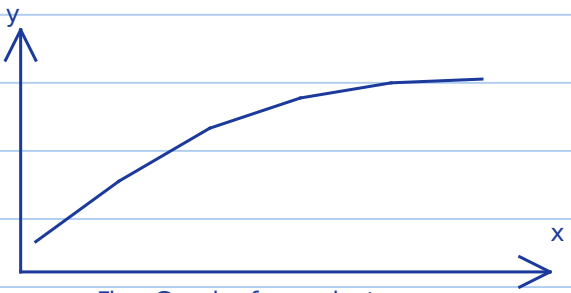


Fig.: Graph of y against x

Question: Derive relation for Youngs Modulus and Shear Modulus.

Answer: Consider a cube of side L fixed at its base and sheared by a tangential force F on its top face, so the top face moves by a small distance x and the shear angle is theta = x / L. The shear stress is F / A and the shear strain is tan theta, approximately theta for small angles, so the shear modulus is G = (F / A) / theta. For an isotropic solid the three moduli are connected because a tensile stress can be resolved into shear components; the standard result is Y = 2 G (1 + sigma), where sigma is Poissons ratio. Equivalently G = Y / (2 (1 + sigma)), and combining with the bulk modulus gives Y = 3 B (1 - 2 sigma) and Y = 9 B G / (3 B + G). For most metals sigma is about 1/3, so Y is roughly 2.67 times G, showing that a material is harder to stretch than to shear. These relations allow any one modulus to be found when the other two are known.

Question: Distinguish between structure of crystalline, glassy, amorphous, and polymeric solids.

Answer: Crystalline solids such as copper, quartz and common salt have atoms arranged in a perfectly regular, repeating three-dimensional pattern extending through the whole solid, so they have a sharp melting point and

definite cleavage planes. Glassy and amorphous solids such as glass, tar and rubber have no long-range order; their atoms are arranged irregularly like a frozen liquid, so they soften gradually over a temperature range instead of melting sharply and are isotropic in their properties. Amorphous is the general class of non-crystalline solids, of which glass is the most familiar example. Polymeric solids such as polythene and nylon consist of very long chain molecules, often partly crystalline and partly amorphous, tangled together; they are light, flexible and can show very large elastic extensions. Thus the four types differ mainly in the degree and pattern of order in their internal structure.

Question: Describe the energy bands in solids.

Answer: When a large number of atoms are brought together in a solid, their discrete energy levels split and merge into continuous ranges called energy bands, because the outer electrons are shared by neighboring atoms. The highest band completely filled at low temperature is the valence band, and the next band above it is the conduction band, in which electrons are free to move and carry current. Between them may lie a forbidden energy gap Eg, in which no electron states exist. In conductors the valence and conduction bands overlap or the conduction band is partly filled, so electrons move freely; in insulators the gap is large, about 5 eV or more; and in semiconductors the gap is small, about 1 eV. The size of this gap explains the electrical behavior of all solids.

Question: Describe superconductivity and its applications.

Answer: Superconductivity is the phenomenon in which certain materials, when cooled below a critical temperature Tc, lose all electrical resistance and also expel magnetic field from their interior (Meissner effect). Mercury becomes superconducting at 4.2 K, while some ceramic materials do so at much higher temperatures above 100 K. A current once started in a

superconducting ring continues for years without any power supply, because there is no I squared R loss. Applications include very powerful electromagnets, MRI scanners in hospitals, magnetic levitation trains, highly sensitive magnetic detectors and lossless power cables. Superconducting magnets also make particle accelerators and future supercomputers with very fast, low-loss switches possible.

Question: Discuss the applications of superconductors for MRI, Maglevs, and supercomputers.

Answer: In MRI (magnetic resonance imaging), superconducting coils cooled by liquid helium produce a strong, steady and uniform magnetic field that allows detailed images of soft tissues inside the body without harmful radiation. In Maglev trains, superconducting magnets lift the train several centimeters above the track and propel it, removing wheel friction so that speeds above 500 km per hour are reached quietly and smoothly in Japan and China. In supercomputers, superconducting Josephson junction switches change state in picoseconds and dissipate almost no heat, promising machines far faster and more energy-efficient than semiconductor computers. Superconductors are also proposed for compact powerful motors, fault-current limiters and energy storage rings. Their wider use is limited mainly by the cost and difficulty of keeping them at very low temperatures.

Question: Describe hysteresis loss.

Answer: When a ferromagnetic material is taken through a complete cycle of magnetization, the graph of flux density B against magnetizing field H forms a closed loop called the hysteresis loop, and B lags behind H. The energy lost as heat in each cycle, called hysteresis loss, is proportional to the area of this loop. This loss arises because energy is spent in repeatedly turning the magnetic domains against internal friction, and it is

not recovered when the field is removed. Soft magnetic materials such as soft iron have a narrow loop, small area and therefore small hysteresis loss, suiting them for transformer cores and electromagnets. Hard materials such as steel have a broad loop and large loss, but they retain strong magnetism and are used for permanent magnets.

Question: Synthesize hysteresis loop for relationship between magnetic field strength and magnetizing current.

Answer: The magnetizing current in a coil wrapped around a ferromagnetic specimen produces the magnetic field strength H, which is proportional to that current, while the flux density B in the specimen is measured as the current is increased, decreased and reversed. Starting from an unmagnetized state, B rises steeply with H and then flattens at saturation, tracing the initial magnetization curve. When H is reduced to zero, B does not return to zero but keeps a residual value called remanence; a reversed field (coercive field) is then needed to bring B back to zero. Continuing the cycle in the opposite direction and back again completes a closed symmetric loop, the hysteresis loop. The shape of this loop thus synthesizes the whole history-dependent relationship between field strength and magnetizing current for that material.

Question: Discuss energy bands and their classification. Explain magnetic properties of soft and hard magnetic materials.

Answer: Energy bands arise when the energy levels of isolated atoms split into continuous allowed bands in a solid, separated by forbidden gaps; the valence band holds the outer bound electrons and the conduction band holds free electrons. On this basis solids are classified as conductors, with overlapping bands and no gap; insulators, with a wide forbidden gap of about 5 eV or more; and semiconductors, with a narrow gap of about 1 eV that a few electrons can cross at room temperature. Soft magnetic

materials such as soft iron are easily magnetized and demagnetized, have high permeability, low coercivity and a narrow hysteresis loop with small loss, so they are used for transformer cores and electromagnets. Hard magnetic materials such as steel and alnico are difficult to magnetize but retain their magnetism strongly, with high remanence, high coercivity and a broad hysteresis loop. They are therefore used for permanent magnets in motors, meters and loudspeakers.

Exercise - Numerical

Question: The lead in pencils is a graphite composition with a Youngs modulus of 1.0 x10^9 N/m squared. Calculate the change in length of the lead in an automatic pencil if you tap it straight into the pencil with a force of 4.0 N. The lead is 0.50 mm in diameter and 60 mm long.

Answer: Step 1: Given: Y = 1.0 x10^9 N/m squared, F = 4.0 N, diameter d = 0.50 mm so radius r = 0.25 x10^-3 m, length L = 60 mm = 0.060 m, change in length delta L is required

Step 2: Formula: Y = (F / A) / (delta L / L), so delta L = F L / (A Y), with A = pi r squared

Step 3: Substitution: A = 3.14 x (0.25 x10^-3) squared = 1.96 x10^-7 m squared, and delta L = (4.0 x 0.060) / (1.96 x10^-7 x 1.0 x10^9)

Step 4: Calculation: delta L = 0.24 / 196 = 1.22 x10^-3 m

Step 5: Result: The lead shortens by about 1.2 mm (the book answer is given as 1.0 mm approximately)

Question: A wire of 2.2 m long and 2.25 mm in diameter, when stretched by a weight of 8.8 kg, its length has been increased by 0.25 mm. Find the stress, strain, and Youngs modulus of the material of the wire. Given g = 9.8 m/s squared.

Answer: Step 1: Given: L = 2.2 m, d = 2.25 mm so r = 1.125 x10^-3 m, mass = 8.8 kg so F = m g = 8.8 x 9.8 = 86.24 N, delta L = 0.25 mm = 0.25 x10^-3 m

Step 2: Formula: A = pi r squared, stress = F / A, strain = delta L / L, Y = stress / strain

Step 3: Substitution: A = 3.14 x (1.125 x10^-3) squared = 3.98 x10^-6 m squared, stress = 86.24 / 3.98 x10^-6, strain = 0.25 x10^-3 / 2.2

Step 4: Calculation: stress = 2.2 x10^7 N/m squared, strain = 1.14 x10^-4, Y = 2.2 x10^7 / 1.14 x10^-4 = 1.9 x10^11 N/m squared, approximately 2 x10^11 N/m squared

Step 5: Result: Stress is 2.2 x10^7 N/m squared, strain is 1.14 x10^-4, and Youngs modulus is about 2 x10^11 N/m squared

Question: A farmer making juice fills a glass bottle to the brim and caps it tightly. The juice expands more than the glass when it warms up, in such a way that the volume increases by 0.2 percent (i.e., delta V / V0 = 2 x10^-3) relative to the space available. Calculate the normal force exerted by the juice per square centimeter, if its bulk modulus is 1.8 x10^9 N/m squared. Assuming that the bottle does not break.

Answer: Step 1: Given: volume strain delta V / V0 = 2 x10^-3, bulk modulus B = 1.8 x10^9 N/m squared, pressure and force per cm squared are required

Step 2: Formula: B = pressure / volume strain, so pressure P = B x (delta V / V0), and force per unit area is the pressure itself

Step 3: Substitution: P = 1.8 x10^9 x 2 x10^-3

Step 4: Calculation: P = 3.6 x10^6 N/m squared; converting to one square centimeter, force = 3.6 x10^6 x 10^-4 = 360 N per cm squared

Step 5: Result: The juice exerts a normal force of 360 N on each square centimeter of the bottle

Question: The elastic limit of copper is 1.5 x10^8 N/m squared. It is to be stretched by a load of 10 kg. Find the diameter of the wire if the elastic limit is not to be exceeded.

Answer: Step 1: Given: elastic limit (maximum stress) = 1.5 x10^8 N/m squared, mass = 10 kg so F = 10 x 9.8 = 98 N, diameter d is required

Step 2: Formula: stress = F / A, so A = F / stress, and d = sqrt(4 A / pi)

Step 3: Substitution: A = 98 / 1.5 x10^8 = 6.53 x10^-7 m squared, d = sqrt(4 x 6.53 x10^-7 / 3.14)

Step 4: Calculation: d = sqrt(8.32 x10^-7) = 9.12 x10^-4 m

Step 5: Result: The diameter of the wire must be at least 0.912 mm

Question: What would be the greatest length of a steel wire which is fixed at one end, and can it be hanged freely without breaking? The breaking stress of steel is 7.8 x10^8 N/m squared, and the density of steel is 7800 kg/m cubed.

Answer: Step 1: Given: breaking stress = 7.8 x10^8 N/m squared, density rho = 7800 kg/m cubed, g = 9.8 m/s squared, greatest length L is required

Step 2: Formula: The wire breaks under its own weight when stress = weight / area = (A L rho g) / A = L rho g, so L = breaking stress / (rho g)

Step 3: Substitution: L = 7.8 x10^8 / (7800 x 9.8)

Step 4: Calculation: L = 7.8 x10^8 / 76440 = 1.02 x10^4 m

Step 5: Result: The greatest length is 1.02 x10^4 m, about 10.2 km

Question: A mild steel wire of radius 0.55 mm and length 3.5 m is stretched by a force of 52 N. Calculate: (a) Longitudinal stress, (b) Longitudinal strain, and (c) Elongation produced in the wire if Youngs

modulus is 2.1 x10^11 N/m squared.

Answer: Step 1: Given: r = 0.55 mm = 0.55 x10^-3 m, L = 3.5 m, F = 52 N, Y = 2.1 x10^11 N/m squared

Step 2: Formula: A = pi r squared, stress = F / A, strain = stress / Y, elongation delta L = strain x L

Step 3: Substitution: A = 3.14 x (0.55 x10^-3) squared = 9.50 x10^-7 m squared, stress = 52 / 9.50 x10^-7, strain = stress / 2.1 x10^11

Step 4: Calculation: (a) stress = 5.47 x10^7 N/m squared, (b) strain = 5.47 x10^7 / 2.1 x10^11 = 2.6 x10^-4, (c) delta L = 2.6 x10^-4 x 3.5 = 9.1 x10^-4 m

Step 5: Result: Longitudinal stress is 5.47 x10^7 N/m squared, longitudinal strain is 2.6 x10^-4, and elongation is 0.91 mm

Question: Calculate the change in volume of a lead block of volume 1.3 m cubed subjected to a pressure of 12 atm. Also, calculate the compressibility of lead. Given the bulk modulus as B = 80 x10^9 N/m squared.

Answer: Step 1: Given: V = 1.3 m cubed, pressure P = 12 atm = 12 x 1.013 x10^5 = 1.216 x10^6 N/m squared, B = 80 x10^9 N/m squared

Step 2: Formula: B = P / (delta V / V), so delta V = P V / B, and compressibility = 1 / B

Step 3: Substitution: delta V = (1.216 x10^6 x 1.3) / 80 x10^9, compressibility = 1 / (80 x10^9)

Step 4: Calculation: delta V = 1.58 x10^6 / 8.0 x10^10 = 1.97 x10^-5 m cubed, compressibility = 1.25 x10^-11 per pascal

Step 5: Result: The volume decreases by 1.97 x10^-5 m cubed and the compressibility of lead is 1.25 x10^-11 m squared per N

Question: The thickness of a metal plate is 0.35 inches. It is drilled to have a hole of radius 0.08 inches on the plate. If the shear strength is 4 x10^4 lbs/in squared, determine the force needed to make that hole.

Answer: Step 1: Given: plate thickness t = 0.35 in, hole radius r = 0.08 in, shear strength = 4 x10^4 lbs/in squared, force F is required

Step 2: Formula: The sheared area is the surface area of the cylinder cut out, A = circumference x thickness = 2 pi r t, and F = shear strength x A

Step 3: Substitution: A = 2 x 3.14 x 0.08 x 0.35 = 0.176 in squared, F = 4 x10^4 x 0.176

Step 4: Calculation: F = 7.04 x10^3 lbs

Step 5: Result: The sheared area is 0.176 in squared and the force needed is about 7 x10^3 lbs

Unit 22: Solid State Electronics

Exercise - Multiple Choice Questions (MCQs)

A semiconductor is an element with a valence electron _____.

✓ (a) Four

(b) Eight

(c) Two

(d) One

A pure semiconductor is known as

(a) Extrinsic

✓ (b) Intrinsic

(c) Transistor

(d) Diode

An acceptor atom is also called

(a) Penta-valent atom

✓ (b) Trivalent atom

(c) Minority carrier

(d) Majority carrier

With which one of the following elements silicon should be doped so as to give p-type of semiconductor?

(a) Germanium

(b) Arsenic

(c) Selenium

✓ (d) Boron

For a full-wave rectifier, the output frequency

(a) Equals one-half the input frequency

(b) Equals the line frequency

(c) Equals two times the input

(d) Is three times the line frequency

LED construction needs a semiconductor material is:

(a) Silicon

(b) Germanium

(c) Gallium

✓ (d) Gallium arsenide

The frequency of a half-wave signal is

(a) Twice the line frequency

✓ (b) Equal to the line frequency

(c) One-half of the line frequency

(d) One-fourth the line frequency

The voltage gain of an emitter follower circuit is

(a) High

✓ (b) Low

(c) Very high

(d) Moderate

What is also called as the conventional amplifier?

(a) Common-collector circuit

(b) Emitter follower circuit

(c) Common base circuit

✓ (d) Common emitter circuit

An op-amp with negative feedback provides _____ output parameter.

(a) Gain

✓ (b) Bandwidth

(c) Input-output impedance

(d) Amplified

✓

Exercise - CRQs (Short Answered Questions)

Question: Describe a p-n junction (diode) and how holes and electrons are produced in semiconductor?

Answer: A p-n junction is formed when a p-type and an n-type semiconductor are joined in a single crystal, and the device so formed is called a junction diode. In a pure (intrinsic) semiconductor, thermal energy breaks some covalent bonds; each freed electron leaves behind a vacant position called a hole, so electrons and holes are produced in equal pairs. In n-type material, penta-valent donor atoms provide extra free electrons, while in p-type material trivalent acceptor atoms create extra holes. At the junction, electrons and holes diffuse across and recombine, forming a thin depletion region with a barrier potential of about 0.7 V for silicon.

Question: Define and distinguish between p-n-p and n-p-n transistors?

Answer: A transistor is a three-layer semiconductor device with emitter, base and collector, used for amplification and switching. In a p-n-p transistor a thin n-type base layer is sandwiched between p-type emitter and collector, and the majority carriers through the device are holes flowing from emitter to collector. In an n-p-n transistor a thin p-type base is sandwiched between n-type emitter and collector, and the majority carriers are electrons. The two types work with opposite supply polarities and opposite directions of conventional current, and their circuit symbols differ in the direction of the arrow on the emitter.

Question: Explain common-base and common collector configurations?

Answer: In the common-base configuration the base is common to the input and output circuits; the signal is applied to the emitter and taken from the collector. It gives voltage gain but a current gain slightly less than one, has low input and high output impedance, and is suited to high-frequency work. In the common-collector configuration the collector

is common; the signal is applied to the base and taken from the emitter, and the circuit is called the emitter follower. It gives a current gain but a voltage gain of just under one, with high input impedance and low output impedance, so it is used as a buffer. The third configuration, common-emitter, is the one most used for general amplification.

Question: Describe the operations of transistors.

Answer: A transistor operates with its emitter-base junction forward biased and its collector-base junction reverse biased, so charge carriers injected by the emitter cross the thin base to the collector. A small base current IB controls a much larger collector current IC, with IC = beta IB, and the total emitter current is IE = IB + IC. When a weak AC signal is applied to the base, it produces large variations of collector current through a load resistor, giving amplification. By driving the transistor fully on (saturation) or fully off (cut-off), the same device also operates as an electronic switch.

Question: Explain the use of transistors as a switch and an amplifier (common-emitter).

Answer: As a switch, a transistor is used in its cut-off state (no base current, transistor OFF) or its saturation state (large base current, transistor fully ON), for example to turn a relay, lamp or motor on and off from a small control signal. As a common-emitter amplifier, the transistor is biased in the active region and a small input signal at the base produces a large amplified output across the collector resistor. The output is 180 degrees out of phase with the input in this configuration. Voltage gain, current gain and power gain are all appreciable, which is why the common-emitter amplifier is the conventional amplifier circuit.

Question: How would you understand the effects of negative feedback on the gain of an operational amplifier?

Answer: Negative feedback returns a part of the output signal to the

inverting input, so it opposes the input and reduces the overall gain of the amplifier. The closed-loop gain then depends almost entirely on the external feedback and input resistors, for an inverting amplifier gain = -Rf / R1, and not on the op-amps own very large but uncertain open-loop gain. In exchange for reduced gain, the amplifier gains stability, wider bandwidth, lower distortion and better control of input and output impedances. Understanding this trade-off, gain exchanged for accuracy and stability, is the key effect of negative feedback on an operational amplifier.

Question: Draw and briefly describe the circuit diagrams for both the inverting and the non-inverting amplifier for single signal input?

Answer: In the inverting amplifier the input signal is applied through resistor R1 to the inverting (-) terminal of the op-amp, the non-inverting (+) terminal is grounded, and feedback resistor Rf connects the output back to the inverting input; the gain is -Rf / R1 and the output is inverted. In the non-inverting amplifier the signal is applied directly to the (+) terminal, while a voltage divider of R1 and Rf feeds part of the output back to the (-) terminal; the gain is 1 + Rf / R1 and the output is in phase with the input. Both circuits use negative feedback to set a precise, stable gain. The inverting circuit also serves as a summing amplifier when several inputs are connected to the same node.

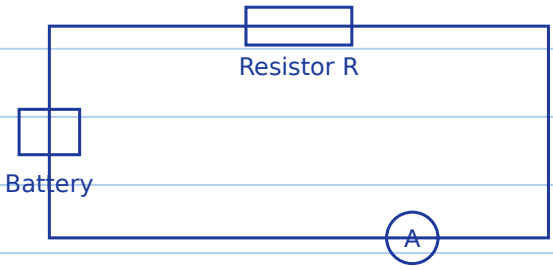


Fig.: Series circuit

Question: Derive an expression for the gain of inverting amplifiers by using virtual earth approximation.

Answer: In an ideal op-amp with negative feedback, the two input terminals are at almost the same potential; since the non-inverting terminal is grounded, the inverting terminal is a virtual earth, at 0 V but drawing no current. The input current through R1 is therefore I = Vi / R1, and because the op-amp takes no input current, this same current must flow through the feedback resistor Rf. The voltage drop across Rf is I Rf, and the output end of Rf is at Vo = - I Rf, the negative sign appearing because the output must drive the current in the opposite sense. Substituting I gives Vo = - (Rf / R1) Vi, so the gain is Av = Vo / Vi = -Rf / R1. The gain is thus set only by the ratio of the two external resistors.

Question: Describe the properties of an ideal operational amplifier.

Answer: An ideal operational amplifier has infinite open-loop voltage gain, so even a microvolt of difference between its inputs would saturate a real device, and feedback is used to control it. It has infinite input impedance, drawing no input current, and zero output impedance, so it can drive any load without loss of signal. Its bandwidth is infinite, its common-mode rejection ratio is infinite, and its output responds instantly with zero noise and zero offset. Real op-amps only approach these values, but the ideal properties make circuit design simple through rules such as the virtual earth approximation.

Exercise - ERQs (Long Answered Questions)

Question: Define Intrinsic (pure) and doped semiconductors. And How the N-type and P-type semiconductors are produced.

Answer: An intrinsic semiconductor is a completely pure semiconductor such as silicon or germanium, in which the numbers of free electrons and holes are equal and conduction is purely due to thermally broken covalent bonds. A doped, or extrinsic, semiconductor is one to which a controlled

small amount of impurity (about 1 atom in 10 million) has been added to greatly increase its conductivity. An N-type semiconductor is produced by doping with a penta-valent donor impurity such as phosphorus, arsenic or antimony; four of its valence electrons bond with silicon atoms and the fifth becomes a free electron, so electrons are the majority carriers. A P-type semiconductor is produced by doping with a trivalent acceptor impurity such as boron, aluminium or gallium; it creates a vacancy or hole for each impurity atom, so holes are the majority carriers. In both types the crystal remains electrically neutral overall, because the numbers of protons and electrons in the material are still balanced.

Question: Describe a P-N junction (diode) discuss its forward and reverse biasing.

Answer: A P-N junction is the boundary formed within a single crystal where p-type material meets n-type material, and it forms the junction diode. At the junction, electrons from the n-side and holes from the p-side diffuse across and recombine, leaving a depletion region containing fixed ions and a barrier potential of about 0.7 V for silicon and 0.3 V for germanium. In forward biasing the positive terminal of the battery is connected to the p-side and the negative to the n-side; the barrier is lowered, the depletion region narrows, and a large current flows once the applied voltage exceeds the barrier potential. In reverse biasing the connections are reversed; the barrier is raised, the depletion region widens, and only a tiny leakage current of minority carriers flows. If the reverse voltage reaches the breakdown value, a large reverse current suddenly flows and may destroy an ordinary diode.

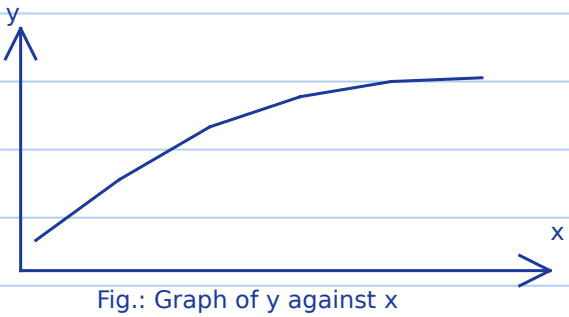


Fig.: Graph of y against x

Question: Describe the I-V characteristic curves of P-N junction.

Answer: The I-V characteristic of a P-N junction is a graph of the current through the diode against the voltage applied across it. In the forward direction the current remains almost zero until the knee voltage (about 0.3 V for germanium and 0.7 V for silicon) is reached, after which the current rises very steeply for small increases of voltage. In the reverse direction only a very small, almost constant leakage current flows as the reverse voltage increases, showing the diode blocking current. At a sufficiently large reverse voltage, breakdown occurs and the reverse current increases sharply, either by the zener or the avalanche mechanism. The curve therefore shows clearly that a diode conducts in one direction only, which is the basis of rectification.

Question: Define rectification and describe the use of diodes for half and full wave rectifications.

Answer: Rectification is the process of converting alternating current into direct current, and a junction diode is used for it because it conducts in one direction only. In half-wave rectification a single diode passes only the half cycles of one polarity and blocks the other half, so the output is pulsating DC with gaps and a ripple frequency equal to the supply frequency. In full-wave rectification, either two diodes with a center-tapped transformer or a four-diode bridge are used, so both half cycles are passed in the same direction through the load. The full-wave output is smoother, has double the ripple frequency and is easier to filter into steady DC with a

capacitor. Rectifiers form the first stage of almost every DC power supply used in electronic equipment.

Question: Describe the function and use of LED, Photodiode and Photo voltaic cell.

Answer: An LED (light emitting diode) is a forward-biased p-n junction made of gallium arsenide type material which emits light when electrons recombine with holes, the color depending on the band gap of the material. It is used in indicator lamps, displays, remote controls (infrared), lighting and optical communication because it is small, fast and long-lasting. A photodiode is a junction diode operated in reverse bias whose reverse current increases when light falls on the junction and creates extra electron-hole pairs; it is used in light sensors, optical receivers, alarms and remote-control receivers. A photovoltaic cell (solar cell) is a large-area p-n junction that converts light energy directly into electrical energy without any external supply, generating a voltage when photons create electron-hole pairs that are separated by the junction field. Solar cells are used in calculators, satellites, street lighting and solar power plants.

Question: Explain the use of transistors as a switch and an amplifier (common-emitter).

Answer: As a switch, the transistor is operated between cut-off, where zero base current makes it behave like an open switch, and saturation, where sufficient base current makes it behave like a closed switch with a very small voltage across it. A small control signal at the base can thus switch a much larger current in the collector circuit, as in relay drivers, logic circuits and motor controls. As a common-emitter amplifier, the emitter is common to input and output, the base-emitter junction is forward biased and the collector junction reverse biased, and the transistor works in the

active region. A small change of base current then produces a beta times larger change of collector current, giving voltage, current and power gain with a 180 degrees phase shift. This dual ability, switching and amplification, makes the transistor the fundamental building block of modern electronics.

Question: Describe the properties of an ideal operational amplifier.

Answer: An ideal op-amp is assumed to have infinite open-loop gain, infinite input impedance, zero output impedance and infinite bandwidth. Because of the infinite gain, in a feedback circuit the difference between its two input terminals is effectively zero, and because of the infinite input impedance it draws no current from the signal source or the feedback network. It rejects any signal common to both inputs completely (infinite CMRR), responds without delay or slew-rate limit, and has no offset voltage, drift or noise. Although no real device is ideal, practical op-amps come close enough that circuits designed with the ideal rules, such as the virtual earth method, work accurately. These properties are what make the op-amp a universal building block for amplifiers, filters, oscillators and analog computing circuits.

Exercise - Numerical

Question: A Ge diode has a voltage drop of 0.4 V when 12 mA flow through it. If the same 470 Ohm is used, what battery voltage is needed?

Answer: Step 1: Given: Diode voltage drop Vd = 0.4 V (book prints 04 V), current I = 12 mA = 0.012 A, series resistor R = 470 ohm, battery voltage V is required

Step 2: Formula: The battery voltage equals the diode drop plus the drop across the series resistor, V = Vd + I R

Step 3: Substitution: V = 0.4 + (0.012 x 470) = 0.4 + 5.64

Step 4: Calculation: V = 6.04 V

Step 5: Result: A battery voltage of 6.04 V is needed

Question: A semiconductor diode laser has a peak emission wavelength of 1.55 micro m. Find its band gap in eV.

Answer: Step 1: Given: Wavelength lambda = 1.55 micro m = 1.55 x10^-6 m = 1550 nm, band gap Eg is required; use h c = 1240 eV nm

Step 2: Formula: Eg = h c / lambda

Step 3: Substitution: Eg = 1240 / 1550

Step 4: Calculation: Eg = 0.80 eV

Step 5: Result: The band gap is 0.8 eV

Question: Calculate the current through 48 ohm resistor in the circuit shown in Fig. (i). Assume the diodes to be of silicon and forward resistance of each diode is 1 ohm.

Answer: Step 1: Given: Supply 10 V, load resistor 48 ohm, silicon diodes with barrier voltage 0.7 V each and forward resistance 1 ohm each; in Fig. (i) the conducting path contains two diodes in series with the 48 ohm resistor, as redrawn in the book Fig. (ii)

Step 2: Formula: Net voltage across the resistive path = supply voltage minus the two diode barrier drops, and current I = net voltage / total resistance

Step 3: Substitution: Net voltage = 10 - 0.7 - 0.7 = 8.6 V, total resistance = 48 + 1 + 1 = 50 ohm, I = 8.6 / 50

Step 4: Calculation: I = 0.172 A = 172 mA

Step 5: Result: The current through the 48 ohm resistor is 172 mA, with

8.6 V across the resistive combination of 50 ohms

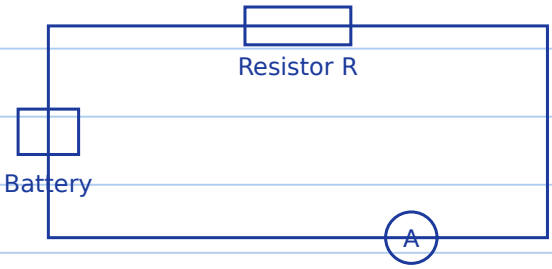


Fig.: Series circuit

Question: Find the voltage VA in the circuit shown in Fig.(i). Use simplified model.

Answer: Step 1: Given: Supply 20 V, a silicon diode (barrier 0.7 V) and a germanium diode (barrier 0.3 V) connected in parallel between the supply and point A, with a 3 kilo-ohm resistor from A to ground; VA is required, using the simplified (barrier voltage) model of Fig. (ii)

Step 2: Formula: The diode with the smaller barrier voltage conducts first and clamps the node, so VA = supply voltage minus the smallest barrier drop that turns on

Step 3: Substitution: The Ge diode needs only 0.3 V, so it conducts and VA = 20 - 0.3; the Si diode then has only 0.3 V across it, less than its 0.7 V barrier, so it stays off

Step 4: Calculation: VA = 20 - 0.3 = 19.7 V

Step 5: Result: The voltage at point A is 19.7 V

Question: In a common base connection, IE = 1 mA, IC = 0.95 mA. Calculate the value of IB.

Answer: Step 1: Given: Emitter current IE = 1 mA, collector current IC = 0.95 mA, base current IB is required

Step 2: Formula: IE = IB + IC, so IB = IE - IC

Step 3: Substitution: IB = 1 - 0.95

Step 4: Calculation: IB = 0.05 mA

Step 5: Result: The base current is 0.05 mA (50 micro A)

Question: Find the value of beta if (i) alpha = 0.9 (ii) alpha = 0.98 (iii) alpha = 0.99.

Answer: Step 1: Given: alpha values 0.9, 0.98 and 0.99, beta is required in each case

Step 2: Formula: beta = alpha / (1 - alpha)

Step 3: Substitution: (i) beta = 0.9 / (1 - 0.9) = 0.9 / 0.1, (ii) beta = 0.98 / (1 - 0.98) = 0.98 / 0.02, (iii) beta = 0.99 / (1 - 0.99) = 0.99 / 0.01

Step 4: Calculation: (i) beta = 9, (ii) beta = 49, (iii) beta = 99

Step 5: Result: beta is 9, 49 and 99 respectively, showing that beta rises very rapidly as alpha approaches 1

Question: Calculate IE in a transistor for which beta = 50 and IB = 20 micro A.

Answer: Step 1: Given: beta = 50, IB = 20 micro A = 0.02 mA, emitter current IE is required

Step 2: Formula: IC = beta IB, and IE = IB + IC = (beta + 1) IB

Step 3: Substitution: IC = 50 x 20 micro A = 1000 micro A = 1.00 mA, IE = 0.02 mA + 1.00 mA

Step 4: Calculation: IE = 1.02 mA

Step 5: Result: The emitter current is 1.02 mA

Question: Find the alpha rating of the transistor shown in Fig. Hence determine the value of IC using both alpha and beta rating of the transistor.

Answer: Step 1: Given: From the figure, IB = 240 micro A = 0.24 mA, IE = 12 mA, beta = 49, alpha and IC are required

Step 2: Formula: alpha = beta / (beta + 1), and IC = beta IB; also IC = alpha IE

Step 3: Substitution: alpha = 49 / (49 + 1) = 49 / 50, IC = 49 x 0.24 mA, check by IC = 0.98 x 12 mA

Step 4: Calculation: alpha = 0.98, IC = 11.76 mA by both methods (49 x 0.24 = 11.76 and 0.98 x 12 = 11.76)

Step 5: Result: The alpha rating is 0.98 and the collector current is 11.76 mA

Unit 23: Digital Electronics

Exercise - Multiple Choice Questions (MCQs)

The voltage level, typically associated with a low state in digital electronic is:

(a) -5 volts

✓ (b) 0 volts

(c) +5 volts

(d) +10 volts

In binary representation, the correspond of 0 in digital electronics is:

✓ (a) Low state

(b) High state

(c) Open circuit

(d) Closed circuit

The primary purpose of a truth table in digital electronics is to:

(a) Determine input voltage

(b) Measure circuit resistance

✓ (c) Analyze output states based on input combinations

(d) Calculate circuit power

Which logic gate combination produces an output of 1 when at least one input is 1?

(a) AND

✓ (b) OR

(c) XOR

(d) NOT

In a circuit with a 2-input OR gate, the status of the lamp if either switch is closed

(a) OFF

✓ (b) ON

(c) Blinking

(d) Flickering

The behavior of 2 input AND gate, when both switches are closed, Lamp is

(a) OFF

✓ (b) ON

(c) blinking

(d) flickering

Question: In a 3-input OR gate, if all inputs are 0 so the output will be: (a) 1 (b) 0 (c) Undefined (d) Both A and B

Answer: (b) An OR gate gives output 0 only in the single case where all of its inputs are 0, and this rule holds for any number of inputs. For a 3-input OR gate the combination 0, 0, 0 therefore produces output 0. Every other combination, having at least one input equal to 1, produces output 1. The output is perfectly defined, so options (c) and (d) do not apply.

If the input is 1, The output of a NOT gate will be:

(a) 1

✓ (b) 0

(c) Undefined

(d) Both A and B

In a 2-input XOR gate, when both inputs are 1 than the output will be

(a) 1

✓ (b) 0

(c) Undefined

(d) Both A and B

Which combination of logic gates is equivalent to an XOR gate?

✓ (a) AND + OR

(b) NOT + NOR

(c) OR + NAND

(d) AND + NAND

Exercise - CRQs (Short Answered Questions)

Question: Explain the significance of signal levels in digital electronics and how they are represented in terms of voltage.

Answer: Signal levels allow information to be represented by just two well-separated voltage ranges, so a digital circuit is almost unaffected by small noise and distortion. The low level, logic 0, is typically 0 V, and the high level, logic 1, is typically +5 V in TTL circuits. A threshold level separates the two ranges: voltages below it are read as 0 and voltages above it as 1, which gives the circuit its noise margin and reliable operation. The difference between the high and low levels is called the voltage swing, and a larger swing improves noise immunity.

Question: Describe the binary representation used in digital electronics. How is this representation related to the high and low states?

Answer: Digital electronics uses the binary system, in which every signal is represented by one of two digits, 0 or 1, called bits. The binary 1 is represented physically by the high state, a higher voltage such as +5 V, corresponding to a closed (ON) switch condition, and the binary 0 is represented by the low state, about 0 V, corresponding to an open (OFF) condition. Combinations of bits in logic gates follow the rules of Boolean algebra, with the output for every input combination given by a truth table. Because only two states are involved, digital systems are simple, reliable and form the basis of computers and modern communication.

Question: Choose any three logic gate symbols and explain their operations.

Answer: The AND gate has a D-shaped symbol with two or more inputs and one output F = A . B; its output is 1 only when all inputs are 1, like two switches in series lighting a lamp only when both are closed. The OR gate has a curved-input, pointed-output symbol and gives F = A + B; its output is 1 when at least one input is 1, like switches in parallel. The NOT gate is a triangle with a small circle at its output and gives F = NOT A; it inverts its single input, producing 1 for input 0 and 0 for input 1. The complete operation of each gate is summarized by its truth table.

Question: Create a combination of logic gates that mimics the behavior of an XOR gate.

Answer: An XOR output is 1 only when the inputs differ, F = A (NOT B) + (NOT A) B, and this expression shows the required combination directly. Take inputs A and B, pass each through a NOT gate to obtain NOT A and NOT B, then feed A with NOT B into one AND gate and NOT A with B into a second AND gate. Finally combine the outputs of the two AND gates in an OR gate; the result is 1 for input combinations 0,1 and 1,0 and 0 for 0,0 and 1,1. This AND-OR-NOT combination therefore has exactly the truth table of a single XOR gate.

Exercise - ERQs (Long Answered Questions)

Question: Design a simple circuit using a 2-input AND gate, a switch, a lamp, and a battery. Explain how the circuit operates and the conditions under which the lamp will be illuminated.

Answer: Connect a battery, two switches A and B and a lamp all in series in a single loop; this series arrangement is the electrical model of a 2-input AND gate, with the switches as inputs and the lamp glow as output 1. When both switches are open, or when only one of them is closed, the loop is broken, no current flows and the lamp remains OFF, corresponding

to outputs 0 for inputs 0,0, 0,1 and 1,0. Only when both switches are closed is the circuit completed, current flows through the lamp and it is illuminated, corresponding to output 1 for input 1,1. The lamp is therefore illuminated under exactly one condition: both inputs high. This simple experiment demonstrates the AND truth table and the idea that an AND gate acts as a logical decision element requiring all conditions to be satisfied together.

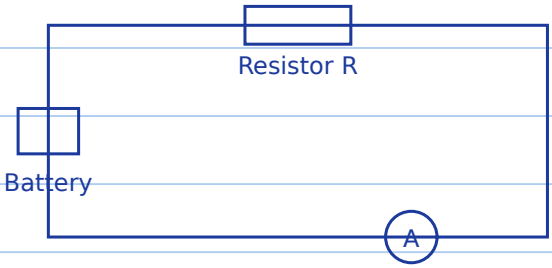


Fig.: Series circuit

Question: Construct a truth table for a 3-input OR gate. Explain how the truth table represents the relationship between input combinations and the resulting output.

Answer: A 3-input OR gate has three inputs A, B, C and one output F = A + B + C, so its truth table has 2 to the power 3 = 8 rows, one for every combination of inputs. The rows are: 0 0 0 gives 0; 0 0 1 gives 1; 0 1 0 gives 1; 0 1 1 gives 1; 1 0 0 gives 1; 1 0 1 gives 1; 1 1 0 gives 1; and 1 1 1 gives 1. The table shows that the output is 0 in only the first row, where every input is 0. In all seven remaining rows at least one input is 1, so the output is 1, which is exactly the OR rule extended to three inputs. The truth table thus represents the complete input-output relationship of the gate in a compact form that can be checked against any real circuit.

Unit 24: Relativity

Exercise - Multiple Choice Questions (MCQs)

The General Theory of Relativity was a new way of understanding of

(a) The speed of light

✓ (b) gravity

(c) mass

(d) force

An object at rest has a mass of 1 kg. What is its mass when it is moving at a speed of 0.9c?

(a) Infinite

(b) 1.2 kg

✓ (c) 2.3 kg

(d) 1 kg

The equivalence of principle in general relativity is:

✓ (a) The equivalence of inertial and gravitational mass

(b) The equivalence of electric and magnetic fields

(c) The equivalence of space and time

(d) The equivalence of matter and energy

Which of the following phenomena is NOT predicted by special relativity?

(a) Time dilation

(b) Length contraction

✓ (c) Gravitational waves

(d) Relativistic mass increase

What does the equivalence principle state about acceleration and gravity?

(a) They are completely different forces

✓ (b) They are indistinguishable for an observer

(c) Gravity is stronger than acceleration

(d) Acceleration cancels out gravity

Light passing near a massive object like a star will bend due to:

✓ (a) Gravitational lensing

(b) refraction

(c) reflection

(d) diffraction

What is the main reason astronomers cannot directly observe black holes?

(a) They are too small

(b) They deflect light

(c) They are too far away

✓ (d) their immense gravity traps light

Objects cannot exceed the speed of light because:

✓ (a) Their mass becomes infinite

(b) Their length becomes zero

(c) Time slows down to zero

(d) they lose all energy

Why is the Galilean transformation not valid at high speeds?

(a) It cannot handle accelerating frames

✓ (b) It violates the constancy of the speed of light

(c) It only works in flat spacetime

(d) It neglects time dilation effects

The example of inertial frame in our everyday life is:

(a) A car accelerating on a highway

✓ (b) A train moving smoothly at constant speed

(c) A person standing on a spinning platform

(d) an airplane encountering turbulence

Exercise - CRQs (Short Answered Questions)

Question: Show that for values of v much less than c, Lorentz transformation reduces to the Galilean transformation.

Answer: The Lorentz transformations are x prime = (x - v t) / sqrt(1 - v squared / c squared) and t prime = (t - v x / c squared) / sqrt(1 - v squared / c squared). When v is much less than c, v squared / c squared is negligible compared with 1, so the square root factor becomes 1 (gamma becomes 1). Also the term v x / c squared in the time equation becomes negligible, so t prime = t. The equations then reduce to x prime = x - v t and t prime = t, which are exactly the Galilean transformations. This shows that Newtonian mechanics is the low-speed limiting case of relativity.

Question: If a particle could move with the velocity of light, how much K.E. would it possess?

Answer: The relativistic kinetic energy is K.E. = (m - m0) c squared, with m = m0 / sqrt(1 - v squared / c squared). If v were equal to c, the denominator would become zero and the relativistic mass m would become infinite. The kinetic energy required would therefore also be infinite, which can never be supplied by any finite source. Hence no material particle can ever be accelerated to the velocity of light, and the question of its kinetic energy at c has no finite answer.

Question: Explain the difference between Special and General Relativity in simple terms.

Answer: Special relativity (1905) deals with objects moving at constant velocity in inertial frames and with the constancy of the speed of light, and it leads to time dilation, length contraction and E = m c squared. It does not include gravity or acceleration. General relativity (1915) extends the ideas to all frames, including accelerating ones, and explains gravity as

the curvature of spacetime produced by mass and energy. In simple words, special relativity tells how space and time behave at very high speeds, while general relativity tells how mass shapes space and time to produce gravity.

Question: Differentiate between Inertial Frames of Reference and Non-Inertial Frames of Reference.

Answer: An inertial frame is at rest or moves with constant velocity in a straight line; in it a body with no net force stays at rest or moves uniformly, and Newtons laws hold in their simple form. A non-inertial frame is accelerating, decelerating or rotating; in it Newtons laws do not hold directly and fictitious (pseudo) forces, such as the backward push felt in an accelerating car, must be introduced. A uniformly moving train is inertial, while a braking train, a turning car and a rotating merry-go-round are non-inertial. Einsteins special relativity applies to inertial frames, while general relativity covers both kinds of frames.

Question: Why cannot any object move at the speed of light?

Answer: According to relativity the mass of a moving object is m = m0 / sqrt(1 - v squared / c squared), which increases without limit as v approaches c. Accelerating the object further therefore demands more and more energy, and reaching c would require infinite energy and would give it infinite mass and zero contracted length, all of which are impossible. Every experiment with particles in accelerators confirms that they approach c ever more closely but never reach it, however much energy is supplied. Only massless particles, such as photons, travel at exactly the speed of light.

Question: What is the limitation in the Galilean Transformation Equation, and how did Lorentz solve it?

Answer: The Galilean equations x prime = x - v t and t prime = t assume

that time is absolute and that velocities simply add, so they predict that a moving observer would measure a different speed of light, contradicting the Michelson-Morley experiment and Einsteins postulate that c is the same for all inertial observers. Lorentz solved this by introducing the factor gamma = 1 / sqrt(1 - v squared / c squared) and making time itself relative, t prime = (t - v x / c squared) / gamma denominator form. In the Lorentz transformation, space and time measurements adjust together in exactly the way needed to keep the speed of light equal to c in every inertial frame. For v much less than c the Lorentz equations reduce to the Galilean ones, so everyday mechanics is preserved.

Question: Calculate the value of gamma (Lorentz factor) if the object is moving at the speed of light.

Answer: Step 1: Given: v = c, Lorentz factor gamma is required

Step 2: Formula: gamma = 1 / sqrt(1 - v squared / c squared)

Step 3: Substitution: With v = c, v squared / c squared = 1, so gamma = 1 / sqrt(1 - 1) = 1 / sqrt(0) = 1 / 0

Step 4: Calculation: Division by zero gives gamma = infinity

Step 5: Result: The Lorentz factor becomes infinite at v = c, which is another way of seeing that a material object would need infinite energy to reach the speed of light, so v = c is impossible for it

Exercise - ERQs (Long Answered Questions)

Question: State and explain the basic postulates of Einstein special theory of relativity. Discuss length-contraction, mass variation and time-dilation.

Answer: The first postulate states that the laws of physics are the same in all inertial frames of reference, so no experiment can detect absolute uniform motion. The second postulate states that the speed of light in

vacuum is the same constant c for all inertial observers, whatever the motion of the source or observer. From these postulates it follows that a moving clock runs slow: time dilation, delta t = delta t0 / sqrt(1 - v squared / c squared), so the moving time interval appears longer to a stationary observer. It also follows that a moving object is measured shorter along its direction of motion: length contraction, L = L0 sqrt(1 - v squared / c squared). Finally the mass increases with speed, m = m0 / sqrt(1 - v squared / c squared), which prevents any object from reaching c. These three effects are negligible at ordinary speeds but have all been confirmed experimentally, for example by the extended lifetimes of fast-moving muons.

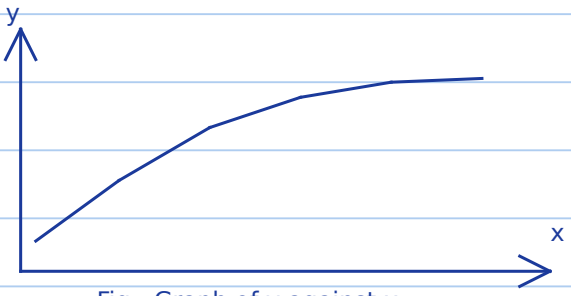


Fig.: Graph of y against x

Question: Explain the concept of mass-energy equivalence. Derive Einstein mass-energy relation and demonstrate that 1 atomic mass unit (u) is equivalent to 931 MeV.

Answer: Mass-energy equivalence states that mass is a concentrated form of energy and that the total energy of a body of relativistic mass m is E = m c squared, so its rest energy is E0 = m0 c squared and its kinetic energy is (m - m0) c squared. In Einsteins derivation, a body emitting two equal light pulses of total energy E in opposite directions loses momentum in a moving frame unless its mass decreases by E / c squared, which forces the conclusion that a loss of energy E corresponds to a loss of mass E / c squared, i.e. E = m c squared. For one atomic mass unit, m = 1.6605 x10^-27 kg, so E = m c squared = 1.6605 x10^-27 x (3 x10^8) squared =

1.494 x10^-10 J. Dividing by 1.602 x10^-13 J per MeV gives E = 931 MeV, approximately. This equivalence is the basis of nuclear binding energy, fission and fusion power, and the energy released in stars.

Question: Discuss the important conclusions derived from General Theory of relativity. What are the experimental observations in favour of these conclusions?

Answer: General relativity concludes that gravity is the curvature of spacetime caused by mass and energy, and it predicts the bending of light by a gravitational field, the gravitational red shift of light climbing out of a field, and the extra precession of the perihelion of Mercury. It also predicts time running slower in stronger gravity (gravitational time dilation) and the existence of gravitational waves and black holes. The bending of starlight by the Sun was observed during the solar eclipse of 1919, and Mercurys anomalous precession of 43 arc-seconds per century matched the theory exactly. Gravitational red shift was confirmed in the Pound-Rebka laboratory experiment, atomic clocks flown around the Earth confirmed gravitational time dilation, and gravitational waves were detected directly by LIGO in 2015. Modern GPS satellites must apply both special and general relativistic corrections daily, providing a routine practical confirmation.

Question: How does it principle of relativity differ from the classical Galilean view?

Answer: The Galilean view assumes absolute space and absolute time: lengths and time intervals are the same for all observers, and velocities add and subtract simply, so the speed of light should depend on the observers motion. The principle of relativity in Einsteins theory keeps the laws of physics, including the speed of light c, the same in all inertial frames, and as a result space and time measurements become relative to

the observer. Events simultaneous in one frame are not necessarily simultaneous in another, moving lengths contract and moving clocks dilate, effects entirely absent in the Galilean picture. Mathematically the Galilean transformation is replaced by the Lorentz transformation, to which it reduces only when v is much less than c. The Galilean view is therefore an approximation valid for everyday speeds, while relativity is the more general and accurate description.

Question: Explain the concept of spacetime curvature in general relativity and how it is used to visualize the effects of gravity.

Answer: In general relativity, space and time are united into a four-dimensional spacetime, and the presence of mass and energy curves this spacetime just as a heavy ball placed on a stretched rubber sheet makes a dent in the sheet. Other objects and even light rays then follow the straightest possible paths (geodesics) in the curved geometry, and we experience this curving of paths as the force of gravity. A small ball rolled across the rubber sheet spirals toward the heavy ball, giving a simple visual model of orbits and of the bending of light near the Sun. Near a very dense object the curvature becomes so extreme that an event horizon forms and a black hole results. This geometric picture explains gravity without any mysterious action at a distance and correctly predicts all the observed tests of the theory.

Question: Derive the basic equations of the Galilean transformation and explain how they relate the positions and velocities of objects in different inertial frames.

Answer: Let frame S prime move with constant velocity v along the x-axis of frame S, with their origins coinciding at t = 0. A point event has coordinates (x, y, z, t) in S; in classical physics time is absolute, so t prime = t, and since the origin of S prime has moved a distance v t in S, the

x-coordinate in S prime is x prime = x - v t. Motion perpendicular to v is unaffected, so y prime = y and z prime = z, completing the Galilean transformation equations. Differentiating x prime = x - v t with respect to the common time gives the velocity rule u prime x = u x - v, while the other components are unchanged, u prime y = u y and u prime z = u z. These equations relate positions and velocities between inertial frames correctly at low speeds, but they fail near the speed of light, where the Lorentz transformation must be used instead.

Question: Discuss the Lorentz transformation equations in special relativity and how they describe the relationship between space and time coordinates in different inertial frames. Give examples to illustrate their application.

Answer: For a frame S prime moving at velocity v along the x-axis of S, the Lorentz transformations are x prime = gamma (x - v t), t prime = gamma (t - v x / c squared), y prime = y and z prime = z, where gamma = 1 / sqrt(1 - v squared / c squared). Unlike the Galilean equations, space and time coordinates are mixed together: the time in one frame depends on both time and position in the other frame, which is the mathematical root of the relativity of simultaneity. Direct consequences are time dilation, delta t = gamma delta t0, and length contraction, L = L0 / gamma, whenever the coordinates of two events are substituted into the equations. For example, fast muons created in the upper atmosphere reach the ground because, in the Earth frame, their lifetime is stretched by gamma, while in the muon frame the atmosphere is length-contracted by the same factor. At ordinary speeds gamma is almost 1 and the equations reduce to the Galilean transformation, showing the agreement of the two theories in the low-speed limit.

Exercise - Numerical

Question: A rod 1 meter long is moving along its length with a velocity 0.6c. Calculate its length as it appears to an observer (a) on the earth (b) moving with the rod itself.

Answer: Step 1: Given: Proper length L0 = 1 m, v = 0.6c, contracted length L is required

Step 2: Formula: L = L0 sqrt(1 - v squared / c squared)

Step 3: Substitution: L = 1 x sqrt(1 - (0.6c) squared / c squared) = sqrt(1 - 0.36) = sqrt(0.64)

Step 4: Calculation: (a) For the observer on Earth, L = 0.8 m; (b) for an observer moving with the rod the relative velocity is zero, so no contraction occurs and L = L0

Step 5: Result: (a) 0.8 m as seen from Earth, (b) 1 m for the observer moving with the rod

Question: How fast would a rocket have to go relative to an observer for its length to be contracted to 99 percent of its length at rest?

Answer: Step 1: Given: L / L0 = 99 percent = 0.99, speed v is required

Step 2: Formula: L = L0 sqrt(1 - v squared / c squared), so v = c sqrt(1 - (L / L0) squared)

Step 3: Substitution: v = c sqrt(1 - (0.99) squared) = c sqrt(1 - 0.9801) = c sqrt(0.0199)

Step 4: Calculation: sqrt(0.0199) = 0.141, so v = 0.141 c = 0.141 x 3 x10^8 = 4.24 x10^7 m/s

Step 5: Result: The rocket must travel at about 42.45 x10^6 m/s (0.141c) relative to the observer

Question: A particle with a proper lifetime of 1 micro s moves through the

laboratory at 2.7 x10^8 m/s. (a) What is its lifetime, as measured by observers in the laboratory? (b) What will be the distance traversed by it before disintegrating?

Answer: Step 1: Given: Proper lifetime delta t0 = 1 micro s = 1 x10^-6 s, v = 2.7 x10^8 m/s = 0.9c, laboratory lifetime and distance are required

Step 2: Formula: delta t = delta t0 / sqrt(1 - v squared / c squared) (time dilation), and distance d = v x delta t

Step 3: Substitution: gamma = 1 / sqrt(1 - 0.81) = 1 / 0.436 = 2.29, so delta t = 2.29 x 1 x10^-6 s, and d = 2.7 x10^8 x 2.29 x10^-6

Step 4: Calculation: (a) delta t = 2.3 x10^-6 s, (b) d = 619 m, approximately 620 m

Step 5: Result: The laboratory lifetime is about 2.3 x10^-6 s and the particle travels about 620 m before disintegrating

Question: At what speed is a particle moving if the mass is equal to three times its rest mass?

Answer: Step 1: Given: m = 3 m0, speed v is required

Step 2: Formula: m = m0 / sqrt(1 - v squared / c squared), so sqrt(1 - v squared / c squared) = m0 / m

Step 3: Substitution: sqrt(1 - v squared / c squared) = 1 / 3, hence 1 - v squared / c squared = 1 / 9 and v squared / c squared = 8 / 9

Step 4: Calculation: v / c = sqrt(8) / 3 = 2 sqrt(2) / 3 = 0.943, so v = 0.943c = 2.83 x10^8 m/s

Step 5: Result: The particle moves at v = (2 sqrt(2) / 3) c, about 0.943c

Question: If 4 kg of a substance is fully converted into energy, how much energy is produced?

Answer: Step 1: Given: Mass converted m = 4 kg, c = 3 x10^8 m/s,

energy E is required

Step 2: Formula: E = m c squared

Step 3: Substitution: E = 4 x (3 x10^8) squared = 4 x 9 x10^16

Step 4: Calculation: E = 3.6 x10^17 J

Step 5: Result: The energy produced is 3.6 x10^17 J

Question: Calculate the rest energy of an electron in joules and in electron volts.

Answer: Step 1: Given: Electron rest mass m0 = 9.11 x10^-31 kg, c = 3 x10^8 m/s, 1 eV = 1.602 x10^-19 J, rest energy E0 is required

Step 2: Formula: E0 = m0 c squared

Step 3: Substitution: E0 = 9.11 x10^-31 x (3 x10^8) squared = 9.11 x10^-31 x 9 x10^16

Step 4: Calculation: E0 = 8.2 x10^-14 J; converting, E0 = 8.2 x10^-14 / 1.602 x10^-19 = 5.11 x10^5 eV

Step 5: Result: The rest energy of an electron is 8.2 x10^-14 J, equal to 0.511 MeV

Question: Calculate the K.E. of an electron moving with a velocity of 0.98 times the velocity of light in the laboratory system.

Answer: Step 1: Given: v = 0.98c, electron rest energy m0 c squared = 8.2 x10^-14 J, kinetic energy is required

Step 2: Formula: gamma = 1 / sqrt(1 - v squared / c squared), and K.E. = (gamma - 1) m0 c squared

Step 3: Substitution: gamma = 1 / sqrt(1 - (0.98) squared) = 1 / sqrt(1 - 0.9604) = 1 / sqrt(0.0396) = 1 / 0.199 = 5.03, so K.E. = (5.03 - 1) x 8.2 x10^-14

Step 4: Calculation: K.E. = 4.03 x 8.2 x10^-14 = 3.3 x10^-13 J

Step 5: Result: The kinetic energy of the electron is 3.3 x10^-13 J (about 2.06 MeV)

Question: At what velocity does the K.E. of a particle equal its rest energy?

Answer: Step 1: Given: K.E. = rest energy m0 c squared, velocity v is required

Step 2: Formula: K.E. = (gamma - 1) m0 c squared, so the condition gives (gamma - 1) m0 c squared = m0 c squared, i.e. gamma = 2

Step 3: Substitution: 1 / sqrt(1 - v squared / c squared) = 2, so sqrt(1 - v squared / c squared) = 1 / 2 and 1 - v squared / c squared = 1 / 4

Step 4: Calculation: v squared / c squared = 3 / 4, so v = (sqrt(3) / 2) c = 0.866c

Step 5: Result: The velocity is (sqrt(3) / 2) c, about 0.866c or 2.6 x10^8 m/s

Question: A particle of rest mass m0 moves with speed c / sqrt(2). Calculate its mass, momentum, total energy and kinetic energy.

Answer: Step 1: Given: Rest mass m0, v = c / sqrt(2), so v squared / c squared = 1 / 2; mass m, momentum p, total energy E and K.E. are required

Step 2: Formula: gamma = 1 / sqrt(1 - v squared / c squared), m = gamma m0, p = gamma m0 v, E = gamma m0 c squared, K.E. = (gamma - 1) m0 c squared

Step 3: Substitution: gamma = 1 / sqrt(1 - 1/2) = 1 / sqrt(1/2) = sqrt(2) = 1.414, so m = sqrt(2) m0, p = sqrt(2) m0 (c / sqrt(2)), E = sqrt(2) m0 c squared, K.E. = (sqrt(2) - 1) m0 c squared

Step 4: Calculation: m = sqrt(2) m0, p = m0 c, E = sqrt(2) m0 c squared =

1.414 m0 c squared, K.E. = 0.414 m0 c squared, approximately 0.41 m0 c squared

Step 5: Result: Mass is sqrt(2) m0, momentum is m0 c, total energy is sqrt(2) m0 c squared, and kinetic energy is 0.41 m0 c squared

Question: The nearest star to Earth is Proximal Centauri, 4.3 light-years away. A spaceship with a constant speed of 0.800c relative to the Earth travels toward the star. (a) How much time would elapse on a clock as measured by travelers on the spacecraft? (b) How long does the trip take according to Earth observers?

Answer: Step 1: Given: Distance in Earth frame L0 = 4.3 light-years, v = 0.800c, times in both frames are required

Step 2: Formula: Earth-frame time t = L0 / v; the travelers time is the proper time t0 = t sqrt(1 - v squared / c squared), equivalently the contracted distance L = L0 sqrt(1 - v squared / c squared) divided by v

Step 3: Substitution: (b) t = 4.3 / 0.800 = 5.38 years; sqrt(1 - 0.64) = sqrt(0.36) = 0.600, so (a) t0 = 5.38 x 0.600, or L = 4.3 x 0.600 = 2.58 light-years and t0 = 2.58 / 0.800

Step 4: Calculation: (b) t = 5.38 years, (a) t0 = 3.22 years

Step 5: Result: (a) 3.22 years elapse on the spacecraft clock, (b) the trip takes 5.38 years according to Earth observers

Unit 25: Quantum Physics

The exercise pages of this unit are missing from the official Sindh Textbook Board scan of this book (the scan jumps from book page 218 to book page 251), so the answers to the questions below have been intentionally left blank. The questions are listed for practice; the solved answers will be added when a complete copy of the book becomes available.

Exercise — Multiple Choice Questions

According to Planck quantum theory, the energy of a photon is:

(a) E = hf

(b) E = mc

(c) E = mv2 / 2

(d) E = hf2

In the photoelectric effect, electrons are emitted from a metal surface only when the incident light:

(a) Is very intense

(b) Has frequency above the threshold frequency

(c) Falls for a long time

(d) Has a long wavelength

If the intensity of incident light in a photoelectric experiment is increased while its frequency is kept constant, then:

(a) The maximum kinetic energy of photoelectrons increases

(b) The stopping potential increases

(c) More photoelectrons are emitted per second

(d) No photoelectrons are emitted

The increase in wavelength in the Compton effect depends upon:

(a) The intensity of X-rays

(b) The scattering angle

(c) The material of the target only

(d) The speed of the target

The minimum energy of a gamma ray photon required for pair production is about:

(a) 0.51 MeV

(b) 1.02 MeV

(c) 2.04 MeV

(d) 13.6 eV

The de Broglie wavelength of a particle is given by:

(a) lambda = mv / h

(b) lambda = h / mv

(c) lambda = hf

(d) lambda = mc / h

The Davisson and Germer experiment confirmed the wave nature of:

(a) Light

(b) Electrons

(c) Protons in nuclei

(d) Sound waves

Heisenberg uncertainty principle states that it is impossible to measure simultaneously, with unlimited accuracy, the:

(a) Charge and mass of a particle

(b) Position and momentum of a particle

(c) Speed and wavelength of light

(d) Energy and charge of a photon

As the temperature of a black body increases, the wavelength at which radiation intensity is maximum:

(a) Increases

(b) Decreases

(c) Remains the same

(d) First increases then decreases

The momentum of a photon of wavelength lambda is:

(a) p = h lambda

(b) p = h / lambda

(c) p = lambda / h

(d) p = hc / lambda

Exercise — Short Answer Questions

Question: What is a black body? State two experimental facts about black body radiation.

Answer:

Question: State Planck hypothesis and explain how it resolved the ultraviolet catastrophe.

Answer:

Question: State the laws of photoelectric emission and explain one law which wave theory could not explain.

Answer:

Question: What is the Compton effect? What does it prove about the nature of light?

Answer:

Question: Describe pair production and annihilation of matter in terms of conservation laws.

Answer:

Question: State de Broglie hypothesis. How was it confirmed by the Davisson and Germer experiment?

Answer:

Question: Why can an electron microscope achieve much higher resolution than an optical microscope?

Answer:

Exercise — Long Answer Questions

Question: Describe black body radiation. What was the ultraviolet

catastrophe and how did Planck quantum hypothesis explain the observed spectrum?

Answer:

Question: Explain the photoelectric effect on the basis of quantum theory and write Einstein photoelectric equation. Mention one use of the effect in solar cells or photocells.

Answer:

Question: Derive the Compton shift expression qualitatively and explain why the effect is observed with X-rays but not with visible light.

Answer:

Question: Explain wave-particle duality and describe how the electron microscope uses the wave nature of electrons to study micro structures of matter.

Answer:

Exercise — Numerical Problems

Question: A black body is at a temperature of 6000 K. Calculate the wavelength at which it radiates most strongly. Wien constant is 2.898x10^-3 m K.

Answer:

Question: Calculate the energy, in joule and in electron volt, of a photon of light of wavelength 500 nm. Take h = 6.63x10^-34 J s, c = 3x10^8 m/s and 1 eV = 1.60x10^-19 J.

Answer:

Question: Light of wavelength 400 nm falls on a metal whose work function is 2.0 eV. Find the maximum kinetic energy of the emitted photoelectrons and the stopping potential.

Answer:

Question: X-rays are scattered from a target at an angle of 90 degrees. Calculate the Compton shift in wavelength. Take h = 6.63x10^-34 J s, m0 = 9.11x10^-31 kg and c = 3x10^8 m/s.

Answer:

Question: An electron is accelerated from rest through a potential difference of 100 V. Calculate its de Broglie wavelength. Take h = 6.63x10^-34 J s, m = 9.11x10^-31 kg and e = 1.60x10^-19 C.

Answer:

Question: The position of an electron is measured with an accuracy of 1x10^-10 m. Estimate the minimum uncertainty in its momentum and in its velocity. Take h = 6.63x10^-34 J s and m = 9.11x10^-31 kg.

Answer:

Unit 26: Atomic Physics

Exercise — Multiple Choice Questions (MCQs)

Atomic spectra are also known as:

(a) Discrete spectra

✓ (b) Line spectra

(c) Emission spectrum

(d) Continuous spectra

The ratio of kinetic energy to total energy for an electron in a Bohr orbit is:

✓ (a) 1:1

(b) 2:3

(c) 1:2

(d) 2:1

The Bohr radius increases as the principal quantum number:

✓ (a) increases

(b) decreases

(c) remains constant

(d) oscillates

Laser beams consist of:

(a) Highly coherent electrons

✓ (b) Highly coherent photons

(c) Highly coherent phonons

(d) Highly coherent neutrons

The ruby laser is an example of:

✓ (a) Optical pumping

(b) Electrical pumping

(c) Chemical pumping

(d) Thermal pumping

Laser action requires a medium with at least:

✓ (a) Three energy levels

(b) Four energy levels

(c) Two energy levels

(d) Five energy levels

Population inversion occurs in:

✓ (a) Active medium

(b) Passive medium

(c) Gaseous medium

(d) Vapour medium

X-rays transfer _____ to metals.

✓ (a) Energy

(b) Force

(c) Pressure

(d) Momentum

X-rays are deflected by:

(a) Magnetic fields

(b) Electric fields

(c) Gravitational fields

✓ (d) No fields

Doubling the voltage of an X-ray tube:

(a) Halves the intensity

(b) Keeps the intensity unchanged

(c) Doubles the intensity

✓ (d) Quadruples the intensity

Exercise — CRQs (Short Answered Questions)

Question: Why do different elements have different spectra?

Answer: Each element has a different number of electrons and a different set of allowed energy levels, fixed by the charge of its nucleus and the arrangement of its electrons. Spectral lines are produced when electrons jump between these levels, and the photon energy hf equals the difference of the two level energies. Since the level differences are different for every element, the wavelengths emitted or absorbed are also different. The spectrum therefore acts as a fingerprint by which an element can be identified.

Question: In the Bohr model, how many times larger is the radius of the fifth Bohr orbit compared to that of the first Bohr orbit?

Answer: Step 1: Given: orbit numbers n = 5 and n = 1, radius of first orbit r1 = r0

Step 2: Formula: the radius of the nth orbit is r n = r0 multiplied by n2

Step 3: Substitution: r5 = r0 x 25 and r1 = r0 x 1

Step 4: Calculation: ratio r5 / r1 = 25 / 1 = 25

Step 5: Result: The fifth Bohr orbit has 25 times the radius of the first orbit.

Question: What is the difference between X-Rays and Gamma Rays?

Answer: X-rays and gamma rays are both high-energy electromagnetic radiations, and they differ mainly in their origin. X-rays are produced outside the nucleus, when fast electrons strike a target or when electrons in inner shells of an atom change levels. Gamma rays are emitted from inside an unstable nucleus when it changes from an excited state to a lower energy state after alpha or beta decay. Gamma rays are generally of higher energy and are more penetrating than X-rays.

Question: State the properties of X-Rays, which makes it possible to detect

cracks in bones.

Answer: X-rays travel in straight lines, are not deflected by electric or magnetic fields, and can pass through flesh and soft tissue much more easily than through dense bone, because bone absorbs them more strongly. They also affect a photographic film and cause fluorescence on suitable screens. When X-rays pass through the body, bones cast darker shadows on the film than the surrounding tissue, so a crack or fracture, where absorption suddenly changes, becomes clearly visible.

Question: What is the energy of a photon that, when absorbed by a hydrogen atom, could cause an electronic transition from (a) the n = 3 state to the n = 5 state and (b) the n = 5 state to the n = 8 state?

Answer: Step 1: Given: ground state energy of hydrogen E1 = -13.6 eV, and the energy of level n is En = -13.6 / n2 eV

Step 2: Formula: photon energy needed = E higher - E lower = 13.6 x (1 / n lower2 - 1 / n higher2) eV

Step 3: Substitution for part (a): E = 13.6 x (1 / 9 - 1 / 25) = 13.6 x 0.0711

Step 4: Calculation for part (a): E = 0.97 eV

Step 5: Substitution for part (b): E = 13.6 x (1 / 25 - 1 / 64) = 13.6 x 0.0244

Step 6: Calculation and Result: E = 0.33 eV, so photons of about 0.97 eV and 0.33 eV respectively can produce the two transitions, and both lie in the infrared region.

Question: What are the (a) wavelength range and (b) frequency range of the Lyman series and the Balmer series?

Answer: The Lyman series is produced by transitions ending at n = 1, and its wavelengths run from about 121.6 nm for the 2 to 1 transition down to the series limit of about 91.2 nm, all in the ultraviolet. Its frequencies

therefore run from about 2.47x10^15 Hz up to about 3.29x10^15 Hz. The Balmer series ends at n = 2, with wavelengths from about 656 nm for the 3 to 2 line down to the series limit of about 365 nm, in the visible and near ultraviolet. Its frequencies run from about 4.57x10^14 Hz up to about 8.22x10^14 Hz.

Question: Distinguish between spontaneous and stimulated emission.

Answer: In spontaneous emission an atom in an excited state falls to a lower state by itself, without any external influence, and emits a photon in a random direction with random phase. In stimulated emission an incoming photon of exactly the transition energy triggers the excited atom to fall and emit a second photon identical in frequency, phase and direction to the first. Spontaneous emission gives the ordinary incoherent light of lamps and discharge tubes. Stimulated emission, by contrast, produces the coherent and amplified light of a laser.

Question: Explain why population inversion is necessary in a laser?

Answer: In normal matter most atoms are in the ground state, so an incoming photon is far more likely to be absorbed than to stimulate emission, and no amplification can occur. Population inversion reverses this situation by keeping more atoms in the upper, metastable state than in the lower state. A passing photon then stimulates more emission than absorption, and one photon quickly multiplies into a large number of identical photons. Without this inversion the beam cannot grow, so inversion is the essential condition for laser action.

Question: In an optically pumped laser, the light that causes optical pumping is always shorter in wavelength than the laser beam. Explain.

Answer: The pumping light must raise atoms from the ground state to a level above the upper laser level, so each pump photon must carry at least as much energy as the full jump, which is more than the energy of

the laser transition itself. Part of the absorbed energy is then lost as heat when the atoms drop without radiation to the metastable laser level. Since photon energy is inversely proportional to wavelength, the more energetic pump light has the shorter wavelength. The laser light, corresponding to the smaller remaining jump, therefore always has the longer wavelength.

Question: A hydrogen atom is in its first excited state (n = 2). Calculate (a) the kinetic energy of the electron, (b) the potential energy of the system, and (c) the total energy of the system.

Answer: Step 1: Given: level n = 2, and for hydrogen the total energy is En = -13.6 / n2 eV

Step 2: Formula: total energy E2 = -13.6 / 4, kinetic energy KE = -E total, and potential energy PE = 2 x E total

Step 3: Substitution: E2 = -13.6 / 4 = -3.4 eV

Step 4: Calculation: KE = -(-3.4) = +3.4 eV, and PE = 2 x (-3.4) = -6.8 eV

Step 5: Check: KE + PE = 3.4 - 6.8 = -3.4 eV, which equals the total energy

Step 6: Result: For n = 2 the kinetic energy is 3.4 eV, the potential energy is -6.8 eV and the total energy is -3.4 eV.

Exercise — ERQs (Long Answered Questions)

Question: What are the postulates of Bohr Model of hydrogen atom? Discuss the importance of this model to explain various series of line spectra in hydrogen atom. Do any of the assumptions of the Bohr theory of hydrogen atom contradict with the classical Physics? Derive the expression for total energy of electron in nth Bohr orbit and show that En is proportional to 1 / n2.

Answer: Bohr postulated that the electron moves in certain stable circular

orbits without radiating energy, that its angular momentum is quantised as mvr = nh / (2 pi), and that a photon is emitted or absorbed only in a jump between orbits, with hf equal to the energy difference. These postulates explain the hydrogen series directly: jumps ending at n = 1 give the Lyman series, at n = 2 the Balmer series, and at n = 3, 4 and 5 the Paschen, Brackett and Pfund series, all from the one formula for En. The model contradicts classical physics because classical theory requires an orbiting, and therefore accelerating, electron to radiate continuously and spiral into the nucleus, which Bohr simply forbade for the allowed orbits. For the energy, the Coulomb force supplies the centripetal force, mv2 / r = e2 / (4 pi epsilon0 r2), and combining this with quantised angular momentum gives KE = +13.6 / n2 eV and PE = -27.2 / n2 eV. Adding them, En = KE + PE = -13.6 / n2 eV, which clearly shows that En is proportional to 1 / n2 and becomes zero at n equal to infinity.

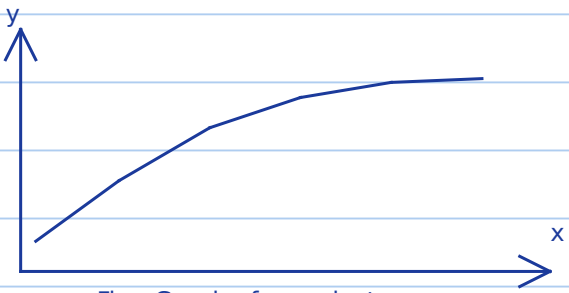


Fig.: Graph of y against x

Question: How X-rays are produced? State the purpose of cooling fins in the X-ray tube.

Answer: X-rays are produced in an evacuated tube when a beam of fast electrons, accelerated through a high voltage from a heated cathode, is suddenly stopped on striking a metal target such as tungsten set in the anode. Most of the kinetic energy of the electrons becomes heat, but a small fraction is converted into X-ray photons by sudden deceleration and by ejecting inner-shell electrons of the target atoms. The shortest wavelength produced is fixed by the tube voltage through eV = hc /

lambda min. Because about 99 percent of the energy appears as heat, the anode would melt without cooling. Cooling fins increase the surface area of the anode and help it lose this heat quickly, often together with oil circulation and a rotating anode, so the tube can work safely for long periods.

Question: Explain why X-rays are appropriate in study of crystalline structure material? Write some main properties of X-rays.

Answer: The wavelength of X-rays, about 10^-10 m, is of the same order as the spacing between atoms in a crystal, so a crystal acts as a natural three-dimensional diffraction grating for them. The regular planes of atoms reflect the X-rays, and the reflected beams reinforce only at angles satisfying the Bragg condition, giving a diffraction pattern from which the arrangement and spacing of atoms can be calculated. The main properties of X-rays are that they travel in straight lines at the speed of light, are not deflected by electric or magnetic fields, and penetrate matter to a degree depending on its density. They affect photographic film, cause fluorescence, and ionise gases through which they pass. They are also harmful to living cells in large doses, so their use needs proper shielding.

Question: What is Laser? Write the characteristics of Laser light. Can a two-level system be used for the production of Laser? Why?

Answer: A laser is a device that produces an intense, coherent beam of light by stimulated emission of radiation, the name being short for Light Amplification by Stimulated Emission of Radiation. Laser light is monochromatic, highly coherent, highly directional with very little spreading, and very intense compared with ordinary light sources. These properties come from photons that are identical in frequency, phase and direction. A two-level system cannot be used for laser production, because pumping can at best make the two populations equal: as soon as atoms

are raised, stimulated emission by the pump light itself sends them down again. Population inversion, the condition needed for net amplification, is therefore impossible in a steady two-level system, and at least three levels with a metastable state are required.

Question: What is pumping? What are the different methods of pumping? Explain optical pumping.

Answer: Pumping is the process by which atoms of the active medium are raised from the ground state to an excited state so that population inversion can be created and maintained. The main methods are optical pumping by a bright light source, electrical pumping by a gas discharge, chemical pumping using the energy of a chemical reaction, and pumping by direct electric current in semiconductor lasers. In optical pumping, used in the ruby laser, a powerful flash tube surrounds the ruby rod and its intense flash is absorbed by chromium atoms in the rod. The atoms rise to a broad upper band, drop quickly without radiation to a metastable level, and collect there. When the metastable level holds more atoms than the lower level, population inversion is achieved and laser action can begin.

Question: What is the principle of Laser? Write the construction and working of Helium-neon laser.

Answer: The principle of the laser is stimulated emission in a medium that has been pumped into population inversion, placed between two mirrors forming an optical resonator so that the light is amplified by repeated passes. A helium-neon laser consists of a long discharge tube filled with helium and neon in the ratio of about 10 to 1, with a fully reflecting mirror at one end and a partly transparent mirror at the other, and electrodes connected to a high voltage supply. An electric discharge excites the helium atoms, which collide with neon atoms and transfer energy to them because their excited levels nearly coincide, producing population

inversion in neon. Neon atoms then emit red laser light of wavelength 632.8 nm by stimulated emission. The mirrors reflect the light back and forth for further amplification, and a small part escapes through the partial mirror as the laser beam.

Question: Give some important properties of lasers. Also write the uses of lasers in the field of medicine, defense and communication.

Answer: Laser light is monochromatic, coherent, highly directional and very intense, and it can be focused to an extremely small spot of very high energy density. It can also be produced in very short pulses and travels without appreciable spreading over long distances. In medicine lasers are used in eye surgery, in removing tumours, in dentistry and as a bloodless scalpel that cuts and seals tissue at the same time. In defence they are used for range finding, target marking, guidance of weapons and measurement of distances. In communication they carry information through optical fibres, where their high frequency allows a very large number of signals to be sent at the same time.

Question: What is wave number? Derive the expression 1 / lambda = RH (1 / p2 - 1 / n2).

Answer: The wave number of radiation is the number of waves contained in unit length, equal to 1 / lambda, and it is directly proportional to the frequency and energy of the photon. In the Bohr model the energy of level n is En = -RH hc / n2, where RH is the Rydberg constant for hydrogen. When an electron jumps from a higher level n to a lower level p, the photon emitted has energy hf = RH hc (1 / p2 - 1 / n2). Dividing both sides by hc and using f / c = 1 / lambda gives 1 / lambda = RH (1 / p2 - 1 / n2), with RH = 1.097x10^7 per metre. Setting p = 1, 2, 3, 4 and 5 in this one expression gives the Lyman, Balmer, Paschen, Brackett and Pfund series respectively.

Question: Derive the expression for Bohr radius and develop a general relation for radii of quantized orbits of hydrogen atom.

Answer: For an electron of mass m and charge -e moving in a circle of radius r around the proton, the Coulomb attraction provides the centripetal force, so mv2 / r = e2 / (4 pi epsilon0 r2). Bohr quantisation of angular momentum gives mvr = nh / (2 pi), from which v = nh / (2 pi m r). Substituting this value of v in the force equation and simplifying removes v and leaves r = epsilon0 n2 h2 / (pi m e2). For the first orbit, n = 1, this gives the Bohr radius r0 = 5.29x10^-11 m = 0.53 angstrom. The general relation is therefore r n = r0 n2, showing that the allowed radii increase as the square of the principal quantum number, as 1, 4, 9, 16 and so on.

Exercise — Numerical

Question: Calculate the energy of an electron in the n = 2 orbit of a hydrogen atom according to the Bohr model. (Ans. -5.447x10^-19 J)

Answer: Step 1: Given: orbit number n = 2, ground state energy of hydrogen E1 = -13.6 eV = -2.176x10^-18 J, and 1 eV = 1.602x10^-19 J

Step 2: Formula: energy of the nth orbit, En = E1 / n2

Step 3: Substitution: E2 = -13.6 / 4 eV = -3.40 eV

Step 4: Calculation in joule: E2 = -3.40 x 1.602x10^-19 = -5.447x10^-19 J

Step 5: Result: The energy of the electron in the n = 2 orbit is -5.447x10^-19 J, the negative sign showing that the electron is bound to the nucleus.

Question: Calculate the speed of the electron if it orbits in (a) the smallest allowed orbit and (b) the second smallest orbit? (c) If the electron moves to larger orbits, does its speed increase, decrease, or stay the same? (Ans. (a) 2.19x10^6 m/s (b) 1.09x10^6 m/s (c) the speed of the electron will

decrease)

Answer: Step 1: Given: the speed in the first Bohr orbit is v1 = 2.19x10^6 m/s, and speed varies as v n = v1 / n

Step 2: Formula: v n = (2.19x10^6 / n) m/s for the nth orbit of hydrogen

Step 3: Substitution for part (a): for the smallest orbit n = 1, v1 = 2.19x10^6 / 1 = 2.19x10^6 m/s

Step 4: Substitution for part (b): for the second orbit n = 2, v2 = 2.19x10^6 / 2 = 1.09x10^6 m/s

Step 5: Reasoning for part (c): speed is inversely proportional to n, so as n increases for larger orbits the speed falls

Step 6: Result: The speeds are 2.19x10^6 m/s in the first orbit and 1.09x10^6 m/s in the second orbit, and the speed decreases as the electron moves to larger orbits.

Question: What are the (a) energy, (b) magnitude of the momentum, and (c) wavelength of the photon emitted when a hydrogen atom undergoes a transition from a state with n = 3 to a state with n = 1? (Ans. (a) 12.1 eV (b) 6.45x10^-27 N s (c) 102 nm)

Answer: Step 1: Given: initial level ni = 3, final level nf = 1, E1 = -13.6 eV, h = 6.63x10^-34 J s, c = 3x10^8 m/s

Step 2: Formula: photon energy E = 13.6 x (1 / nf2 - 1 / ni2) eV, momentum p = E / c, and wavelength lambda = hc / E

Step 3: Substitution: E = 13.6 x (1 / 1 - 1 / 9) = 13.6 x 8 / 9

Step 4: Calculation of energy: E = 12.1 eV = 12.1 x 1.602x10^-19 = 1.938x10^-18 J

Step 5: Calculation of momentum: p = 1.938x10^-18 / 3x10^8 = 6.45x10^-27 N s

Step 6: Calculation of wavelength: lambda = (6.63x10^-34 x 3x10^8) / 1.938x10^-18 = 1.026x10^-7 m = 102.6 nm

Step 7: Result: The emitted photon has energy 12.1 eV, momentum 6.45x10^-27 N s and wavelength about 102 nm, in the Lyman series.

Question: What is the energy of the photon emitted by hydrogen atom when the hydrogen atom changes directly from the n = 5 state to the n = 2 state? (Ans. 2.85 eV)

Answer: Step 1: Given: initial level ni = 5, final level nf = 2, E1 = -13.6 eV

Step 2: Formula: photon energy E = 13.6 x (1 / nf2 - 1 / ni2) eV

Step 3: Substitution: E = 13.6 x (1 / 4 - 1 / 25) = 13.6 x (0.25 - 0.04)

Step 4: Calculation: E = 13.6 x 0.21 = 2.856 eV

Step 5: Result: The photon emitted in the direct 5 to 2 transition has an energy of about 2.85 eV and belongs to the Balmer series.

Question: How much work must be done to pull apart the electron and the proton that make up the hydrogen atom if the atom is initially in (a) its ground state and (b) the state with n = 3? (Ans. (a) 13.6 eV (b) 1.51 eV)

Answer: Step 1: Given: the total energy of the atom in level n is En = -13.6 / n2 eV, and to free the electron its total energy must be raised to zero

Step 2: Formula: work required W = 0 - En = +13.6 / n2 eV, which is the ionisation energy from that level

Step 3: Substitution for part (a): for the ground state n = 1, W = 13.6 / 1 = 13.6 eV

Step 4: Substitution for part (b): for n = 3, W = 13.6 / 9 = 1.51 eV

Step 5: Result: Work of 13.6 eV is needed from the ground state, but only 1.51 eV from the n = 3 state, because the electron is already less tightly bound there.

Question: (a) What is the wavelength of light for the least energetic photon emitted in the Balmer series of the hydrogen atom spectrum lines? (b) What is the wavelength of the series limit? ((a) 659 nm (b) 366 nm)

Answer: Step 1: Given: the Balmer series ends at nf = 2; the least energetic photon comes from the smallest jump, ni = 3 to nf = 2, and the series limit comes from ni = infinity to nf = 2

Step 2: Formula: E = 13.6 x (1 / nf2 - 1 / ni2) eV and lambda = hc / E, with hc = 1240 eV nm

Step 3: Substitution for part (a): E = 13.6 x (1 / 4 - 1 / 9) = 13.6 x 0.1389 = 1.89 eV

Step 4: Calculation for part (a): lambda = 1240 / 1.89 = 656 nm, about 659 nm as printed in the book

Step 5: Substitution for part (b): at the limit E = 13.6 x (1 / 4 - 0) = 3.40 eV

Step 6: Calculation for part (b): lambda = 1240 / 3.40 = 365 nm, about 366 nm

Step 7: Result: The least energetic Balmer photon has wavelength about 659 nm (red light) and the Balmer series limit is about 366 nm.

Question: A laser emits light with a wavelength of 632.8 nm and has a power output of 55 mW. Calculate the energy of one photon emitted by this laser. (Ans. 3.14x10^-19 J)

Answer: Step 1: Given: wavelength lambda = 632.8 nm = 632.8x10^-9 m, h = 6.63x10^-34 J s, c = 3x10^8 m/s, E = ?

Step 2: Formula: energy of one photon, E = hc / lambda; the power output is not needed for the energy of a single photon

Step 3: Substitution: E = (6.63x10^-34 x 3x10^8) / 632.8x10^-9

Step 4: Calculation: numerator = 1.989x10^-25, so E = 1.989x10^-25 /

6.328x10^-7 = 3.14x10^-19 J

Step 5: Result: Each photon of the helium-neon laser light carries an energy of 3.14x10^-19 J, about 1.96 eV.

Question: Calculate the wavelength of X-rays if the energy of one photon emitted by the X-ray machine is 1.9878x10^-15 joules. (Ans. 0.1 nm)

Answer: Step 1: Given: photon energy E = 1.9878x10^-15 J, h = 6.63x10^-34 J s, c = 3x10^8 m/s, lambda = ?

Step 2: Formula: E = hc / lambda, rearranged as lambda = hc / E

Step 3: Substitution: lambda = (6.63x10^-34 x 3x10^8) / 1.9878x10^-15

Step 4: Calculation: numerator = 1.989x10^-25, so lambda = 1.989x10^-25 / 1.9878x10^-15 = 1.0x10^-10 m

Step 5: Result: The wavelength of the X-rays is 1.0x10^-10 m, that is 0.1 nm, typical of X-rays used in diffraction and medical work.

Unit 27: Nuclear Physics

Exercise — Multiple Choice Questions (MCQs)

Identify particle x in the following nuclear reaction Be-9 (Z = 4) + He-4 (Z = 2) to C-12 (Z = 6) + x:

(a) Electron

(b) proton

✓ (c) neutron

(d) photon

In the equation Al-27 (Z = 13) + He-4 (Z = 2) to P-30 (Z = 15) + X, the correct symbol for X is:

(a) electron, A = -1 and Z = 0

(b) hydrogen, A = 1 and Z = 1

(c) helium, A = 4 and Z = 2

✓ (d) neutron, A = 1 and Z = 0

Beta rays emitted by a radioactive material are:

(a) Electromagnetic radiations

(b) The electrons orbiting around the nucleus

✓ (c) Charged particles emitted by the nucleus

(d) Neutral particles

The number of alpha and beta particles emitted in the following radioactive decay is: X-200 (Z = 90) to Y-168 (Z = 80) + ?X

✓ (a) 8 and 6

(b) 6 and 8

(c) 8 and 8

(d) 6 and 6

If radium has half-life of 5 years. Thus for a nucleus in a sample of radium,

the probability of decay in ten years is:

(a) 50%

✓ (b) 75%

(c) 100%

(d) 60%

Emission of beta particles by an element affects its mass number A in the following way:

(a) Increases by 1

(b) decreases by 1

(c) Increases by 2

✓ (d) remains the same

As the number of nucleons in a nucleus increases, the binding energy per nucleon:

(a) Increases continuously with mass number

(b) Decreases continuously with mass number

(c) remains constant with mass number

✓ (d) First increases and then decreases with increase of mass number

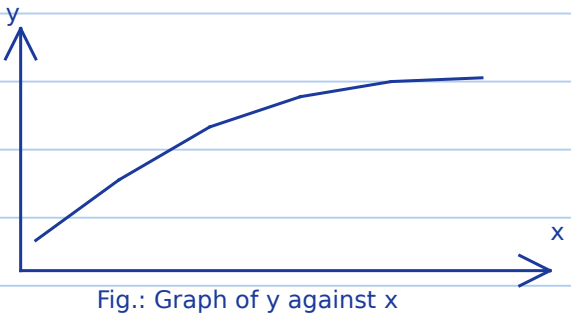


Fig.: Graph of y against x

Moderator in a nuclear reactor slows down the neutrons to:

(a) Decrease the probability of escape

✓ (b) Increase the probability of nuclear fission

(c) Decrease the probability of absorption

(d) all of the above

Emission of beta-plus particles by an element affects its atomic number Z

in the following way:

(a) Increases by 1

✓ (b) decreases by 1

(c) Increases by 2

(d) remains the same

The half life period of a radioactive element is 100 days. After 400 days, 16 g of the element will be reduced to

(a) 8g

(b) 4g

(c) 2g

✓ (d) 1g

Exercise — CRQs (Short Answered Questions)

Question: Why are protons and neutrons necessary for the stability of an atomic nucleus?

Answer: Protons repel one another strongly because they all carry positive charge, so a nucleus of protons alone would fly apart at once. Neutrons carry no charge but attract both protons and neutrons through the strong nuclear force, which acts over the very short range inside the nucleus. By adding attraction without adding repulsion, and by keeping the protons slightly farther apart, neutrons hold the nucleus together. A suitable balance of protons and neutrons is therefore necessary, and nuclei with a badly unbalanced ratio are unstable and radioactive.

Question: How do isotopes of an element differ and why are these differences significant?

Answer: Isotopes of an element have the same number of protons and electrons but different numbers of neutrons, so they have the same atomic number Z but different mass numbers A. Their chemical properties

are almost identical, because chemistry depends on the electrons, but their nuclear properties differ greatly: some isotopes are stable while others are radioactive with different half-lives. These differences are significant because radioactive isotopes are used as tracers in medicine and industry, in radiotherapy, and in dating old materials, while the heavy isotope uranium-235, unlike uranium-238, can sustain a fission chain reaction.

Question: Name the two methods of controlling a chain reaction.

Answer: The first method is the use of control rods made of a neutron-absorbing material such as boron or cadmium, which are pushed into or pulled out of the reactor core to absorb more or fewer neutrons and so control the rate of fission. The second method is the use of a moderator, such as ordinary water, heavy water or graphite, which controls the speed of the neutrons and hence the probability that they will cause further fission. Together these methods keep the multiplication factor at exactly one for steady working. In an emergency the control and safety rods are dropped fully in to stop the reaction quickly.

Question: Why is nuclear decay described as spontaneous and random?

Answer: Nuclear decay is called spontaneous because it happens by itself, without any external cause, and it is not affected by temperature, pressure or chemical state of the material. It is called random because it is impossible to predict which particular nucleus will decay next or exactly when any one nucleus will decay. We can only state the probability that a nucleus decays in a given time, expressed through the decay constant and the half-life. For a very large number of nuclei this probability gives the accurate average behaviour described by the decay law.

Question: Explain what happens during the nuclear fusion process.

Answer: In nuclear fusion two light nuclei, such as hydrogen isotopes,

come so close together that the strong nuclear force joins them into a single heavier nucleus. To overcome the repulsion between their positive charges the nuclei must move at very high speed, which needs temperatures of millions of kelvin as in the interior of the Sun. Because the product nucleus has a smaller mass than the two original nuclei together, the mass lost appears as energy according to E = mc2. Fusion of hydrogen into helium is the source of energy of the Sun and the stars, and of the hydrogen bomb.

Question: How are activity and decay constant related in radioactive materials?

Answer: The activity of a sample is the number of nuclei decaying per second, and it is directly proportional to the number of undecayed nuclei N present at that time. The constant of proportionality is the decay constant lambda, so activity A = lambda N, measured in becquerel. The decay constant is the probability that one nucleus decays in one second, so a large lambda means a rapidly decaying, short-lived material. As N decreases with time, the activity decreases in the same exponential way, A = A0 e raised to -lambda t.

Question: What is meant by binding energy and binding energy per nucleon?

Answer: The binding energy of a nucleus is the energy that must be supplied to break the nucleus completely into its separate protons and neutrons, and it equals the mass defect of the nucleus multiplied by c2. It is a measure of how strongly the nucleus is held together by the nuclear force. The binding energy per nucleon is this total binding energy divided by the mass number A. It tells the average energy needed to remove one nucleon, and the higher its value, the more stable the nucleus; it is maximum, about 8.8 MeV, near iron.

Question: Why is the concept of half-life important in studying radioactive substances?

Answer: Half-life gives a simple and fixed measure of how fast a radioactive substance decays: it is the time in which half of the nuclei present, and hence half of the activity, disappear. It allows us to calculate how much of a sample will remain after any time and how active a source will be when it is used. Choosing an isotope with a suitable half-life is essential in medicine, where the source must work long enough for treatment but not remain active in the body for too long. Half-life is also the basis of radioactive dating, for example carbon-14 dating of fossils.

Question: Explain the process of positron emission.

Answer: Positron emission, or beta-plus decay, occurs in nuclei that have too many protons compared with neutrons. A proton inside the nucleus changes into a neutron, a positron and a neutrino, and the positron and neutrino are emitted from the nucleus. As a result the atomic number Z decreases by 1 while the mass number A stays the same, and the nucleus moves towards stability. The emitted positron soon meets an ordinary electron and the two annihilate, producing two gamma ray photons of 0.51 MeV each, which is the effect used in PET scanning in medicine.

Exercise — ERQs (Long Answered Questions)

Question: What is radioactivity? State the law of radioactive disintegration. Show that radioactive decay is exponential in nature.

Answer: Radioactivity is the spontaneous disintegration of an unstable nucleus with the emission of alpha, beta or gamma radiation, forming a new element. The law of radioactive disintegration states that the rate at which nuclei disintegrate at any instant is directly proportional to the number of undecayed nuclei present at that instant, that is -dN / dt =

lambda N, where lambda is the decay constant. Integrating this relation from N0 nuclei at t = 0 to N at time t gives ln (N / N0) = -lambda t, or N = N0 e raised to -lambda t. Because N contains the exponential factor e raised to -lambda t, the number of nuclei, and with it the activity, falls exponentially and never quite reaches zero. Putting N = N0 / 2 gives the half-life T half = 0.693 / lambda, which is constant for a given substance.

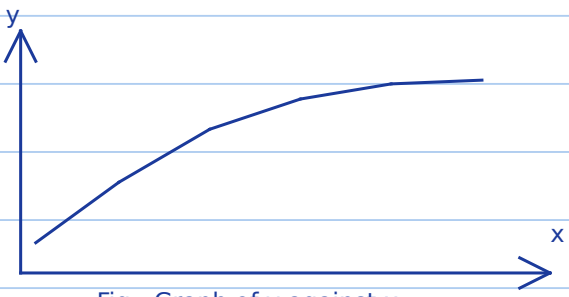


Fig.: Graph of y against x

Question: Explain the main differences between alpha, beta, and gamma emissions.

Answer: Alpha emission is the loss of a helium nucleus, with A = 4 and Z = 2, from a heavy nucleus; it decreases A by 4 and Z by 2, has a range of only a few centimetres in air, is stopped by paper, and is strongly ionising. Beta emission is the loss of a fast electron or positron created inside the nucleus; it leaves A unchanged but changes Z by 1, travels up to a few metres in air, is stopped by a few millimetres of aluminium, and is moderately ionising. Gamma emission is electromagnetic radiation of very short wavelength emitted when an excited nucleus falls to a lower energy state; it changes neither A nor Z, travels at the speed of light with a long range, is stopped only by thick lead, and is weakly ionising but highly penetrating. Alpha and beta particles are deflected by electric and magnetic fields in opposite senses appropriate to their charges, while gamma rays are not deflected at all.

Question: A fission reactor produces energy to drive a generator. Describe briefly how this energy is produced.

Answer: In the reactor core, a slow neutron striking a uranium-235 nucleus makes it unstable, and it splits into two lighter nuclei of comparable mass, called fission fragments, together with two or three fresh neutrons and a large amount of energy, about 200 MeV per fission. The mass of the products is slightly less than the original mass, and this mass defect appears mainly as the kinetic energy of the fragments and neutrons. Collisions of these particles inside the fuel rods convert the kinetic energy into heat. This heat is carried away by a coolant to a heat exchanger, where it raises steam that drives a turbine and generator in the usual way. Control rods and a moderator keep the chain reaction steady at the required power level.

Question: Define Q-value of a nuclear reaction and its significance.

Answer: The Q-value of a nuclear reaction is the net energy released in the reaction, equal to the total kinetic energy of the products minus the total kinetic energy of the reacting particles. By the mass-energy relation it is also Q = (total mass of reactants - total mass of products) multiplied by c2, expressed in MeV when masses are in atomic mass units using 1 u = 931.5 MeV. A positive Q means the reaction is exothermic and releases energy, while a negative Q means the reaction is endothermic and can occur only if the incoming particles bring at least that much kinetic energy. The Q-value therefore tells us whether a proposed reaction can supply energy, as in reactors and stars, and how much energy each reaction will give.

Question: Describe the construction and working of GM counter.

Answer: A Geiger-Muller counter consists of a metal tube acting as the cathode, filled with a low-pressure gas such as argon with a small amount of a quenching vapour, and a fine wire along its axis acting as the anode. A potential difference of about 400 to 500 V, just below the breakdown

voltage, is applied between the wire and the tube, and radiation enters through a thin mica window at one end. When an ionising particle enters, it ionises some gas atoms; the electrons produced rush towards the wire and, in the strong field near it, ionise many more atoms by collision, causing an avalanche and a sudden pulse of current. This pulse is amplified, counted by a scaler and can also produce a click in a loudspeaker, so the counter records the number of ionising events per unit time. The quenching vapour prevents continuous discharge, so after a short dead time the tube is ready for the next particle.

Question: Discuss nuclear reactions induced by neutrons. Why are neutrons preferred to other particles?

Answer: Neutrons induce many important nuclear reactions because, having no charge, they can enter a nucleus without being repelled by its positive charge, and they produce reactions such as fission of uranium-235, capture reactions forming new isotopes, and the production of artificial radioisotopes. Slow, thermal neutrons are especially effective for fission, while fast neutrons are used in other reactions and in breeding new fuel. Charged particles such as protons and alpha particles must be given high energy in accelerators to overcome the Coulomb barrier before they can reach the nucleus. Neutrons face no such barrier and can react even at very low energies. For this reason neutron-induced reactions are the working basis of nuclear reactors and of most practical production of radioisotopes.

Question: What is a nuclear reaction? Explain nuclear fission and fusion.

Answer: A nuclear reaction is a process in which a nucleus is changed by striking it with a particle such as a neutron, proton or alpha particle, producing new nuclei and usually releasing or absorbing energy; both A and Z are balanced on the two sides of the reaction equation. In fission a

heavy nucleus such as uranium-235 splits, after absorbing a neutron, into two medium nuclei with the release of more neutrons and about 200 MeV of energy, and the escaping neutrons can produce a chain reaction. In fusion two light nuclei join to form a heavier nucleus, and the mass lost appears as energy; this needs extremely high temperature to overcome the repulsion between the nuclei. Fission is used in nuclear reactors and the atomic bomb, while fusion powers the Sun and the hydrogen bomb and is being developed for future power stations.

Question: How do the Sun and stars produce energy? What is the proton-proton cycle? Explain with details.

Answer: The Sun and stars produce energy by nuclear fusion in their extremely hot, dense cores, where hydrogen nuclei are converted into helium and the mass lost appears as radiant energy according to E = mc2. In the proton-proton cycle, which dominates in the Sun, two protons first combine to form a deuteron with the emission of a positron and a neutrino. The deuteron then captures another proton to form light helium-3, and finally two helium-3 nuclei combine to form ordinary helium-4 and release two protons back. The net result is that four protons are converted into one helium-4 nucleus, two positrons, two neutrinos and about 25 MeV of energy, the positrons annihilating with electrons to add further energy. This energy slowly travels to the surface of the Sun and is radiated into space as light and heat, sustaining life on Earth.

Exercise — Numerical

Question: In 9.0 days the number of radioactive nuclei decreases to one-eighth the number present initially. What is the half-life (in days) of the material? [Ans: 3 days]

Answer: Step 1: Given: time t = 9.0 days, final number N = N0 / 8, half-life

T = ?

Step 2: Formula: after each half-life the number halves, so N = N0 x (1/2) raised to (t / T), that is 1/8 = (1/2) raised to (t / T)

Step 3: Substitution: (1/2)^3 = (1/2) raised to (9.0 / T), because 1/8 = (1/2)^3

Step 4: Calculation: comparing powers, 9.0 / T = 3, so T = 9.0 / 3 = 3.0 days

Step 5: Result: The half-life of the material is 3 days.

Question: The P-32 (Z = 15) isotope of phosphorus has a half-life of 14.28 days. What is its decay constant in units of s-1? [Ans: 5.62x10^-7 s-1]

Answer: Step 1: Given: half-life T = 14.28 days, decay constant lambda = ?

Step 2: Formula: lambda = 0.693 / T, with T expressed in seconds

Step 3: Substitution for T in seconds: T = 14.28 x 24 x 3600 = 1.234x10^6 s

Step 4: Calculation: lambda = 0.693 / 1.234x10^6 = 5.62x10^-7 per second

Step 5: Result: The decay constant of phosphorus-32 is 5.62x10^-7 s-1, meaning each nucleus has this probability of decay in one second.

Question: Find the binding energy (in MeV) for lithium Li-7 (Z = 3) (atomic mass = 7.016003 u). [Ans: 39.2 MeV]

Answer: Step 1: Given: Li-7 has Z = 3 protons and N = 4 neutrons, atomic mass of Li-7 = 7.016003 u, mass of a hydrogen atom = 1.007825 u, mass of neutron = 1.008665 u, and 1 u = 931.5 MeV

Step 2: Formula: mass defect dm = (Z x mass of hydrogen atom + N x mass of neutron) - atomic mass, and binding energy BE = dm x 931.5 MeV

Step 3: Substitution: dm = (3 x 1.007825 + 4 x 1.008665) - 7.016003 = (3.023475 + 4.034660) - 7.016003

Step 4: Calculation: dm = 7.058135 - 7.016003 = 0.042132 u, so BE = 0.042132 x 931.5 = 39.25 MeV

Step 5: Result: The binding energy of lithium-7 is about 39.2 MeV, or about 5.6 MeV per nucleon.

Question: The binding energy of a nucleus is 225.0 MeV. What is the mass defect of the nucleus in atomic mass units? [Ans: 0.2415 u]

Answer: Step 1: Given: binding energy BE = 225.0 MeV, mass defect dm = ?, and 1 u of mass corresponds to 931.5 MeV

Step 2: Formula: BE = dm multiplied by c2, so dm = BE / 931.5 in atomic mass units

Step 3: Substitution: dm = 225.0 / 931.5

Step 4: Calculation: dm = 0.2415 u

Step 5: Result: The mass defect of the nucleus is 0.2415 u, that is, the nucleus is lighter than its separate nucleons by this amount.

Question: A copper penny has a mass of 3.0 g. Determine the energy (in MeV) that would be required to break all the copper nuclei into their constituent protons and neutrons. Ignore the energy that binds the electrons to the nucleus and the energy that binds one atom to another in the structure of the metal. For simplicity, assume that all the copper nuclei are Cu-63 (Z = 29) (atomic mass = 62.939598 u). [Ans: 1.51249x10^25 MeV]

Answer: Step 1: Given: mass of penny = 3.0 g, Cu-63 has Z = 29 and N = 34, atomic mass = 62.939598 u, hydrogen atom mass = 1.007825 u, neutron mass = 1.008665 u, Avogadro number = 6.022x10^23 per mol, 1

u = 931.5 MeV

Step 2: Formula: mass defect per nucleus dm = (29 x 1.007825 + 34 x 1.008665) - 62.939598, binding energy per nucleus = dm x 931.5 MeV, number of nuclei = (3.0 / 62.939598) x 6.022x10^23, and total energy = binding energy per nucleus x number of nuclei

Step 3: Substitution for mass defect: dm = (29.226925 + 34.294610) - 62.939598 = 63.521535 - 62.939598 = 0.581937 u

Step 4: Calculation of binding energy per nucleus: BE = 0.581937 x 931.5 = 542.1 MeV

Step 5: Calculation of number of nuclei: N = (3.0 / 62.939598) x 6.022x10^23 = 2.87x10^22 nuclei

Step 6: Calculation of total energy: E total = 542.1 x 2.87x10^22 = about 1.5x10^25 MeV

Step 7: Result: About 1.5x10^25 MeV (the book gives 1.51249x10^25 MeV) would be needed to break all the copper nuclei in the penny into separate protons and neutrons.

Question: Write the beta-plus decay process for each of the following nuclei with their proper chemical symbols including Z and A for each daughter nucleus: (a) F-18 (b) O-15. [Ans: (a) F-18 (Z = 9) to O-18 (Z = 8) + e+ + neutrino (b) O-15 (Z = 8) to N-15 (Z = 7) + e+ + neutrino]

Answer: Step 1: Given: in beta-plus decay a proton changes into a neutron, so A stays the same and Z decreases by 1, and a positron e+ and a neutrino are emitted

Step 2: Rule applied to part (a): fluorine F-18 has Z = 9, so the daughter has Z = 8 and A = 18, which is oxygen

Step 3: Equation for part (a): F-18 (Z = 9) changes to O-18 (Z = 8) + e+ +

neutrino

Step 4: Rule applied to part (b): oxygen O-15 has Z = 8, so the daughter has Z = 7 and A = 15, which is nitrogen

Step 5: Equation for part (b): O-15 (Z = 8) changes to N-15 (Z = 7) + e+ + neutrino

Step 6: Result: The two decay processes are F-18 to O-18 plus a positron and a neutrino, and O-15 to N-15 plus a positron and a neutrino, with A unchanged and Z reduced by 1 in each case.

Question: A device used in radiation therapy for cancer contains 0.50 g of cobalt Co-60 (Z = 27) (59.933819 u). The half-life of Co-60 (Z = 27) is 5.27 years. Determine the activity of the radioactive material. [Ans: 2.10x10^13 decays/second (Bq)]

Answer: Step 1: Given: mass = 0.50 g, molar mass of Co-60 = 59.933819 g per mol, half-life T = 5.27 years, Avogadro number = 6.022x10^23 per mol, activity A = ?

Step 2: Formula: number of nuclei N = (mass / molar mass) x Avogadro number, decay constant lambda = 0.693 / T, and activity A = lambda N

Step 3: Substitution for N: N = (0.50 / 59.933819) x 6.022x10^23 = 5.02x10^21 nuclei

Step 4: Substitution for lambda: T = 5.27 x 365 x 24 x 3600 = 1.66x10^8 s, so lambda = 0.693 / 1.66x10^8 = 4.17x10^-9 per second

Step 5: Calculation of activity: A = 4.17x10^-9 x 5.02x10^21 = 2.09x10^13 decays per second

Step 6: Result: The activity of the cobalt source is about 2.10x10^13 Bq, which is why cobalt-60 is a strong and useful source for radiotherapy.

Unit 28: Particle Physics

Exercise — Multiple Choice Questions (MCQs)

The Standard Model classifies elementary particles into two main groups:

(a) Baryons and Leptons

✓ (b) Fermions and Bosons

(c) Quarks and Gluons

(d) Hadrons and Mesons

The following is NOT a flavor of quark:

(a) Up

(b) Down

✓ (c) Electron

(d) Top

The charge of an up quark is:

(a) +1/2 e

(b) -1/2 e

✓ (c) +2/3 e

(d) -2/3 e

The concept of quark confinement implies that:

✓ (a) Quarks cannot exist as free particles outside of hadrons.

(b) Quarks are always found in pairs with opposite charges.

(c) Quarks have a strong affinity for gluons.

(d) Quarks are the fundamental building blocks of all matter.

The primary role of the Higgs boson in the Standard Model is:

(a) Mediating electromagnetic interactions

✓ (b) Providing mass to other particles

(c) Transmitting the strong nuclear force

(d) Creating dark matter

According to the Standard Model, the term color refers to:

(a) Visible light spectrum

✓ (b) Charge property of quarks

(c) Mass of particles

(d) Spin of particles

A particle made up of a quark and an antiquark is called:

(a) Lepton

(b) Baryon

✓ (c) Meson

(d) Neutrino

An elementary particle that feels all three fundamental forces (electromagnetic, weak, and strong nuclear forces) is:

(a) Lepton

✓ (b) Quark

(c) Electron

(d) Neutrino

The primary function of a Geiger-Muller counter in particle physics is to:

(a) Measure the velocity of particles

✓ (b) Detect and count ionizing radiation

(c) Create antimatter particles

(d) Generate magnetic fields

In a Geiger-Muller counter, the gas commonly used to detect ionizing radiation is:

(a) Oxygen

(b) Neon

✓ (c) Argon

(d) Helium

Exercise — CRQs (Short Answered Questions)

Question: Explain the difference between bosons and fermions and their roles in mediating fundamental forces.

Answer: Fermions are particles with half-integer spin, such as quarks, electrons and neutrinos, which obey the Pauli exclusion principle and form the matter of the universe. Bosons are particles with integer spin, such as the photon, gluon and W and Z particles, which do not obey the exclusion principle in the same way. The fundamental forces arise from the exchange of bosons between fermions: photons carry the electromagnetic force, gluons the strong force, and W and Z bosons the weak force. In short, fermions are the particles of matter and bosons are the carriers, or field particles, of the forces between them.

Question: Compare and contrast the properties of quarks and leptons, the two main categories of fermions in the Standard Model.

Answer: Quarks and leptons are both elementary fermions with spin 1/2, and each family has six members arranged in three generations. Quarks carry fractional electric charge, +2/3 e or -1/3 e, and also carry color charge, so they feel all three forces: strong, electromagnetic and weak. Leptons carry whole-number charge, such as -e for the electron or zero for the neutrinos, and they carry no color charge, so they do not feel the strong force. Quarks are always confined inside hadrons, while leptons such as electrons and neutrinos can exist freely on their own.

Question: Define the term lepton and provide examples of leptons. Explain their fundamental properties and role in the Standard Model of particle physics.

Answer: A lepton is an elementary fermion that does not take part in the strong nuclear interaction; the name comes from a word meaning light or small. The leptons are the electron, muon and tau, together with their

three matching neutrinos: the electron neutrino, muon neutrino and tau neutrino. They have spin 1/2, the charged leptons each carry charge -e, and the neutrinos are neutral and have extremely small mass. In the Standard Model leptons form, together with quarks, the basic matter content of the universe, and the electron is essential to the structure of every atom.

Question: Explain the concept of color charge in quarks and its significance in the strong nuclear force. How does the combination of quarks contribute to the color-neutral nature of protons and neutrons?

Answer: Color charge is the charge of the strong force, carried by quarks in three forms called red, green and blue, and exchanged between quarks by gluons, which carry the strong interaction. It is significant because the strong force binds quarks into hadrons and, as a leftover effect, binds protons and neutrons together inside the nucleus. A proton or a neutron contains three quarks, one of each color, and the three colors combine to give a color-neutral, or white, total. In the same way a meson combines a color with its anticolor, so all particles observed freely in nature are color neutral, as confinement requires.

Question: Describe the structure of a proton and neutron in terms of its quark composition. How do quarks combine to form a proton and a neutron, and what are the specific types of quarks involved?

Answer: A proton is a baryon made of three quarks: two up quarks and one down quark, written uud, held together by gluons through the strong force. Its charge is +2/3 + 2/3 - 1/3 = +1 e, exactly the proton charge. A neutron is made of one up quark and two down quarks, written udd, and its charge is +2/3 - 1/3 - 1/3 = 0, so it is electrically neutral. Only the lightest quarks, up and down, are involved in ordinary protons and neutrons, and the three colors of the three quarks combine to make each

particle color neutral.

Exercise — ERQs (Long Answered Questions)

Question: Explain the structure of the Standard Model, including the different types of particles and their relationships. How does the model classify fundamental forces?

Answer: The Standard Model organises all known elementary particles into fermions, which form matter, and bosons, which carry forces. The fermions consist of six quarks (up, down, charm, strange, top and bottom) and six leptons (electron, muon, tau and three neutrinos), arranged in three generations of increasing mass, each quark and lepton having an antiparticle. Quarks combine as baryons of three quarks and mesons of a quark-antiquark pair to form the hadrons, while leptons exist freely. The gauge bosons are the photon, the eight gluons, and the W, Z and Higgs bosons, the Higgs giving mass to other particles through the Higgs field. The model classifies three fundamental forces in this scheme: the electromagnetic force carried by photons, the strong force carried by gluons between color charges, and the weak force carried by W and Z bosons, responsible for beta decay. Gravity is not included in the Standard Model, which is its best known limitation.

Question: Describe in detail all fundamental forces and their associated field particles. What is a boson, and why are bosons referred to as field particles?

Answer: The gravitational force, the weakest but longest-range force, acts between all masses and is described by general relativity; its proposed carrier, the graviton, has not been detected and lies outside the Standard Model. The electromagnetic force acts between electric charges, has infinite range, and is carried by the massless photon. The strong nuclear

force acts between color charges of quarks, is the strongest force but has a very short range, and is carried by the eight gluons, binding quarks in hadrons and nucleons in the nucleus. The weak force acts on quarks and leptons over an extremely short range and is carried by the massive W and Z bosons; it is responsible for beta decay and for changing quark flavour. A boson is a particle of integer spin, 0, 1 or 2. Bosons are called field particles because in quantum field theory every force field has its own boson, and the force between two particles is pictured as the exchange of these bosons between them.

Question: Describe the operating principle of a Geiger-Muller counter. How does it detect and quantify ionizing radiation, and what are its limitations in terms of measurement range and types of radiation detected?

Answer: The Geiger-Muller counter works on the ionisation of a gas by radiation: a metal tube filled with argon at low pressure has a central wire anode held a few hundred volts positive, just below the self-discharge voltage. Radiation entering through a thin mica window ionises gas atoms, and the freed electrons, accelerated in the intense field near the wire, produce an avalanche of further ionisation, giving one large standard pulse for each entering particle. The pulses are amplified and counted by a scaler, so the count rate measures the activity or intensity of the radiation reaching the tube. Its limitations are that every pulse has the same size, so it gives no information about the energy or type of the radiation, and it cannot distinguish alpha, beta and gamma events by itself. It also has a dead time after each pulse during which new events are missed, so it undercounts at very high rates, and weakly penetrating alpha particles can enter only through the thin window.

Question: Discuss the uses of a Wilson cloud chamber in particle physics experiments. How does it visualize charged particle tracks, and what

factors can affect its performance?

Answer: The Wilson cloud chamber makes the paths of charged particles visible and was used in classic discoveries such as the positron and several particles from cosmic rays. The chamber contains air saturated with water or alcohol vapour, and a sudden expansion cools the vapour into a supersaturated state in which it is ready to condense but needs ions as centres. A passing charged particle ionises the vapour along its path, droplets condense on these ions, and a thin white track of droplets appears and can be photographed. In a magnetic field the curvature of the track shows the charge and momentum of the particle, and the thickness of the track indicates its ionising power. Its performance is affected by the degree of supersaturation and the speed and timing of the expansion, by temperature and cleanliness of the chamber, and by the need to photograph the tracks in the brief moment before the droplets grow and fall.